
Basics of Long-Range Dependence and Self-Similarity

This chapter serves as a general introduction. It is necessary for understanding the more advanced topics covered in subsequent chapters. We introduce here basic concepts and results involving long-range dependence and self-similarity.

There are various definitions of long-range dependence, not all equivalent, because of the presence of slowly varying functions. These functions vary slower than a power; for example, they can be a constant or a logarithm. We give five definitions of long-range dependence. As discussed in Section 2.2, these definitions can be equivalent depending on the choice of the slowly varying functions. The reader should read the statement of the various propositions of that section, but their proofs which are somewhat technical may be skipped in a first reading.

Short-range dependent time series are introduced in Section 2.3. A typical example is the $\text{ARMA}(p, q)$. Examples of time series with long-range dependence, such as $\text{FARIMA}(0, d, 0)$ and $\text{FARIMA}(p, d, q)$ are given in Section 2.4. Self-similarity is defined in Section 2.5. Examples of self-similar processes are given in Section 2.6. These include:

- Fractional Brownian motion
- Bifractional Brownian motion
- The Rosenblatt process
- $S\alpha S$ Lévy motion
- Linear fractional stable motion
- Log-fractional stable motion
- The Telecom process
- Linear fractional Lévy motion

The Lamperti transformation, relating self-similarity to stationarity is stated in Section 2.7. Section 2.8 deals with the connection between long-range dependence and self-similarity. The infinite variance case is considered in Section 2.9. Section 2.10 focuses on heuristic methods of estimation of the self-similarity parameter. These include:

- The R/S method
- The aggregated variance method
- Regression in the spectral domain
- Wavelet-based estimation

Finally, we indicate in Section 2.11 how to generate long- and short-range dependent series, using the

- Cholesky decomposition

whose complexity is $O(N^3)$ and hence relatively slow, and the

- Circulant matrix embedding

which is an exact method with the complexity $O(N \log N)$.

2.1 Definitions of Long-Range Dependent Series

We start with the notion of *long-range dependence*, also called *long memory* or *strong dependence*. It is commonly defined for second-order stationary time series. Recall from Sections 1.1.3 and 1.2 that a second-order stationary time series $X = \{X_n\}_{n \in \mathbb{Z}}$ has a constant mean $\mu_X = \mathbb{E}X_n$ and an autocovariance function

$$\gamma_X(n - m) = \mathbb{E}X_n X_m - \mathbb{E}X_n \mathbb{E}X_m = \mathbb{E}X_n X_m - \mu_X^2,$$

where “ n ”, “ m ” are viewed as time and γ_X depends only on the time lag $|m - n|$. One can characterize the time series X by specifying its mean μ_X and autocovariance $\gamma_X(n)$, $n \in \mathbb{Z}$, or one can take a “spectral domain” perspective by focusing on the spectral density $f_X(\lambda)$, $\lambda \in (-\pi, \pi]$, of the time series, if it exists (see Section 1.3). It is defined as

$$\int_{-\pi}^{\pi} e^{in\lambda} f_X(\lambda) d\lambda = \gamma_X(n), \quad n \in \mathbb{Z},$$

that is, the spectral density $f_X(\lambda)$ is a function whose Fourier coefficients are the autocovariances $\gamma_X(n)$ (up to a multiplicative constant following the convention of Appendix A.1.1).

A *white noise* time series $Z = \{Z_n\}_{n \in \mathbb{Z}}$, defined in Example 1.2.1 and denoted $\{Z_n\} \sim \text{WN}(0, \sigma_Z^2)$, is a trivial example of a second-order stationary time series. It consists of uncorrelated random variables Z_n with zero mean and common variance σ_Z^2 , and for which

$$\gamma_Z(n) = \begin{cases} \sigma_Z^2, & \text{if } n = 0, \\ 0, & \text{if } n \neq 0, \end{cases} \quad f_Z(\lambda) = \frac{\sigma_Z^2}{2\pi}, \text{ for } \lambda \in (-\pi, \pi].$$

White noise time series are used in the construction of correlated time series. We are interested, in particular, in time series with long-range dependence. These involve a parameter

$$d \in (0, 1/2) \tag{2.1.1}$$

and a slowly varying function L .

Definition 2.1.1 A function L is *slowly varying at infinity* if it is *positive* on $[c, \infty)$ with $c \geq 0$ and, for any $a > 0$,

$$\lim_{u \rightarrow \infty} \frac{L(au)}{L(u)} = 1. \tag{2.1.2}$$

A function L is *slowly varying at zero* if the function $L(1/u)$ is slowly varying at infinity. A function R is *regularly varying at infinity* with exponent $\rho \in \mathbb{R}$ if

$$R(u) = u^\rho L(u), \tag{2.1.3}$$

where L is slowly varying at infinity. It is *regularly varying at zero* if $R(1/u)$ is regularly varying at infinity. A slowly varying function is a regularly varying function with $\rho = 0$.

(See Bingham, Goldie, and Teugels [157] for a comprehensive treatment of slowly varying functions.) Since $R(au)/R(u) \rightarrow a^\rho$, as $u \rightarrow \infty$, the slowly varying function can typically be ignored when considering the limit. For example, the functions $L(u) = \text{const} > 0$ and $L(u) = \log u$, $u > 0$, are slowly varying at infinity.

The presence of a slowly varying function is important in practice as it yields significantly greater flexibility. Thus, in condition IV below, we consider the spectral density $f_X(\lambda) = L_4(\lambda)\lambda^{-2d}$, where L_4 is slowly varying at zero. Without L_4 , we would have a very specific spectral density, namely, $f_X(\lambda) = \lambda^{-2d}$, $0 < \lambda \leq \pi$, but by choosing L_4 accordingly, we can have $f_X(\lambda) = e^{-\lambda}\lambda^{-2d}$ or even $f_X(\lambda) = |\log \lambda|\lambda^{-2d}$. But, for example, we cannot have $f_X(\lambda) = (2 + \cos(1/\lambda))\lambda^{-2d}$ since the function $2 + \cos(1/\lambda)$ is not slowly varying. The requirement that a slowly varying function is ultimately positive is part of the theory of such functions. Here we will allow a slowly varying function to be negative on $(0, c)$, $c > 0$, in order to gain flexibility in some of the conditions below.

Having defined the notion of slowly and regularly varying functions, we now turn to the various definitions of long-range dependence that can be found in the literature. Each definition involves a different condition which is stated below. Conditions I, II, III and V are expressed in the time domain and condition IV is expressed in the spectral domain. These conditions are, in general, not equivalent, as will be discussed below. They are, however, equivalent in particular cases, for example, for the FARIMA(0, d , 0) series defined in Section 2.4.

Condition I. The time series $X = \{X_n\}_{n \in \mathbb{Z}}$ has a linear representation

$$X_n = \mu_X + \sum_{k=0}^{\infty} \psi_k Z_{n-k} = \mu_X + \sum_{j=-\infty}^n \psi_{n-j} Z_j$$

with $Z = \{Z_n\}_{n \in \mathbb{Z}} \sim \text{WN}(0, \sigma_Z^2)$ and a sequence $\{\psi_k\}_{k \geq 0}$ satisfying

$$\psi_k = L_1(k)k^{d-1}, \quad k = 0, 1, \dots, \quad (2.1.4)$$

where L_1 is a slowly varying function at *infinity* and d is as in (2.1.1).

Condition II. The autocovariance function of the time series $X = \{X_n\}_{n \in \mathbb{Z}}$ satisfies

$$\gamma_X(k) = L_2(k)k^{2d-1}, \quad k = 0, 1, \dots, \quad (2.1.5)$$

where L_2 is a slowly varying function at *infinity*.

Condition III. The autocovariances of the time series $X = \{X_n\}_{n \in \mathbb{Z}}$ are not absolutely summable; that is,

$$\sum_{k=-\infty}^{\infty} |\gamma_X(k)| = \infty. \quad (2.1.6)$$

Condition IV. The time series $X = \{X_n\}_{n \in \mathbb{Z}}$ has a spectral density satisfying

$$f_X(\lambda) = L_4(\lambda)\lambda^{-2d}, \quad 0 < \lambda \leq \pi, \quad (2.1.7)$$

where L_4 is a slowly varying function at zero.

Condition V. The time series $X = \{X_n\}_{n \in \mathbb{Z}}$ satisfies

$$\text{Var}(X_1 + \cdots + X_N) = L_5(N)N^{2d+1}, \quad N = 1, 2, \dots, \quad (2.1.8)$$

where L_5 is a slowly varying function at infinity.

Remark 2.1.2 Since a spectral density f satisfies $f_X(\lambda) = f_X(-\lambda)$, it is enough to focus only on positive $\lambda > 0$. The convergence $\lambda \rightarrow 0$ in (2.1.7) as well as in other places below is understood to be for $\lambda > 0$. Alternatively, one could express (2.1.7) as $f_X(\lambda) = L_4(|\lambda|)|\lambda|^{-2d}$ and consider both positive and negative λ .

Remark 2.1.3 The slowly varying functions $L_1(k)$ and $L_2(k)$ in conditions I and II can be negative or zero for a finite number of k s. The slowly varying functions L_4 and L_5 in conditions IV and V are nonnegative on their domain and (strictly) positive for small and large enough arguments, respectively, by definition.

Remark 2.1.4 In many instances, for example in statistical estimation, the slowly varying functions in (2.1.4)–(2.1.7) are such that¹

$$L(u) \sim \text{const} > 0, \quad (2.1.9)$$

at either infinity or zero. Much of the mathematical theory, however, can be developed for general, or fairly general, slowly varying functions. When assuming (2.1.9) in connection to the conditions above, we shall write $L_j(u) \sim c_j$, $j = 1, 2, 4, 5$, especially to emphasize that estimating c_j s might be of interest as well, in addition to the estimation of d .

Definition 2.1.5 A second-order stationary time series $X = \{X_n\}_{n \in \mathbb{Z}}$ is called *long-range dependent* (LRD, in short) if one of the non-equivalent conditions I–V above holds. The parameter $d \in (0, 1/2)$ is called a long-range dependence (LRD) parameter. One also says that the series X has *long memory*.

As stated above, the conditions I–V are not equivalent in general and hence there is not just one accepted definition of LRD series. Which definition is to be used depends on the context. We will specify in the sequel, when necessary, the definition which is used.

We show in Section 2.2 that the conditions I–V are related as in Figure 2.1, where “ $\implies$ ” means implication and “ $\overset{*}{\implies}$ ” supposes that the slowly varying function is quasi-monotone (see Definition 2.2.11 below). The proofs and discussions in Section 2.2 shed further light on long-range dependence, and should be particularly useful to those unfamiliar with the area. The more technical parts of the proofs of the implications above are moved to Appendix A.2 in order for Section 2.2 to be more accessible.

Remark 2.1.6 The conditions above extend naturally to continuous-time stationary processes $\{X(t)\}_{t \in \mathbb{R}}$ (see Remarks 1.2.6 and 1.3.8). Condition I is stated for linear processes

¹ We write $a(u) \sim b(u)$ to signify $a(u)/b(u) \rightarrow 1$ as $u \rightarrow \infty$.

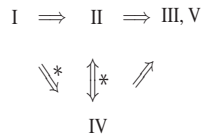


Figure 2.1 The relations between the various definitions of LRD.

$X(t) = \int_{-\infty}^t a(t-u)Z(du)$ in Remark 1.2.6 requiring that $a(u) = L_1(u)u^{d-1}$. Condition II remains the same, with the only difference that $h \in \mathbb{R}$. Condition III requires $\int_{\mathbb{R}} |\gamma_X(h)|dh = \infty$. Condition IV remains exactly the same. Finally, condition V is stated for $\text{Var}(\int_0^N X(u)du)$.

2.2 Relations Between the Various Definitions of Long-Range Dependence

We first list some important properties of slowly and regularly varying functions. We then use these properties to show how the various definitions of long-range dependence are related.

2.2.1 Some Useful Properties of Slowly and Regularly Varying Functions

We consider slowly and regularly varying functions at infinity. A useful property of slowly varying functions is *Potter's bound* (Theorem 1.5.6, (i), in Bingham et al. [157]): for every $\delta > 0$, there is $u_0 = u_0(\delta)$ such that

$$\frac{L(u)}{L(v)} \leq 2 \max \left\{ \left(\frac{u}{v} \right)^\delta, \left(\frac{u}{v} \right)^{-\delta} \right\} \quad (2.2.1)$$

for $u, v \geq u_0$. Note that, without loss of generality, since it is ultimately positive, L may be replaced by $|L|$ in (2.2.1). Moreover, if $|L|$ is bounded away from 0 and infinity on every compact subset of $[0, \infty)$, then (2.2.1) implies that there is $C > 1$ such that

$$\left| \frac{L(u)}{L(v)} \right| \leq C \max \left\{ \left(\frac{u}{v} \right)^\delta, \left(\frac{u}{v} \right)^{-\delta} \right\} \quad (2.2.2)$$

for $u, v > 0$ (Theorem 1.5.6, (ii), in Bingham et al. [157] and Exercise 2.1 below). A practical consequence of (2.2.1) is that, for any arbitrarily small $\delta > 0$, there are constants C_1, C_2 and $u_1 > 0$ such that

$$C_1 u^{-\delta} \leq L(u) \leq C_2 u^\delta, \quad (2.2.3)$$

for $u \geq u_1$.

Another useful property is the so-called *uniform convergence theorem*, according to which, if R is regularly varying with exponent $\rho \in \mathbb{R}$, then

$$\sup_{a \in A} \left| \frac{R(au)}{R(u)} - a^\rho \right| \rightarrow 0, \quad \text{as } u \rightarrow \infty, \quad (2.2.4)$$

for

$$A = \begin{cases} [u_0, v_0], & 0 < u_0 \leq v_0 < \infty, & \text{if } \rho = 0, \\ (0, v_0], & 0 < v_0 < \infty, & \text{if } \rho > 0, \\ [u_0, \infty), & 0 < u_0 < \infty, & \text{if } \rho < 0, \end{cases}$$

with the assumption, when $\rho > 0$, that R is bounded on each interval of $(0, v_0]$ (Theorem 1.5.2 in Bingham et al. [157]).

The following result, known as Karamata's theorem, is very handy.

Proposition 2.2.1 *Let L be a slowly varying function at infinity and $p > -1$. Then, for*

$$c_k = L(k)k^p, \quad k \geq 1,$$

we have that

$$\sum_{k=1}^n c_k \sim \frac{L(n)n^{p+1}}{p+1}, \quad \text{as } n \rightarrow \infty.$$

Proof This is a standard result in the theory of regular variation. Observe that, for large enough n so that $L(n) \neq 0$,

$$S_n = \frac{1}{L(n)n^{p+1}} \sum_{k=1}^n c_k = \sum_{k=1}^n \frac{L(k)}{L(n)} \left(\frac{k}{n}\right)^p \frac{1}{n} = \int_0^1 f_n(x) dx,$$

where

$$f_n(x) = \sum_{k=1}^n \frac{L(k)}{L(n)} \left(\frac{k}{n}\right)^p 1_{[\frac{k-1}{n}, \frac{k}{n})}(x) = \frac{L(\lceil nx \rceil)}{L(n)} \left(\frac{\lceil nx \rceil}{n}\right)^p, \quad x \in (0, 1),$$

$1_A(x) = 1$ if $x \in A$, and $= 0$ if $x \notin A$, is the indicator function of a set A , and $\lceil a \rceil$ is the (ceiling) integer part of $a \in \mathbb{R}$.² We now want to apply the dominated convergence theorem. For fixed $x \in (0, 1)$, by using (2.2.4) with $\rho = 0$, $a = \lceil nx \rceil/n$ and $u = n$, we have $L(\lceil nx \rceil)/L(n) \rightarrow 1$ and hence also $f_n(x) \rightarrow x^p =: f(x)$. Since $L(k)$ defines c_k on positive integers only, we may suppose without loss of generality that L is defined on $(0, \infty)$ and satisfies the assumptions for Potter's bound (2.2.2). (If $L(k) = 0$, the Potter's bound for $L(k)/L(n) = 0$ obviously holds.) Then, by choosing a small enough δ so that $p - \delta + 1 > 0$ in Potter's bound (2.2.2), we obtain

$$|f_n(x)| \leq C \left(\frac{\lceil nx \rceil}{n}\right)^{p-\delta} \leq \begin{cases} Cx^{p-\delta}, & p - \delta < 0, \\ C, & p - \delta \geq 0. \end{cases}$$

The function in the bound is integrable on $(0, 1)$. Hence, the dominated convergence theorem yields

$$S_n \rightarrow \int_0^1 f(x) dx = \int_0^1 x^p dx = \frac{1}{p+1}. \quad \square$$

A continuous version of the above result (Karamata's theorem) is given in the next proposition. A proof can be found in Bingham et al. [157], Proposition 1.5.8.

² That is, $\lceil a \rceil$ is the smallest integer which is greater than or equal to $a \in \mathbb{R}$.

Proposition 2.2.2 *Let L be a slowly varying function at infinity which is locally bounded in $[x_0, \infty)$, and $p > -1$. Then*

$$\int_{x_0}^x u^p L(u) du \sim \frac{L(x)x^{p+1}}{p+1}, \quad \text{as } x \rightarrow \infty.$$

Remark 2.2.3 We considered above slowly and regularly varying functions at infinity. If the functions are slowly and regularly varying at zero, the results above can still be used by transforming the functions to slowly and regularly varying functions at infinity, applying the results above and finally translating the results back to the zero range. For example, if $L(u)$ is a slowly varying function at zero, then $L(1/u)$ is a slowly varying function at infinity. Potter's bound (2.2.1) for $L(1/u)$ becomes: for every $\delta > 0$, there is u_0 such that

$$\frac{L(1/u)}{L(1/v)} \leq 2 \max \left\{ \left(\frac{u}{v} \right)^\delta, \left(\frac{u}{v} \right)^{-\delta} \right\} = 2 \max \left\{ \left(\frac{1/u}{1/v} \right)^\delta, \left(\frac{1/u}{1/v} \right)^{-\delta} \right\}$$

for $u, v \geq u_0$ or $1/u, 1/v \leq 1/u_0$. This is the same as: for every $\delta > 0$, there is x_0 such that

$$\frac{L(x)}{L(y)} \leq 2 \max \left\{ \left(\frac{x}{y} \right)^\delta, \left(\frac{x}{y} \right)^{-\delta} \right\}$$

for $x, y \leq x_0$.

In some instances, several of the conditions above characterizing long-range dependence may hold at the same time. To understand the various conditions I–V, we will compare them through a sequence of propositions. Their proofs are somewhat technical and may be skipped in a first reading.

2.2.2 Comparing Conditions II and III

Proposition 2.2.4 *Condition II implies III.*

Proof By applying Proposition 2.2.1 with $c_k = \gamma_X(k)$, $L = L_2$, $p = 2d - 1$, and using $\gamma_X(k) = \gamma_X(-k)$, $k \geq 1$, we have

$$\sum_{k=-n}^n |\gamma_X(k)| = |\gamma_X(0)| + 2 \sum_{k=1}^n |\gamma_X(k)| \sim \frac{L_2(n)n^{2d}}{d}, \quad \text{as } n \rightarrow \infty.$$

By (2.2.3), $L_2(n) \geq Cn^{-\delta}$ for large enough n , where $\delta > 0$ is arbitrarily small and C is a constant. Then, for another constant C and large enough n , $\sum_{k=-n}^n |\gamma_X(k)| \geq Cn^{2d-\delta} \rightarrow \infty$ by choosing $\delta < 2d$. The result now follows. $\square$

2.2.3 Comparing Conditions II and V

Another interesting consequence of Proposition 2.2.1 is related to the sample mean of the series,

$$\bar{X}_N = \frac{X_1 + \cdots + X_N}{N}. \quad (2.2.5)$$

Proposition 2.2.5 *If condition II holds, then*

$$\text{Var}(\bar{X}_N) \sim \frac{L_2(N)N^{2d-1}}{d(2d+1)}, \quad \text{as } N \rightarrow \infty. \quad (2.2.6)$$

Proof Observe that, by using Proposition 2.2.1,

$$\begin{aligned} \text{Var}(\bar{X}_N) &= \frac{1}{N^2} \sum_{j,k=1}^N \gamma_X(j-k) = \frac{1}{N^2} \sum_{h=-(N-1)}^{N-1} (N-|h|) \gamma_X(h) \\ &= \frac{1}{N} \sum_{h=-(N-1)}^{N-1} \gamma_X(h) - \frac{1}{N^2} \sum_{h=-(N-1)}^{N-1} |h| \gamma_X(h) \\ &\sim \frac{2L_2(N)N^{2d-1}}{2d} - \frac{2L_2(N)N^{2d-1}}{2d+1} = \frac{L_2(N)N^{2d-1}}{d(2d+1)}. \end{aligned}$$

Another way to prove the result is to write

$$\text{Var}(\bar{X}_N) = \frac{1}{N} \gamma_X(0) + \frac{2}{N^2} \sum_{k=1}^{N-1} \sum_{h=1}^k \gamma_X(h)$$

and apply Proposition 2.2.1 twice to the last term. $\square$

Remark 2.2.6 A wrong way would be to write

$$\text{Var}(\bar{X}_N) = \frac{1}{N} \sum_{h=-(N-1)}^{N-1} \left(1 - \frac{|h|}{N}\right) \gamma_X(h) \sim \frac{1}{N} \sum_{h=-(N-1)}^{N-1} \gamma_X(h), \quad (2.2.7)$$

which would lead to the wrong asymptotics $L_2(N)N^{2d-1}/d$. The asymptotic equivalence in (2.2.7) cannot be justified by the dominated convergence theorem because $\sum_{h=-\infty}^{\infty} |\gamma_X(h)| = \infty$.

An immediate consequence of Proposition 2.2.5 is that condition II implies V.

Corollary 2.2.7 *Condition II implies V with the slowly varying functions related as*

$$L_5(N) \sim \frac{L_2(N)}{d(2d+1)}, \quad \text{as } N \rightarrow \infty. \quad (2.2.8)$$

Remark 2.2.8 Proposition 2.2.1 is related to another interesting perspective on LRD. For each $N \geq 1$, form a series $X^{(N)} = \{X_n^{(N)}\}_{n \in \mathbb{Z}}$ obtained by taking averages over blocks of size N ,

$$X_n^{(N)} = \frac{1}{N} \sum_{k=N(n-1)+1}^{Nn} X_k, \quad n \in \mathbb{Z}.$$

Each series $X^{(N)}$ is second order stationary. Denote its autocovariance function by $\gamma_X^{(N)}(k)$. One can show that the condition (2.1.8) with $d \in (0, 1/2)$ is equivalent to

$$\gamma_X^{(N)}(k) \sim L_5(N)N^{2d-1} \frac{1}{2} \left(|k+1|^{2d+1} - 2|k|^{2d+1} + |k-1|^{2d+1} \right), \quad \text{as } N \rightarrow \infty, \quad (2.2.9)$$

for each fixed $k \in \mathbb{Z}$. This is stated as Exercise 2.3 below. The expression $(|k+1|^{2d+1} - 2|k|^{2d+1} + |k-1|^{2d+1})/2$ on the right-hand side of (2.2.9) is the autocovariance function of the so-called fractional Gaussian noise (see Section 2.8), which is the series of the increments of fractional Brownian motion.

Note that the series $\{X_n\}_{n \in \mathbb{Z}}$ consisting of uncorrelated variables satisfies

$$\text{Var}(\bar{X}_N) \sim \text{Var}(X_0)N^{-1}.$$

This stands in stark contrast with Proposition 2.2.5, where the variance of the sample mean of LRD series behaves like $N^{-(1-2d)}$ as $N \rightarrow \infty$. Since $1 - 2d < 1$, this sample mean can generally be estimated less reliably than in the case of uncorrelated variables. Proposition 2.2.5 also suggests that $\bar{X}_N$ should be normalized by $N^{d-1/2}$, instead of the common normalization $N^{-1/2}$, in order to converge to a limit. If $L_2(u) \sim c_2$ as in statistical applications, then (2.2.6) becomes

$$\text{Var}(\bar{X}_N) \sim \frac{c_2 N^{2d-1}}{d(2d+1)}, \quad \text{as } N \rightarrow \infty. \quad (2.2.10)$$

Note that, for example, if the right-hand side of (2.2.10) is to be used in the construction of confidence intervals for μ_X , then both d and c_2 need to be estimated.

Note also that (2.2.10) implies

$$\text{Var}(N^{1/2-d}\bar{X}_N) \rightarrow \frac{c_2}{d(2d+1)}, \quad \text{as } N \rightarrow \infty. \quad (2.2.11)$$

The constant $c_2/d(2d+1)$ is known as the *long-run variance* of the LRD series X . (See also Remark 2.2.20 below.)

2.2.4 Comparing Conditions I and II

We now turn to showing that condition I implies II. In this regard, several observations should be made about condition I. First, the series satisfying condition I is well-defined in the $L^2(\Omega)$ -sense (see the footnote to Example 1.2.3). Indeed, note that

$$\mathbb{E} \left(\sum_{k=m_1}^{m_2} \psi_k Z_{n-k} \right)^2 = \sigma_Z^2 \sum_{k=m_1}^{m_2} \psi_k^2,$$

where $\psi_k^2 = (L_1(k))^2 k^{2d-2}$. Using (2.2.3), we have $0 < L_1(k) \leq Ck^\delta$ for large enough k where $\delta > 0$ is arbitrarily small and C is a constant. Then, for large enough m_1, m_2 , the latter sum is bounded by

$$C^2 \sigma_Z^2 \sum_{k=m_1}^{m_2} k^{2d-2+2\delta}$$

and converges to 0 as $m_1, m_2 \rightarrow \infty$ for small enough δ such that $2d - 1 + 2\delta < 0$. Second, observe that, for $k \geq 0$,

$$\gamma_X(k) = \sigma_Z^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+k}$$

(see Example 1.2.5). We will use:

Proposition 2.2.9 *Let L be a slowly varying function and $-1 < p < -1/2$. Then,*

$$c_j = L(j)j^p, \quad j \geq 0,$$

implies that

$$\sum_{j=0}^{\infty} c_j c_{j+k} \sim (L(k))^2 k^{2p+1} B(p+1, -2p-1), \quad \text{as } k \rightarrow \infty,$$

where

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = \int_0^{\infty} y^{a-1} (1+y)^{-a-b} dy = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \quad a, b > 0, \quad (2.2.12)$$

*is the beta function.*³

The function $\Gamma(z)$ in (2.2.12) is the gamma function defined by

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt, \quad z > 0.$$

We shall often use its following two properties:

$$\Gamma(z+1) = z\Gamma(z), \quad z > 0, \quad (2.2.13)$$

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}, \quad 0 < z < 1. \quad (2.2.14)$$

The relation (2.2.13), in particular, is used to extend the definition of the gamma function to $z \leq 0$ in a recursive way through $\Gamma(z) = \Gamma(z+1)/z$, $z \leq 0$. Note that with this definition, $\Gamma(k) = \infty$ when $k = 0, -1, -2, \dots$

Proof of Proposition 2.8 The proof is similar to that of Proposition 2.2.1. Observe that, for large enough k so that $L(k) \neq 0$,

$$\begin{aligned} \frac{1}{(L(k))^2 k^{2p+1}} \sum_{j=0}^{\infty} c_j c_{j+k} &= \sum_{j=0}^{\infty} \frac{L(j)}{L(k)} \frac{L(j+k)}{L(k)} \left(\frac{j}{k}\right)^p \left(1 + \frac{j}{k}\right)^p \frac{1}{k} \\ &\sim \int_0^{\infty} u^p (1+u)^p du = \int_0^1 (1-v)^p v^{-2-2p} dv = B(p+1, -2p-1), \end{aligned}$$

³ To go from the first to the second integral in (2.2.12), set $x = y/(1+y)$.

where the last asymptotic relation follows by using Potter's bound and the dominated convergence theorem as in the proof of Proposition 2.2.1, and where the last two integrals are equal in view of the second equality in (2.2.12) with $a = p + 1$ and $b = -1 - 2p$. $\square$

Corollary 2.2.10 *Condition I implies II. In particular,*

$$L_2(k) \sim \sigma_Z^2(L_1(k))^2 B(d, 1 - 2d). \quad (2.2.15)$$

2.2.5 Comparing Conditions II and IV

We next turn to the comparison of conditions II and IV, which is probably the most delicate one. To be able to go from condition II to IV, we would like to be able to write

$$f_X(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} e^{-ik\lambda} \gamma_X(k) \quad (2.2.16)$$

for an LRD series satisfying condition II (cf. (1.3.1)). Note that, since condition II implies III, one may expect at most that (2.2.16) converges conditionally; that is, it converges but does not converge absolutely. The most general condition for this and for relating conditions II and IV involves the following definition.

Definition 2.2.11 A slowly varying function L on $[0, \infty)$ is called *quasi-monotone* if it is of bounded (finite) variation on any compact interval of $[0, \infty)$ and if, for some $\delta > 0$,

$$\int_0^x u^\delta |dL(u)| = O(x^\delta L(x)), \quad \text{as } x \rightarrow \infty. \quad (2.2.17)$$

It is known (but not obvious) that the condition (2.2.17) can be replaced by

$$\int_x^\infty u^{-\delta} |dL(u)| = O(x^{-\delta} L(x)), \quad \text{as } x \rightarrow \infty \quad (2.2.18)$$

(Corollary 2.8.2 in Bingham et al. [157]), and that “for some $\delta > 0$ ” can be replaced by “for all $\delta > 0$ ” (see the bottom of p. 108 in Bingham et al. [157]). Another popular class is the *Zygmund class* (or the class of normalized slowly varying functions) of slowly varying functions L such that, for any (equivalently, one) $\delta > 0$, $u^{-\delta} L(u)$ is ultimately decreasing⁴ and $u^\delta L(u)$ is ultimately increasing. It is known (but not obvious) that the Zygmund class is a proper subset of the class of quasi-monotone slowly varying functions (Theorem 1.5.5 and Corollary 2.7.4 in Bingham et al. [157]).

Example 2.2.12 (*Quasi-monotonicity*) The slowly varying functions $\log u$, $1/\log u$ and $\log |\log u|$, $u > 0$, can be verified to be quasi-monotone. For example, for $L(u) = \log u$, $u > 0$, the integral in (2.2.18) becomes

$$\int_x^\infty u^{-\delta} d \log u = \int_x^\infty u^{-\delta-1} du = \frac{x^{-\delta}}{\delta} \leq \frac{x^{-\delta} \log x}{\delta} = \frac{x^{-\delta} L(x)}{\delta},$$

for $x > e$. Hence, the condition (2.2.18) holds.

⁴ This means that there is a u_0 such that $u^{-\delta} L(u)$ is decreasing for $u > u_0$.

Example 2.2.13 (*Quasi-monotonicity*) Let $a > 0$ and $p \in \mathbb{R}$. The function $L(u) = (1 + a/u)^p$ is slowly varying since $L(u) \rightarrow 1$ as $u \rightarrow \infty$. The function L is also quasi-monotone since it verifies the condition (2.2.18). Indeed, observe that

$$\begin{aligned} \int_x^\infty u^{-\delta} \left| d \left(1 + \frac{a}{u} \right)^p \right| &= |p|a \int_x^\infty u^{-\delta-2} \left(1 + \frac{a}{u} \right)^{p-1} du \leq C \int_x^\infty u^{-\delta-2} du \\ &= \frac{C}{\delta+1} x^{-\delta-1} \leq C' x^{-\delta} = O \left(x^{-\delta} \left(1 + \frac{a}{x} \right)^p \right), \quad \text{as } x \rightarrow \infty, \end{aligned}$$

where C, C' are two constants which depend on a, p , and where we used the fact that $(1 + a/x)^q \rightarrow 1$ as $x \rightarrow \infty$ for any $q \in \mathbb{R}$. This example is used in the proof of Proposition 2.2.14, which is given in Appendix A.2.

The proof of the following proposition is technical in nature and can be found in Appendix A.2. An informal derivation can be found in Remark 2.2.18 below.

Proposition 2.2.14 Suppose L_2 in (2.1.5) is a quasi-monotone slowly varying function. Then, the series X , with autocovariance satisfying (2.1.5), has a spectral density

$$f_X(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} e^{-ik\lambda} \gamma_X(k) = \frac{1}{2\pi} \left\{ \gamma_X(0) + 2 \sum_{k=1}^{\infty} \cos(k\lambda) \gamma_X(k) \right\}, \quad (2.2.19)$$

where the series converges conditionally for every $\lambda \in (-\pi, \pi] \setminus \{0\}$. Moreover, we have that

$$f_X(\lambda) \sim \lambda^{-2d} L_2 \left(\frac{1}{\lambda} \right) \frac{1}{\pi} \Gamma(2d) \cos(d\pi), \quad \text{as } \lambda \rightarrow 0. \quad (2.2.20)$$

As a consequence, one has the following result.

Corollary 2.2.15 If L_2 is a quasi-monotone slowly varying function, then condition II implies IV. In particular,

$$L_4(\lambda) \sim L_2 \left(\frac{1}{\lambda} \right) \frac{1}{\pi} \Gamma(2d) \cos(d\pi), \quad \text{as } \lambda \rightarrow 0. \quad (2.2.21)$$

The converses of Proposition 2.2.14 and Corollary 2.2.15 can also be proved. The statements are as follows. An informal derivation can be found in Remark 2.2.18 below.

Proposition 2.2.16 Suppose $L_4(1/u)$ is a quasi-monotone slowly varying function on $[1/\pi, \infty)$, where L_4 appears in (2.1.7). Then, the autocovariance function γ_X of the series X with the spectral density (2.1.7) satisfies

$$\gamma_X(n) = 2 \int_0^\pi \cos(n\lambda) f_X(\lambda) d\lambda \sim n^{2d-1} L_4 \left(\frac{1}{n} \right) 2\Gamma(1-2d) \sin(d\pi), \quad \text{as } n \rightarrow \infty. \quad (2.2.22)$$

Proof This follows from $\gamma_X(n) = \int_{-\pi}^\pi e^{in\lambda} f_X(\lambda) d\lambda$ and from Proposition A.2.2 in Appendix A.2 with $p = 1 - 2d$, noting that $\cos((1 - 2d)\pi/2) = \sin(d\pi)$. $\square$

Corollary 2.2.17 *If the equality in (2.1.7) takes place on $(0, \pi]$ with a quasi-monotone slowly varying function $L_4(1/u)$ on $[1/\pi, \infty)$, then condition IV implies II. In particular,*

$$L_2(n) \sim L_4\left(\frac{1}{n}\right) 2\Gamma(1-2d) \sin(d\pi), \quad \text{as } n \rightarrow \infty. \quad (2.2.23)$$

Note that Corollaries 2.2.15 and 2.2.17 are consistent because of (2.2.14) and $2 \sin(x) \cos(x) = \sin(2x)$.

Remark 2.2.18 The relation (2.2.20) can informally be argued as follows: by using (2.2.19) and discretization of the integral below as $\lambda \rightarrow 0$,

$$\begin{aligned} f_X(\lambda) &= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} e^{-ik\lambda} \gamma_X(k) = \frac{1}{2\pi} \left\{ \gamma_X(0) + 2 \sum_{k=1}^{\infty} \cos(k\lambda) \gamma_X(k) \right\} \\ &\approx \frac{1}{\pi} \sum_{k=1}^{\infty} \cos(k\lambda) L_2(k) k^{2d-1} = \lambda^{-2d} L_2\left(\frac{1}{\lambda}\right) \frac{1}{\pi} \sum_{k=1}^{\infty} \cos(k\lambda) \frac{L_2(k\lambda/\lambda)}{L_2(1/\lambda)} (k\lambda)^{2d-1} \lambda \\ &\approx \lambda^{-2d} L_2\left(\frac{1}{\lambda}\right) \frac{1}{\pi} \int_0^{\infty} \cos(u) u^{2d-1} du = \lambda^{-2d} L_2\left(\frac{1}{\lambda}\right) \frac{1}{\pi} \Gamma(2d) \cos(d\pi), \end{aligned}$$

where the last equality follows from Gradshteyn and Ryzhik [421], p. 437, Formula 3.761.9 (see also the relation (2.6.6) below). Similarly, the asymptotic relation in (2.2.22) can informally be derived as:

$$\begin{aligned} \gamma_X(n) &= 2 \int_0^{\pi} \cos(n\lambda) L_4(\lambda) \lambda^{-2d} d\lambda = 2 \int_0^{n\pi} \cos(u) L_4\left(\frac{u}{n}\right) \left(\frac{u}{n}\right)^{-2d} \frac{1}{n} du \\ &= n^{2d-1} L_4\left(\frac{1}{n}\right) 2 \int_0^{n\pi} \cos(u) \frac{L_4(u/n)}{L_4(1/n)} u^{-2d} du \approx n^{2d-1} L_4\left(\frac{1}{n}\right) 2 \int_0^{\infty} \cos(u) u^{-2d} du \\ &= n^{2d-1} L_4\left(\frac{1}{n}\right) 2\Gamma(1-2d) \cos\left(\frac{(1-2d)\pi}{2}\right) = n^{2d-1} L_4\left(\frac{1}{n}\right) 2\Gamma(1-2d) \sin(d\pi), \end{aligned}$$

by using the same relation in Gradshteyn and Ryzhik [421] as above.

Remark 2.2.19 It is known that conditions II and IV are not equivalent in general. See Examples 5.2 and 5.3 in Samorodnitsky [879], or Exercise 2.6 below. Another important related note is the following. It has sometimes been claimed in the literature that conditions II and IV are equivalent when

$$L_2(u) \sim c_2, \quad L_4(\lambda) \sim c_4, \quad (2.2.24)$$

where $c_2, c_4 > 0$ are two constants. In fact, even in this case, the two conditions are not equivalent and counterexamples can be provided for either direction. See Gubner [436]. In other words, as shown in the results above, the exact behavior of the slowly varying functions $\gamma_X(k)k^{1-2d} = L_2(k)$ and $f_X(\lambda)\lambda^{2d} = L_4(\lambda)$ is important in going from one condition to the other. This is also why, for example, condition II is written as $\gamma_X(k) = L_2(k)k^{2d-1}$ and not as $\gamma_X(k) \sim L_2(k)k^{2d-1}$.

Remark 2.2.20 If (2.2.24) holds and the slowly varying functions can be related as in (2.2.23), then

$$c_2 = c_4 2\Gamma(1 - 2d) \sin(d\pi). \quad (2.2.25)$$

In estimation (see Section 2.10 and Chapter 10), for example, it is more advantageous to estimate c_4 in the spectral domain, for which c_2 can then be deduced from (2.2.25) and used e.g., in (2.2.10). In particular, the long-run variance of X appearing in (2.2.11) can be expressed as

$$\frac{c_2}{d(2d + 1)} = \frac{c_4 2\Gamma(1 - 2d) \sin(d\pi)}{d(2d + 1)}. \quad (2.2.26)$$

Estimation of the long-run variance expressed through c_4 is discussed in Abadir, Distaso, and Giraitis [3]. See also Section 2.3 for the definition of *long-run variance* in the case of short-range dependence.

2.2.6 Comparing Conditions I and IV

We briefly discuss here conditions I and IV. Under condition I, the sequence $\{\psi_k\}_{k \geq 0}$ is in $\ell^2(\mathbb{Z})$ and hence has a Fourier transform

$$\psi(e^{-i\lambda}) := \widehat{\phi}(\lambda) := \sum_{k=0}^{\infty} \psi_k e^{-ik\lambda} \quad (2.2.27)$$

convergent in $L^2(-\pi, \pi]$ (see Appendix A.1.1). Then, by Theorem 4.10.1 in Brockwell and Davis [186], the series X has a spectral density given by

$$f_X(\lambda) = \frac{\sigma_Z^2}{2\pi} |\psi(e^{-i\lambda})|^2.$$

Moreover, if the slowly varying function L_1 in (2.1.4) is quasi-monotone, Proposition A.2.1 in Appendix A.2 implies that the Fourier series (2.2.27) above converges conditionally and has a power-law like behavior around zero. This immediately yields the following result.

Proposition 2.2.21 *If L_1 is a quasi-monotone slowly varying function, then condition I implies IV. In particular,*

$$L_4(\lambda) \sim \left(L_1\left(\frac{1}{\lambda}\right)\right)^2 \frac{\Gamma(d)^2 \sigma_Z^2}{2\pi}, \quad \text{as } \lambda \rightarrow 0. \quad (2.2.28)$$

Note that Corollary 2.2.15 and Proposition 2.2.21 imply that

$$L_2\left(\frac{1}{\lambda}\right) \sim \left(L_1\left(\frac{1}{\lambda}\right)\right)^2 \frac{\Gamma(d)^2 \sigma_Z^2}{2\Gamma(2d) \cos(d\pi)}. \quad (2.2.29)$$

This is consistent with Corollary 2.2.10 if $2B(d, 1 - 2d)\Gamma(2d) \cos(d\pi) = \Gamma(d)^2$. The latter equality follows by writing $B(d, 1 - 2d) = \Gamma(d)\Gamma(1 - 2d)/\Gamma(1 - d)$ and using $\Gamma(1 - d)\Gamma(d) = \pi/\sin(d\pi)$ and $\Gamma(1 - 2d)\Gamma(2d) = \pi/\sin(2d\pi) = \pi/(2\sin(d\pi) \cos(d\pi))$ by (2.2.14).

2.2.7 Comparing Conditions IV and III

We show that condition IV implies III. To do so, we use the following lemma.

Lemma 2.2.22 *If $\sum_{k=-\infty}^{\infty} |\gamma_X(k)| < \infty$, then the series $X = \{X_n\}_{n \in \mathbb{Z}}$ has a continuous spectral density $f_X(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} e^{-ik\lambda} \gamma_X(k)$.*

Proof Since $f_X(\lambda) \leq \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} |\gamma_X(k)| < \infty$, it follows from the dominated convergence theorem that $\lim_{\lambda \rightarrow \lambda_0} f_X(\lambda) = f_X(\lambda_0)$. In particular, $f_X(0) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_X(k) < \infty$. $\square$

Suppose now that condition III fails; that is, $\sum_{k=-\infty}^{\infty} |\gamma_X(k)| < \infty$. It follows from Lemma 2.2.22 that $f_X(\lambda)$ is continuous and $f_X(0) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_X(k) < \infty$. But by condition IV and (2.1.1), $\lim_{\lambda \rightarrow 0} f_X(\lambda) = \lim_{\lambda \rightarrow 0} L_4(\lambda) \lambda^{-2d} = \infty$. Hence a contradiction, showing that condition IV implies III.

Corollary 2.2.23 *Condition IV implies III.*

2.2.8 Comparing Conditions IV and V

Finally, condition IV also implies V as shown in the following proposition.

Proposition 2.2.24 *Condition IV implies V. In particular,*

$$L_5(N) \sim L_4\left(\frac{1}{N}\right) \frac{2\Gamma(1-2d) \sin(d\pi)}{d(2d+1)}, \quad \text{as } N \rightarrow \infty. \quad (2.2.30)$$

Proof Write the series $\{X_n\}$ in its spectral representation as $X_n = \int_{(-\pi, \pi]} e^{in\lambda} Z(d\lambda)$, where $Z(d\lambda)$ is a complex-valued Hermitian random measure with control measure $f_X(\lambda) d\lambda$ (see Sections 1.3.4 and Appendix B.1). Then, by using $e^{i\lambda} + \dots + e^{iN\lambda} = e^{i\lambda}(e^{iN\lambda} - 1)/(e^{i\lambda} - 1)$,

$$X_1 + \dots + X_N = \int_{(-\pi, \pi]} e^{i\lambda} \frac{e^{iN\lambda} - 1}{e^{i\lambda} - 1} Z(d\lambda)$$

and hence

$$\text{Var}(X_1 + \dots + X_N) = \int_{-\pi}^{\pi} \frac{|e^{iN\lambda} - 1|^2}{|e^{i\lambda} - 1|^2} f_X(\lambda) d\lambda.$$

Then,

$$\begin{aligned} \frac{N^{-1-2d}}{L_4(1/N)} \text{Var}(X_1 + \dots + X_N) &= \frac{2N^{-1-2d}}{L_4(1/N)} \int_0^{\pi} \frac{|e^{iN\lambda} - 1|^2}{|e^{i\lambda} - 1|^2} f_X(\lambda) d\lambda \\ &= \frac{2N^{-1-2d}}{L_4(1/N)} \left(\int_0^{\epsilon} + \int_{\epsilon}^{\pi} \right) \frac{|e^{iN\lambda} - 1|^2}{|e^{i\lambda} - 1|^2} f_X(\lambda) d\lambda =: I_{1,N} + I_{2,N}, \end{aligned}$$

for $\epsilon > 0$. Note that, since the integrand above is bounded on $[\epsilon, \pi]$,

$$I_{2,N} \leq \frac{CN^{-1-2d}}{L_4(1/N)} \int_{\epsilon}^{\pi} f_X(\lambda) d\lambda \rightarrow 0,$$

as $N \rightarrow \infty$. On the other hand, by condition IV,

$$\begin{aligned} I_{1,N} &= \frac{2N^{-1-2d}}{L_4(1/N)} \int_0^{\epsilon} \frac{|e^{iN\lambda} - 1|^2}{|e^{i\lambda} - 1|^2} \lambda^{-2d} L_4(\lambda) d\lambda \\ &= 2 \int_0^{N\epsilon} \frac{|e^{ix} - 1|^2}{|e^{ix/N} - 1|^2} \left(\frac{x}{N}\right)^2 x^{-2d-2} \frac{L_4(x/N)}{L_4(1/N)} dx =: \int_0^{\infty} g_N(x) dx. \end{aligned}$$

We have $g_N(x) \rightarrow 2|e^{ix} - 1|^2 x^{-2d-2}$, $x > 0$, as $N \rightarrow \infty$, and by using Potter's bound (2.2.1), $|g_N(x)| \leq C|e^{ix} - 1|^2 x^{-2d-2\pm\delta} =: g(x)$, $x > 0$, for any fixed $\delta > 0$. Since the function $g(x)$ is integrable on $(0, \infty)$ for sufficiently small $\delta > 0$, and $\delta \in (0, 1/2)$, the dominated convergence theorem yields

$$\begin{aligned} I_{1,N} &\rightarrow 2 \int_0^{\infty} |e^{ix} - 1|^2 x^{-2d-2} dx = 4 \int_0^{\infty} (1 - \cos x) x^{-2d-2} dx = \frac{4}{2d+1} \int_0^{\infty} \sin(x) x^{-2d-1} dx \\ &= \frac{2}{d(2d+1)} \int_0^{\infty} \cos(x) x^{-2d} dx = \frac{2\Gamma(1-2d) \sin(d\pi)}{d(2d+1)}, \end{aligned}$$

where we used integration by parts twice and Formula 3.761.9 in Gradshteyn and Ryzhik [421], p. 437 (see also (2.6.6)). This shows that condition IV implies V, and that (2.2.30) holds. $\square$

2.3 Short-Range Dependent Series and their Several Examples

According to condition III in Section 2.1, the autocovariances of long-range dependent series are not (absolutely) summable. This is why time series with absolutely summable autocovariances are often called *short-range dependent*.

Definition 2.3.1 A second-order stationary time series $X = \{X_n\}_{n \in \mathbb{Z}}$ is called *short-range dependent* (SRD, in short) if its autocovariances are absolutely summable; that is, if

$$\sum_{k=-\infty}^{\infty} |\gamma_X(k)| < \infty. \quad (2.3.1)$$

It follows from Lemma 2.2.22 that SRD series have a continuous spectral density $f_X(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} e^{-ik\lambda} \gamma_X(k)$ and, in particular, that

$$f_X(0) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_X(k) < \infty. \quad (2.3.2)$$

This is in contrast to the spectral densities of LRD series which blow up at the origin as in condition IV. Note also that, for SRD series (2.2.7) is correct and thus arguing as in the proof of Proposition 2.2.5,

$$\text{Var}(\bar{X}_N) \sim \frac{1}{N} \sum_{k=-\infty}^{\infty} \gamma_X(k), \quad (2.3.3)$$

where $\bar{X}_N$ is the sample mean (Exercise 2.5). Since (2.3.2) and (2.3.3) for SRD series are obtained from the analogous expressions for LRD series by setting

$$d = 0,$$

this value of d is often associated with SRD. We also note that the constant $\sum_{k=-\infty}^{\infty} \gamma_X(k)$ appearing on the right-hand side of (2.3.3) is known as the *long-run variance* of the SRD series X . The long-run variance of LRD series was defined in (2.2.11) and (2.2.26).

There is an interesting special type of SRD series, whose autocovariances are absolutely summable as in (2.3.1), but they sum up to zero. These series have the following special name.

Definition 2.3.2 A SRD time series $X = \{X_n\}_{n \in \mathbb{Z}}$ is called *antipersistent* if

$$\sum_{k=-\infty}^{\infty} \gamma_X(k) = 0. \quad (2.3.4)$$

Observe that, for antipersistent series, (2.3.2) implies

$$f_X(0) = 0. \quad (2.3.5)$$

Examples are series with spectral density

$$f_X(\lambda) = L_4(\lambda) \lambda^{-2d}, \quad 0 < \lambda \leq \pi, \quad (2.3.6)$$

where L_4 is a slowly varying function at zero, and

$$d < 0.$$

This is the same exact expression as in condition IV of LRD series but with d being negative. As for LRD series, one could expect to relate (2.3.6) to conditions as in I, II and V for LRD series but with negative d . This is stated as Exercise 2.7.

We conclude this section with examples of several popular SRD series. In order to show that they are SRD, one just has to check that they satisfy condition (2.3.1). Examples of antipersistent series appear in Section 2.4 below.

Example 2.3.3 (ARMA(p, q) series) Recall from Example 1.2.4 that ARMA(p, q) time series are defined as second-order stationary series X satisfying the equation

$$X_n - \varphi_1 X_{n-1} - \cdots - \varphi_p X_{n-p} = Z_n + \theta_1 Z_{n-1} + \cdots + \theta_q Z_{n-q}, \quad (2.3.7)$$

where the coefficients $\varphi_1, \dots, \varphi_p$ satisfy conditions given below, and $\{Z_n\}_{n \in \mathbb{Z}} \sim \text{WN}(0, \sigma_Z^2)$. It is convenient to express the conditions in terms of the so-called *backshift operator* B , defined as

$$B^k X_n = X_{n-k}, \quad B^0 = I, \quad k \in \mathbb{Z}. \quad (2.3.8)$$

With this notation, the ARMA(p, q) equation becomes

$$\varphi(B)X_n = \theta(B)Z_n,$$

where

$$\varphi(z) = 1 - \varphi_1 z - \cdots - \varphi_p z^p, \quad \theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$$

are the so-called *characteristic polynomials*. The polynomial $\varphi(z)$ can be factored out as $\varphi(z) = (1 - r_1^{-1}z) \cdot \dots \cdot (1 - r_p^{-1}z)$. Suppose that the polynomial roots are such that $|r_m| \neq 1, m = 1, \dots, p$.

Writing the ARMA(p, q) equation as

$$(I - r_1^{-1}B) \cdot \dots \cdot (I - r_p^{-1}B)X_n = \theta(B)Z_n,$$

it is tempting to interpret the solution to this equation as

$$X_n = (I - r_1^{-1}B)^{-1} \cdot \dots \cdot (I - r_p^{-1}B)^{-1} \theta(B)Z_n$$

with

$$(I - r^{-1}B)^{-1} = \begin{cases} \sum_{k=0}^{\infty} r^{-k} B^k, & \text{if } |r|^{-1} < 1, \\ -\frac{rB^{-1}}{I - rB^{-1}} = -rB^{-1} \sum_{k=0}^{\infty} r^k B^{-k}, & \text{if } |r|^{-1} > 1. \end{cases}$$

This time series can be written as

$$X_n = \sum_{j=-\infty}^{\infty} \psi_j B^j Z_n = \sum_{j=-\infty}^{\infty} \psi_j Z_{n-j}, \quad (2.3.9)$$

where $\psi(z) = \sum_{j=-\infty}^{\infty} \psi_j z^j$ satisfies the formal relation $\psi(z) = \theta(z)\varphi(z)^{-1}$. In fact, one can show that the series (2.3.9) is well-defined, second-order stationary and satisfies the ARMA(p, q) equation (2.3.7) (Brockwell and Davis [186]).

To show that the series (2.3.9) is SRD, observe that the function $\psi(z) = \theta(z)\varphi(z)^{-1}$ is analytic on $r^{-1} < |z| < r$ with some $r > 1$. Therefore, its Laurent series⁵ $\psi(z) = \sum_{j=-\infty}^{\infty} \psi_j z^j$ converges absolutely on $r^{-1} < |z| < r$. In particular, there is $\epsilon > 0$ such that $|\psi_j|(1 + \epsilon)^{|j|} \rightarrow 0$, as $|j| \rightarrow \infty$. Then, for a constant K ,

$$|\psi_j| \leq \frac{K}{(1 + \epsilon)^{|j|}}$$

and for $h > 0$,

$$\begin{aligned} |\gamma_X(h)| &= \sigma_Z^2 \left| \sum_{k=-\infty}^{\infty} \psi_k \psi_{k+h} \right| \leq \sigma_Z^2 \sum_{k=-\infty}^{\infty} |\psi_k| |\psi_{k+h}| \leq \sigma_Z^2 K^2 \sum_{k=-\infty}^{\infty} \frac{1}{(1 + \epsilon)^{|k|+|k+h|}} \\ &= \sigma_Z^2 K^2 \left(\sum_{k=0}^{\infty} \frac{1}{(1 + \epsilon)^{2k}} \right) \frac{1}{(1 + \epsilon)^h} + \sigma_Z^2 K^2 \sum_{k=1}^h \frac{1}{(1 + \epsilon)^h} + \sigma_Z^2 K^2 \sum_{k=h+1}^{\infty} \frac{1}{(1 + \epsilon)^{2k-h}} \\ &= \sigma_Z^2 K^2 \left(\sum_{k=0}^{\infty} \frac{1}{(1 + \epsilon)^{2k}} + h + \sum_{k=1}^{\infty} \frac{1}{(1 + \epsilon)^{2k}} \right) s^h = (C_1 + C_2 h) s^h, \end{aligned}$$

⁵ A complex-valued function $\psi(z)$ may fail to be analytic at a point $z_0 \in \mathbb{C}$, and therefore cannot have a Taylor expansion around that point. It may possess an expansion in positive and negative powers of $z - z_0$ in an annular domain $r^{-1} < |z - z_0| < r$ for some $r > 1$. Such an expansion is called a *Laurent series*.

where $s = 1/(1+\epsilon) \in (0, 1)$. Thus, autocovariances of $\text{ARMA}(p, q)$ are not only absolutely summable but also at least dominated by a function with an exponential decay.

As illustration, consider the special case of the $\text{ARMA}(1, 0)$ series $X = \{X_n\}_{n \in \mathbb{Z}}$, namely, the *autoregressive series of order one* or $\text{AR}(1)$ in Example 1.2.3. It satisfies the equation

$$X_n = \varphi X_{n-1} + Z_n, \quad (2.3.10)$$

where $|\varphi| < 1$ and $\{Z_n\}_{n \in \mathbb{Z}} \sim \text{WN}(0, \sigma_Z^2)$. Here $\varphi_1 = \varphi$. The stationary solution of (2.3.10) is

$$X_n = \sum_{j=0}^{\infty} \varphi^j Z_{n-j},$$

with the series converging in the $L^2(\Omega)$ sense since $\sum_{j=0}^{\infty} |\varphi^j|^2 < \infty$ (Example 1.2.3). Moreover,

$$\mathbb{E}X_n = 0, \quad \text{Var}(X_n) = \frac{\sigma_Z^2}{1 - \varphi^2},$$

and the autocovariance function and the spectral density of $\{X_n\}$ are given by

$$\gamma_X(n) = \sigma_Z^2 \frac{\varphi^n}{1 - \varphi^2}, \quad f_X(\lambda) = \frac{\sigma_Z^2}{2\pi} \frac{1}{|1 - \varphi e^{-i\lambda}|^2} = \frac{\sigma_Z^2}{2\pi} \frac{1}{(1 - 2\varphi \cos \lambda + \varphi^2)} \quad (2.3.11)$$

(see also Examples 1.3.2 and 1.3.3).

In general, for a stationary $\text{ARMA}(p, q)$ series (2.3.7), the spectral density is

$$f_X(\lambda) = \frac{\sigma_Z^2}{2\pi} \frac{|\theta(e^{-i\lambda})|^2}{|\phi(e^{-i\lambda})|^2} \quad (2.3.12)$$

(see Brockwell and Davis [186], Theorem 4.4.2). This is a direct consequence of the relation (2.3.9) which describes the ARMA time series $\{X_n\}$ as the output of a linear filter with white noise input and transfer function $\psi(z) = \varphi(z)^{-1}\theta(z)$ (see also Section 1.3.2). The spectral density of $\{X_n\}$ is then $f_X(\lambda) = |\psi(e^{-i\lambda})|^2 f_Z(\lambda)$ where $f_Z(\lambda) = \sigma_Z^2/2\pi$ is the spectral density of the white noise input series $\{Z_n\}$.

Example 2.3.4 (Markov chains) Consider a Markov chain $X = \{X_n\}_{n \geq 0}$ with a finite number of states $i = 1, \dots, m$, and characterized by transition probabilities

$$\begin{aligned} p_{ij} &= \mathbb{P}(X_{n+1} = j | X_n = i) \\ &= \mathbb{P}(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0), \quad i, j = 1, \dots, m. \end{aligned}$$

Denote the transition matrices by

$$P = (p_{ij})_{1 \leq i, j \leq m}, \quad P^n = (p_{ij}^{(n)})_{1 \leq i, j \leq m},$$

where $p_{ij}^{(n)}$ denotes the probability of moving from state i to state j in n steps. Assume, in addition, that X is an irreducible Markov chain (i.e., it is possible to reach any state from any

state) and aperiodic (its period equals 1). Then, there are the so-called invariant probabilities $\pi_i > 0$, $i = 1, \dots, m$, such that

$$\sum_{i=1}^m \pi_i = 1, \quad \lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j, \quad \pi_i = \sum_{j=1}^m \pi_j p_{ji}. \quad (2.3.13)$$

In particular, a Markov chain X with the initial distribution $\mathbb{P}(X_0 = i) = \pi_i$ is (strictly) stationary. Consider now

$$Y_n = g(X_n)$$

for some function g , which is also a stationary Markov chain under the initial distribution π for X .

We will show next that, if the matrix P has distinct characteristic roots (eigenvalues), then

$$|\gamma_Y(k)| \leq Cs^k, \quad \text{with some } 0 < s < 1. \quad (2.3.14)$$

That is, the Markov chain has at least exponentially decreasing autocovariances and hence is SRD. Turning to the proof of (2.3.14), if the characteristic roots $\lambda_1, \dots, \lambda_m$ (which are complex-valued when P is not symmetric) are distinct, then P can be diagonalized as

$$P = E \Lambda F' = \begin{pmatrix} e'_1 \\ \vdots \\ e'_m \end{pmatrix} \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_m \end{pmatrix} (f_1 \dots f_m),$$

where $E = (e_1 \dots e_m)$ consists of right eigenvectors e_i of P (possibly complex-valued) associated with eigenvalue λ_i ; that is, $P e_i = \lambda_i e_i$ and where $F = (f_1 \dots f_m)$ consists of left eigenvectors of P satisfying $f'_i P = \lambda_i f'_i$. In matrix notation, we have $P E = E \Lambda$ and $F' P = \Lambda F'$ as well as $F' = E^{-1}$. All vectors are column vectors and prime denotes a transpose.

Then,

$$P^n = E \Lambda^n F' = \sum_{i=1}^m \lambda_i^n e_i f'_i.$$

Using $\sum_{j=1}^m p_{ij} = 1$ and (2.3.13),

$$P \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}, \quad \bar{\pi}' = \bar{\pi}' P, \quad \text{with } \bar{\pi} = \begin{pmatrix} \pi_1 \\ \vdots \\ \pi_m \end{pmatrix},$$

implying that $\lambda_1 = 1$, $e'_1 = (1, \dots, 1)$, $f'_1 = \bar{\pi}'$. Hence, setting

$$\Pi = (\bar{\pi} \dots \bar{\pi})' = \begin{pmatrix} \pi_1 & \cdots & \pi_m \\ \vdots & \ddots & \vdots \\ \pi_1 & \cdots & \pi_m \end{pmatrix},$$

one gets

$$P^n = \Pi + \sum_{i=2}^m \lambda_i^n e_i f'_i.$$

Since $P^n \rightarrow \Pi$ and λ_i s are distinct, we have $\lambda_i^n \rightarrow 0$ and hence $|\lambda_i| < 1$, $i = 2, \dots, m$. Then,

$$|p_{ij}^{(n)} - \pi_j| \leq Cs^n, \quad \text{for some } 0 < s < 1, \quad (2.3.15)$$

which can be taken as $s = \max\{|\lambda_i| : i = 2, \dots, m\}$. (Feller [361], Chapter XVI, refers to the analysis above as an algebraic treatment of Markov chains.)

Finally, observe that

$$\begin{aligned} \gamma_Y(n) &= \mathbb{E}_{\bar{\pi}} g(X_n)g(X_0) - (\mathbb{E}_{\bar{\pi}} g(X_0))^2 \\ &= \sum_{i,j=1}^m g(i)g(j)\pi_i p_{ij}^{(n)} - \sum_{i,j=1}^m g(i)g(j)\pi_i \pi_j = \sum_{i,j=1}^m g(i)g(j)\pi_i \{p_{ij}^{(n)} - \pi_j\} \end{aligned}$$

and the result (2.3.14) follows from (2.3.15).

Relation (2.3.14) shows that Markov chains (at least with a finite number of states) are SRD. For this reason, LRD series are often called non-Markovian.

2.4 Examples of Long-Range Dependent Series: FARIMA Models

Examples of LRD series can be readily constructed by taking series satisfying condition I or IV in Section 2.1. We consider here one such celebrated class of series, namely, the FARIMA(p, d, q) series. We shall begin by introducing FARIMA(0, d , 0) series, and then discuss the more general case of FARIMA(p, d, q) series. These series are LRD for $d \in (0, 1/2)$ as in Section 2.1. For completeness sake, we shall define these series for other values of d as well. For some of these values, the series is SRD.

2.4.1 FARIMA(0, d , 0) Series

FARIMA(0, d , 0) series will be defined as

$$X_n = (I - B)^{-d} Z_n = \sum_{j=0}^{\infty} b_j B^j Z_n = \sum_{j=0}^{\infty} b_j Z_{n-j}, \quad (2.4.1)$$

where B is the backshift operator, $\{Z_n\}_{n \in \mathbb{Z}} \sim \text{WN}(0, \sigma_Z^2)$ and b_j s are coefficients in the Taylor expansion

$$\begin{aligned} (1 - z)^{-d} &= 1 + dz + \frac{d(d+1)}{2!} z^2 + \frac{d(d+1)(d+2)}{3!} z^3 + \dots \\ &= \sum_{j=0}^{\infty} \left(\prod_{k=1}^j \frac{k-1+d}{k} \right) z^j = \sum_{j=0}^{\infty} \frac{\Gamma(j+d)}{\Gamma(j+1)\Gamma(d)} z^j =: \sum_{j=0}^{\infty} b_j z^j. \end{aligned} \quad (2.4.2)$$

(In some instances below, b_j will be denoted $b_j^{(-d)}$ in order to indicate the dependence on d , and where $(-d)$ refers to the exponent of $(I - B)$.)

The time series (2.4.1) is well-defined when $\sum_{j=0}^{\infty} b_j^2 < \infty$, which depends on the behavior of b_j as $j \rightarrow \infty$. Observe that, by Stirling's formula $\Gamma(p) \sim \sqrt{2\pi} e^{-(p-1)} (p-1)^{p-1/2}$, as $p \rightarrow \infty$, we have: for $a, b \in \mathbb{R}$,

$$\frac{\Gamma(j+a)}{\Gamma(j+b)} \sim \frac{e^{-(j+a-1)}(j+a-1)^{j+a-\frac{1}{2}}}{e^{-(j+b-1)}(j+b-1)^{j+b-\frac{1}{2}}} \quad (2.4.3)$$

$$= e^{b-a} \frac{j^{a-\frac{1}{2}} (1 + \frac{a-1}{j})^{a-\frac{1}{2}} (1 + \frac{a-1}{j})^j}{j^{b-\frac{1}{2}} (1 + \frac{b-1}{j})^{b-\frac{1}{2}} (1 + \frac{b-1}{j})^j} \sim e^{b-a} j^{a-b} \frac{e^{a-1}}{e^{b-1}} = j^{a-b}, \quad (2.4.4)$$

as $j \rightarrow \infty$. This yields

$$b_j \sim \frac{j^{d-1}}{\Gamma(d)}, \quad (2.4.5)$$

as $j \rightarrow \infty$. The formula (2.4.5) is valid for any $d \in \mathbb{R}$, but when d is zero or a negative integer, $\Gamma(d) = \infty$ and hence $1/\Gamma(d)$ should be interpreted as 0.

The time series (2.4.1) is therefore well-defined if $2(d-1) + 1 = 2d - 1 < 0$ or

$$d < 1/2.$$

Definition 2.4.1 A time series given by (2.4.1) is called *FARIMA(0, d, 0) series* when $d < 1/2$.

The “F” and “I” in FARIMA stand for “Fractionally” and “Integrated”, respectively. The reason for this terminology will become apparent below. A FARIMA sequence is sometimes called ARFIMA, and the differencing operator $I - B$ is denoted by Δ .

Remark 2.4.2 When d is zero or a negative integer, (2.4.1) reduces to a finite sum involving the iterated differences $\Delta^{-d} = (I - B)^{-d}$. Thus,

$$\begin{aligned} d = 0 : X_n &= Z_n, \\ d = -1 : X_n &= \Delta^1 Z_n = Z_n - Z_{n-1}, \\ d = -2 : X_n &= \Delta^2 Z_n = Z_n - 2Z_{n-1} + Z_{n-2}. \end{aligned}$$

We provide next several properties of FARIMA(0, d, 0) series. It follows from its definition that FARIMA(0, d, 0), $d < 1/2$, series is stationary. Since $b_j \sim j^{d-1}/\Gamma(d)$, it is LRD when $0 < d < 1/2$, in the sense of condition I and hence II, III and V. When $d = 0$, the FARIMA(0, d, 0) series is a white noise and hence SRD. Suppose now $d < 0$ and not equal to a negative integer. Observe that

$$\sum_h |\gamma_X(h)| \leq \sigma_Z^2 \sum_h \sum_j |b_j b_{j+h}| = \sigma_Z^2 \left(\sum_j |b_j| \right)^2 < \infty,$$

since $|b_j| \sim j^{d-1}/|\Gamma(d)|$ are absolutely summable for $d < 0$. If d is a negative integer, then $\Gamma(d) = \infty$ and we must consider this case separately. If $d = -1$, for example, $X_n = Z_n - Z_{n-1} = \Delta Z_n$ and thus $X_n = \Delta^{-d} Z_n$ for $d = -1, -2, \dots$. In all these cases $\gamma_X(h) = 0$

for $|h| > |d|$ and hence the series is still SRD. Thus, FARIMA(0, d , 0) is SRD when $d < 0$. A more precise asymptotics for $\gamma_X(h)$ will be obtained below. The next result provides the spectral density of the series.

Proposition 2.4.3 *FARIMA(0, d , 0), $d < 1/2$, series has spectral density*

$$f_X(\lambda) = \frac{\sigma_Z^2}{2\pi} \left(2 \sin \frac{\lambda}{2}\right)^{-2d} = \frac{\sigma_Z^2}{2\pi} |1 - e^{-i\lambda}|^{-2d} \sim \frac{\sigma_Z^2}{2\pi} |\lambda|^{-2d}, \quad \text{as } \lambda \rightarrow 0.$$

Proof Note that, since $\sum_{j=0}^{\infty} b_j^2 < \infty$, the series $\sum_{j=0}^{\infty} b_j e^{-ij\lambda}$ converges in $L^2(-\pi, \pi]$. In fact,

$$\sum_{j=0}^{\infty} b_j e^{-ij\lambda} = (1 - e^{-i\lambda})^{-d}, \quad \text{a.e. } d\lambda. \quad (2.4.6)$$

For $d < 0$, this follows directly from the facts that the Taylor series $(1 - z)^{-d} = \sum_{j=0}^{\infty} b_j z^j$ converges for $|z| = 1$ (and hence for $z = e^{-i\lambda}$) and that the $L^2(-\pi, \pi]$ and pointwise limits of a series must be the same a.e. When $0 < d < 1/2$, this argument does not apply because $\sum_{j=0}^{\infty} |b_j| = \infty$. In this case, to show (2.4.6), take $0 < r < 1$ and consider the sequence of functions $(1 - re^{-i\lambda})^{-d}$. Observe that

$$(1 - re^{-i\lambda})^{-d} = \sum_{j=0}^{\infty} b_j r^j e^{-ij\lambda} \rightarrow \sum_{j=0}^{\infty} b_j e^{-ij\lambda}$$

in $L^2(-\pi, \pi]$ as $r \rightarrow 1$, because

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \sum_{j=0}^{\infty} b_j r^j e^{-ij\lambda} - \sum_{j=0}^{\infty} b_j e^{-ij\lambda} \right|^2 d\lambda = \sum_{j=0}^{\infty} b_j^2 (1 - r^j)^2 \rightarrow 0,$$

as $r \rightarrow 1$, by using $\sum_{j=0}^{\infty} b_j^2 < \infty$. On the other hand, $(1 - re^{-i\lambda})^{-d} \rightarrow (1 - e^{-i\lambda})^{-d}$ pointwise for all $\lambda \in (-\pi, \pi] \setminus \{0\}$. Since the $L^2(-\pi, \pi]$ and pointwise limits of a sequence must be the same a.e., we deduce (2.4.6). Finally, by using (2.4.6), Theorem 4.10.1 in Brockwell and Davis [186] yields

$$f_X(\lambda) = \frac{\sigma_Z^2}{2\pi} \left| \sum_{j=0}^{\infty} b_j e^{-ij\lambda} \right|^2 = \frac{\sigma_Z^2}{2\pi} |1 - e^{-i\lambda}|^{-2d}. \quad \square$$

Corollary 2.4.4 *The FARIMA(0, d , 0) series, with $d < 1/2$ but not a negative integer or zero, has autocovariances*

$$\gamma_X(k) = \sigma_Z^2 \frac{(-1)^k \Gamma(1 - 2d)}{\Gamma(k - d + 1) \Gamma(1 - k - d)} = \sigma_Z^2 \frac{\Gamma(1 - 2d)}{\Gamma(1 - d) \Gamma(d)} \frac{\Gamma(k + d)}{\Gamma(k - d + 1)} \quad (2.4.7)$$

$$\sim k^{2d-1} \sigma_Z^2 \frac{\Gamma(1 - 2d)}{\Gamma(1 - d) \Gamma(d)} = k^{2d-1} \sigma_Z^2 \frac{\Gamma(1 - 2d) \sin(d\pi)}{\pi}, \quad \text{as } k \rightarrow \infty. \quad (2.4.8)$$

When d is a negative integer or zero, the first equality in (2.4.7) continues to hold with the convention $\Gamma(0) = \Gamma(-1) = \dots = \infty$. In particular, in this case, $\gamma_X(k) = 0$ for $k \geq 1 - d$.

Proof The first equality in (2.4.7), for any $d < 1/2$, follows from

$$\gamma_X(k) = \int_{-\pi}^{\pi} e^{ik\lambda} f_X(\lambda) d\lambda = \frac{\sigma_Z^2}{2\pi} \int_0^{2\pi} \cos(k\lambda) (2 \sin(\lambda/2))^{-2d} d\lambda = \frac{\sigma_Z^2}{\pi} \int_0^{\pi} \cos(2kx) (2 \sin x)^{-2d} dx$$

by using the following identity valid for $\Re \nu > 0$:

$$\int_0^{\pi} \cos(ax) (\sin x)^{\nu-1} dx = \frac{\pi \cos(a\pi/2)}{2^{\nu-1} \nu B(\frac{\nu+a+1}{2}, \frac{\nu-a+1}{2})} = \frac{\pi \cos(a\pi/2) \Gamma(\nu) 2^{1-\nu}}{\Gamma((\nu+a+1)/2) \Gamma((\nu-a+1)/2)}$$

(Gradshteyn and Ryzhik [421], p. 397, Formula 3.631.8). When d is not a negative integer or zero, the second equality in (2.4.7) follows by using (2.2.13) and writing

$$\begin{aligned} \Gamma(1-k-d) &= \frac{\Gamma(2-k-d)}{1-k-d} = \frac{\Gamma(3-k-d)}{(1-k-d)(2-k-d)} = \frac{\Gamma(1-d)}{(1-k-d)(2-k-d) \dots (-d)} \\ &= \frac{\Gamma(1-d)}{(-1)^k (d+k-1)(d+k-2) \dots d} = \frac{\Gamma(1-d)\Gamma(d)}{(-1)^k (d+k-1)(d+k-2) \dots d\Gamma(d)} = \frac{\Gamma(1-d)\Gamma(d)}{(-1)^k \Gamma(k+d)}. \end{aligned}$$

The asymptotic relation in (2.4.8) follows from (2.4.4) and the equality in (2.4.8) follows from (2.2.14). $\square$

Having an explicit form for autocovariances is useful in many instances, for example, in simulations (see Section 2.11 below). The autocorrelations $\rho_X(k) = \gamma_X(k)/\gamma_X(0)$ can be expressed as

$$\rho_X(k) = \prod_{j=1}^k \frac{j-1+d}{j-d}, \quad k \geq 1. \quad (2.4.9)$$

An explicit form of the partial autocorrelation coefficients is given in Exercise 2.8. These coefficients enter in the Durbin–Levinson algorithm and are useful for prediction. See also the related Exercise 2.9.

By (2.4.5), Proposition 2.4.3 and Corollary 2.4.4, the FARIMA(0, d , 0), $0 < d < 1/2$, series is LRD in the sense of conditions I, IV and II. These series are thus LRD in the sense of all five conditions found in Section 2.1. Note also that, when $d < 0$, Proposition 2.4.3 shows that the series is antipersistent because in this case $\sum_{k=-\infty}^{\infty} \gamma_X(k) = 2\pi f_X(0) = 0$.

The FARIMA(0, d , 0) series was defined as $X_n = (I - B)^{-d} Z_n$ in (2.4.1). Can we invert the operator $(I - B)^{-d}$ and write $(I - B)^d X_n = Z_n$? The following result provides an answer.

Proposition 2.4.5 *Let $X = \{X_n\}_{n \in \mathbb{Z}}$ be a FARIMA(0, d , 0) series. If $-1 < d < 1/2$, then*

$$(I - B)^d X_n = Z_n. \quad (2.4.10)$$

The range $-1 < d < 1/2$ is motivated by the fact that the coefficients in $(I - B)^d = \sum_{j=0}^{\infty} b_j^{(d)} B^j$ satisfy

$$b_j^{(d)} \sim \frac{j^{-d-1}}{\Gamma(-d)} \quad (2.4.11)$$

in view of (2.4.5) and hence decay to zero only when $-1 < d$. The proof of the proposition is based on the fact that $(I - B)^{-d}$ and $(I - B)^d$ are linear filters operating on second-order time series.

Proof We want to show that the FARIMA series $X_n = (I - B)^{-d} Z_n$, $n \in \mathbb{Z}$, satisfies (2.4.10). Let $b_j^{(d)}$ be coefficients in $(1 - z)^d = \sum_{j=0}^{\infty} b_j^{(d)} z^j$. Write also the white noise Z_n in its spectral representation as

$$Z_n = \int_{(-\pi, \pi]} e^{in\lambda} W(d\lambda),$$

where W is a random measure with orthogonal increments having control measure $\mathbb{E}|W(d\lambda)|^2 = (\sigma_Z^2/2\pi)d\lambda$ (see Section 1.3.4 and Appendix B.1).

By Theorem 4.10.1 in Brockwell and Davis [186], concerning linear filters,

$$X_n = (I - B)^{-d} Z_n = \int_{(-\pi, \pi]} e^{in\lambda} (1 - e^{-i\lambda})^{-d} dW(\lambda).$$

(i) Consider first the case $0 < d < 1/2$. Since the coefficients $b_j^{(d)}$ in $(I - B)^d$ are absolutely summable in view of (2.4.5), Remark 1 following Theorem 4.10.1 in Brockwell and Davis [186] implies that

$$(I - B)^d X_n = \int_{(-\pi, \pi]} e^{in\lambda} (1 - e^{-i\lambda})^d (1 - e^{-i\lambda})^{-d} W(d\lambda) = \int_{(-\pi, \pi]} e^{in\lambda} W(d\lambda) = Z_n. \quad (2.4.12)$$

(ii) When $d = 0$, the result is trivial.

(iii) When $-1/2 < d < 0$, we can use the convergence of the series $\sum_{j=0}^{\infty} b_j^{(d)} e^{ij\lambda}$ in $L^2(-\pi, \pi]$:

$$\int_{-\pi}^{\pi} \left| \sum_{j=0}^{\infty} b_j^{(d)} e^{ij\lambda} \right|^2 d\lambda = 2\pi \sum_{j=0}^{\infty} |b_j^{(d)}|^2 < \infty,$$

which holds because of (2.4.11). We also have

$$\int_{-\pi}^{\pi} \left| \sum_{j=0}^{\infty} b_j^{(d)} e^{ij\lambda} \right|^2 |1 - e^{-i\lambda}|^{-2d} d\lambda < \infty,$$

since $|1 - e^{-i\lambda}|^{-2d}$ is bounded for $d < 0$. Then, (2.4.12) holds as well by using Theorem 4.10.1 in Brockwell and Davis [186].

(iv) When $-1 < d \leq -1/2$, we need a more refined argument because $\sum_{j=0}^{\infty} |b_j^{(d)}|^2 = \infty$ by (2.4.11). This argument also applies to the case $-1 < d < 1/2$ with $d \neq 0$. We will show that

$$\lim_{k \rightarrow \infty} \mathbb{E} \left(\sum_{j=0}^k b_j^{(d)} X_{n-j} - Z_n \right)^2 = 0.$$

Observe that (2.4.5) is valid in the range $-1 < d < 1/2$, $d \neq 0$, with $1/\Gamma(d) \neq 0$. The relation $(1-z)^d(1-z)^{-d} = 1$ yields

$$b_0^{(d)}b_0^{(-d)} = 1, \quad \sum_{j=0}^m b_j^{(d)}b_{m-j}^{(-d)} = 0, \quad m \geq 1, \quad (2.4.13)$$

by identifying the coefficients of z with identical powers. Then,

$$\begin{aligned} \sum_{j=0}^k b_j^{(d)} X_{n-j} - Z_n &= \sum_{j=0}^k b_j^{(d)} \sum_{l=0}^{\infty} b_l^{(-d)} Z_{n-j-l} - Z_n \\ &= \sum_{m=0}^{\infty} \left(\sum_{j=0}^{\min\{k,m\}} b_j^{(d)} b_{m-j}^{(-d)} \right) Z_{n-m} - Z_n = \sum_{m=k+1}^{\infty} \left(\sum_{j=0}^k b_j^{(d)} b_{m-j}^{(-d)} \right) Z_{n-m} \end{aligned}$$

by using (2.4.13). It is then enough to show that

$$I_k := \sum_{m=k+1}^{\infty} \left(\sum_{j=0}^k b_j^{(d)} b_{m-j}^{(-d)} \right)^2 \rightarrow 0,$$

as $k \rightarrow \infty$. By using (2.4.11), for large k and a generic constant C_j ,

$$\begin{aligned} I_k &\leq C_1 \sum_{m=k+1}^{\infty} \left(\sum_{j=0}^k (j+1)^{-d-1} (m-j)^{d-1} \right)^2 \\ &= \frac{C_1}{k} \sum_{m=k+1}^{\infty} \left(\sum_{j=0}^k \left(\frac{j+1}{k} \right)^{-d-1} \left(\frac{m-j}{k} \right)^{d-1} \frac{1}{k} \right)^2 \frac{1}{k} \leq \frac{C_2}{k} \int_{1+\frac{1}{k}}^{\infty} \left(\int_0^1 u^{-d-1} (x-u)^{d-1} du \right)^2 dx \\ &= \frac{C_2}{k} \int_{1+\frac{1}{k}}^{\infty} x^{-2} \left(\int_0^{\frac{1}{x}} z^{-d-1} (1-z)^{d-1} dz \right)^2 dx \leq C_3 \frac{k^{-2d-1}}{k} = C_3 k^{-2d-2} \rightarrow 0, \end{aligned}$$

where we used $\int_0^{1/x} z^{-d-1} (1-z)^{d-1} dz \sim Cx^d$, as $x \rightarrow \infty$, $\int_0^{1/x} z^{-d-1} (1-z)^{d-1} dz \sim C(x-1)^d$, as $x \rightarrow 1$, and the fact that $-1 < d < 1/2$. $\square$

Property (2.4.10) is referred to as the *invertibility* of the series X . The proposition states that X is invertible in the range $d \in (-1, 1/2)$.

In Figure 2.2, we provide several realizations of Gaussian FARIMA(0, d , 0) series generated through the circulant matrix embedding method discussed in Section 2.11.2. The series in the left, middle and right plots are, respectively, antipersistent, uncorrelated (independent) and long-range dependent. Note that the realizations are consistent with the spectral domain perspective of time series outlined in Section 1.3.4. For LRD series, in particular, the spectral density diverges at the origin and hence low frequencies (long waves) dominate in the spectral representation of the series. Indeed, the LRD series in the right plot of the figure tends to stay away from the zero mean for extended periods of time. In fact, this is one of the characteristic features of LRD series. In contrast, the antipersistent series in the left plot fluctuates even more than the uncorrelated series in the middle plot.

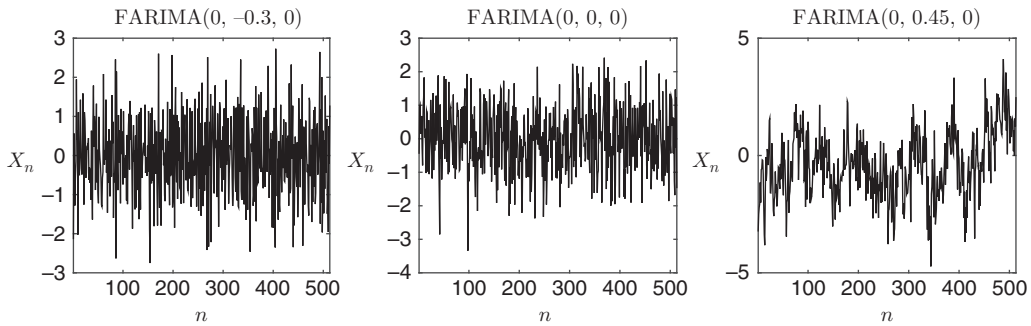


Figure 2.2 Several realizations of Gaussian FARIMA(0, d , 0) series. Left: $d = -0.3$. Middle: $d = 0$. Right: $d = 0.45$.

The sequence $X_n = (I - B)^{-d} Z_n$, $n \in \mathbb{Z}$, can be viewed as the transform of the FARIMA(0, 0, 0) series $\{Z_n\}$ into a FARIMA(0, d , 0) series. How can we obtain the FARIMA(0, d , 0) series $\{X_n\}$ from a FARIMA(0, d' , 0) series $\{X'_n\}$? The following proposition shows how this can be done.

Proposition 2.4.6 Let $d < 1/2$ and $X' = \{X'_n\}_{n \in \mathbb{Z}}$ be a FARIMA(0, d' , 0) series with

$$d' = d - [d + 1/2] \in \left[-\frac{1}{2}, \frac{1}{2}\right),$$

where $[\cdot]$ denotes the integer part.⁶ Then,

$$X_n = (I - B)^{-[d+1/2]} X'_n, \quad n \in \mathbb{Z},$$

is a FARIMA(0, d , 0) series.

Proof Since $d < 1/2$ and $[d + 1/2]$ is integer, $-[d + 1/2]$ is a nonnegative integer and therefore the expansion of $(1 - z)^{-[d+1/2]}$ has only a finite number of non-zero coefficients which are obviously summable. Therefore, this proposition follows as in Proposition 2.4.5 (case $0 < d < 1/2$). $\square$

$\Delta = I - B$ is called the *differencing operator* since $\Delta X_n = (I - B)X_n = X_n - X_{n-1}$. In view of the preceding proposition, any FARIMA(0, d , 0), $d < 1/2$, series can be obtained from a FARIMA(0, d' , 0) series with

$$-1/2 \leq d' < 1/2$$

by differencing it an integer number of times. This is why the range $[-1/2, 1/2)$ is called the *principal range* of d .

Example 2.4.7 (FARIMA(0, d , 0) series with $d < -1/2$) Since $d = -5.7 = 0.3 - 6$, a FARIMA(0, -5.7 , 0) series X can be obtained by differencing $-[-5.7 + 1/2] = -[-5.2] = -(-6) = 6$ times a FARIMA(0, 0.3, 0) series X' ; that is, $X_n = \Delta^6 X'_n$.

⁶ That is, $[a]$ is the largest integer which is less than or equal to $a \in \mathbb{R}$.

Similarly, since $-5.3 = -0.3 - 5$, a FARIMA(0, -5.3 , 0) series X can be obtained from a FARIMA(0, -0.3 , 0) series X' by differencing it five times; that is, $X_n = \Delta^5 X'_n$.

The preceding discussion and Proposition 2.4.6 also suggest the following natural extension of the definition of FARIMA(0, d , 0) series to $d \geq 1/2$.

Definition 2.4.8 A series $X = \{X_n\}_{n \in \mathbb{Z}}$ is called FARIMA(0, d , 0) for $d \geq 1/2$ if it is obtained by taking the $[d + 1/2]$ th order partial sum series of FARIMA(0, d' , 0) series with parameter $d' = d - [d + 1/2]$ in the principal range $-1/2 \leq d' < 1/2$.

By definition, FARIMA(0, d , 0) is not stationary when $d \geq 1/2$. It satisfies the relation

$$(I - B)^{[d+1/2]} X_n = (I - B)^{-(d-[d+1/2])} Z_n,$$

where $(I - B)^{[d+1/2]}$ is the differencing operator $\Delta = I - B$ iterated $[d + 1/2]$ times.

Example 2.4.9 (FARIMA(0, d , 0) series with $d \geq 1/2$) Let Z_n , $n \in \mathbb{Z}$, be a FARIMA(0, 0, 0); that is, a white noise series. The series X_n , $n \in \mathbb{Z}$, defined as $X_n = \sum_{k=1}^n Z_k$ for $n > 0$, $X_0 = 0$, $X_n = \sum_{k=n}^{-1} Z_k$ for $n < 0$, is a FARIMA(0, 1, 0) series.

2.4.2 FARIMA(p , d , q) Series

Finally, we briefly discuss the class of stationary FARIMA(p , d , q) series. This class combines ARMA(p , q) series in Example 2.3.3 and FARIMA(0, d , 0) series. As in Example 2.3.3, consider polynomials

$$\varphi(z) = 1 - \varphi_1 z - \cdots - \varphi_p z^p, \quad \theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q \quad (2.4.14)$$

and assume that they do not have common zeros. Suppose that φ does not have roots on the unit circle. Consider the series

$$X_n = \varphi^{-1}(B)\theta(B)(I - B)^{-d} Z_n =: \sum_{j=-\infty}^{\infty} b_j Z_{n-j}, \quad (2.4.15)$$

where $\{Z_n\}_{n \in \mathbb{Z}} \sim \text{WN}(0, \sigma_Z^2)$ and $\{b_k\}_{k \in \mathbb{Z}}$ are coefficients in the Laurent expansion $\varphi^{-1}(z)\theta(z)(1 - z)^{-d} = \sum_{k=-\infty}^{\infty} b_k z^k$. Note that the sequence $\{b_k\}_{k \in \mathbb{Z}} \in \ell^2(\mathbb{Z})$ since b_k can be viewed as Fourier coefficients of the function $\varphi^{-1}(e^{-i\lambda})\theta(e^{-i\lambda})(1 - e^{-i\lambda})^{-d}$.

Definition 2.4.10 The series $X = \{X_n\}_{n \in \mathbb{Z}}$ in (2.4.15) is called FARIMA(p , d , q), $d < 1/2$, series.

The basic idea behind FARIMA(p , d , q) models is that the operator $(I - B)^{-d}$ is responsible for LRD features of the series while $\varphi^{-1}(B)\theta(B)$ accounts for SRD behavior.

A number of properties can be proved about FARIMA(p, d, q) series. As in Proposition 2.4.3, one can show that this series has a spectral density

$$f_X(\lambda) = \frac{\sigma_Z^2}{2\pi} \frac{|\theta(e^{-i\lambda})|^2}{|\varphi(e^{-i\lambda})|^2} |1 - e^{-i\lambda}|^{-2d}. \quad (2.4.16)$$

This expression is rather obvious since the relation (2.4.15) indicates that a FARIMA(p, d, q) series can be viewed as the output of a linear filter with the transfer function $(1 - z)^{-d}$ and, as input, an ARMA(p, q) series with the spectral density given in (2.3.12).

Proposition 2.4.11 *Suppose that $\varphi(z)$ defined in (2.4.14) has roots outside the unit circle. Then, the FARIMA(p, d, q) series, $0 < d < 1/2$, is LRD in the sense of conditions I, II, III, IV and V.*

Proof Since the FARIMA(p, d, q) has spectral density (2.4.16) with $0 < d < 1/2$, it is LRD in the sense of condition IV. Since the roots of $\varphi(z)$ are outside the unit circle, the coefficients b_k in (2.4.15) are zero for $k \leq -1$ (cf. (2.3.9) and the relation preceding it). One can then show that

$$b_k \sim \frac{\theta(1)}{\varphi(1)} \frac{k^{d-1}}{\Gamma(d)} \quad (2.4.17)$$

(Exercise 2.10; compare with (2.4.5)). Hence, FARIMA(p, d, q), $0 < d < 1/2$, series is also LRD in the sense of condition I. By Corollaries 2.2.10 and 2.2.7, and Proposition 2.2.4, the times series is LRD in the sense of conditions II, III and V as well. $\square$

Explicit autocovariance formulas are also known, for example, for FARIMA(1, d , 0) series (Hosking [484]). (In the case of FARIMA(0, d , q), they can be obtained directly from the case of FARIMA(0, d , 0) series.) When $\theta(z)$ does not have zeroes on the unit circle and $-1 < d < 1/2$, one can show that FARIMA(p, d, q) satisfies the relation

$$\theta^{-1}(B)\varphi(B)(I - B)^d X_n = Z_n$$

(e.g., Theorem 3.4, (c), in Palma [789]); that is, $\{X_n\}$ is invertible.

2.5 Definition and Basic Properties of Self-Similar Processes

We now turn to self-similar processes. These processes are closely related to long-range dependent series though the discussion of exact relationships is postponed till Section 2.8. We shall, in the meantime, consider self-similar processes somewhat separately from long-range dependent series.

Definition 2.5.1 A stochastic process $\{X(t)\}_{t \in \mathbb{R}}$ is called *self-similar* (SS) or *H-self-similar* (H-SS) if there is $H > 0$ such that, for all $c > 0$,

$$\{X(ct)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{c^H X(t)\}_{t \in \mathbb{R}}, \quad (2.5.1)$$

where $\stackrel{d}{=}$ denotes equality of the finite-dimensional distributions.⁷ Thus, for any $c > 0$ and times $t_1, \dots, t_m$ in $\mathbb{R}$ and integer $m \geq 1$, the vector

$$(X(ct_1), \dots, X(ct_m))'$$

has the same distribution as the vector

$$(c^H X(t_1), \dots, c^H X(t_m))'.$$

The parameter H is called the *self-similarity parameter* (also *Hurst index* or *Hurst parameter*).

Since H needs not to be equal to 1, self-similarity should be called *self-affinity*. Since, moreover, it is not the path of $X(t)$ but its distribution which scales, one should say “statistical self-similarity” or better, “statistical self-affinity.” We will continue to use “self-similarity” because this term is now widely used.

Intuitively, self-similarity means that a stochastic process scaled in time (that is, plotted with a different time scale) looks statistically the same as the original process when properly rescaled in space. Physically, expanding the time scale (large c) should require an expansion of the space scale, that is, a large factor c^H and hence the self-similarity parameter H should be positive. One also has reasons to exclude $H \leq 0$ on theoretical grounds. Note that (2.5.1) implies that, for any $c > 0$,

$$X(c) \stackrel{d}{=} c^H X(1).$$

One expects the left-hand side to converge to $X(0)$ as $c \rightarrow 0$ under mild assumptions. If $H < 0$, however, the right-hand side diverges. The case $H = 0$ could be included in Definition 2.5.1 but, under milder assumptions, it leads to the trivial stochastic process $X(t) = X(0)$ a.s. for all t (see Embrechts and Maejima [350], Theorem 1.1.2).

A self-similar process cannot be (strictly) stationary. If it were stationary, then, for any $t \in \mathbb{R}$ and $c > 0$,

$$X(t) \stackrel{d}{=} X(ct) \stackrel{d}{=} c^H X(t).$$

But the right-hand side of this relation diverges as $c \rightarrow \infty$. Even though a self-similar process is not (strictly) stationary, it can have stationary increments. In fact, our focus will often be on self-similar processes with stationary increments. One reason is that increments of self-similar processes provide models of long-range dependent series. Though the definition of stationarity of increments was given in Section 1.1.2, we recall it next to emphasize its important role in the study of SS processes.

Definition 2.5.2 A process $\{X(t)\}_{t \in \mathbb{R}}$ has *stationary increments* if, for any $h \in \mathbb{R}$,

$$\{X(t+h) - X(h)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{X(t) - X(0)\}_{t \in \mathbb{R}} \quad (2.5.2)$$

⁷ Depending on the context $\stackrel{d}{=}$ may refer either to equality of the finite-dimensional distributions or of the marginal distributions. The relation $\{X_1(t)\} \stackrel{d}{=} \{X_2(t)\}$ always means equality of the finite-dimensional distributions.

or, equivalently, if for any $s \in \mathbb{R}$,

$$\{X(t) - X(s)\}_{t \in \mathbb{R}} \stackrel{d}{=} \{X(t-s) - X(0)\}_{t \in \mathbb{R}}.$$

If $\{X(t)\}_{t \in \mathbb{R}}$ is Gaussian with mean 0 and $X(0) = 0$, then stationarity of the increments is equivalent to

$$\mathbb{E}(X(t) - X(s))^2 = \mathbb{E}(X(t-s))^2, \quad \text{for all } s, t \in \mathbb{R} \quad (2.5.3)$$

(Exercise 2.11).

Remark 2.5.3 The relation (2.5.2) implies that, for a process $\{X(t)\}_{t \in \mathbb{R}}$ with stationary increments and any $h \in \mathbb{R}$, the process

$$Y(t) = X(t+h) - X(t), \quad t \in \mathbb{R}, \quad (2.5.4)$$

is stationary. To see that Definition 2.5.2 implies (2.5.4), note that, for any $s \in \mathbb{R}$,

$$\begin{aligned} \{Y(t+s)\}_{t \in \mathbb{R}} &= \{X(t+s+h) - X(t+s)\}_{t \in \mathbb{R}} \\ &= \{(X(t+s+h) - X(s)) - (X(t+s) - X(s))\}_{t \in \mathbb{R}} \\ &\stackrel{d}{=} \{(X(t+h) - X(0)) - (X(t) - X(0))\}_{t \in \mathbb{R}} = \{Y(t)\}_{t \in \mathbb{R}}. \end{aligned}$$

If $\mathbb{R}$ is replaced by $\mathbb{Z}$, then the stationarity of $\{Y(t)\}$ in (2.5.4) implies that $\{X(t)\}$ has stationary increments. Indeed for $h \in \mathbb{Z}$, say $h > 0$, observe that

$$\{X(t+h) - X(h)\}_{t \in \mathbb{Z}} = \left\{ \sum_{k=0}^{t-1} (X(k+1+h) - X(k+h)) \right\}_{t \in \mathbb{Z}} \quad (2.5.5)$$

$$\stackrel{d}{=} \left\{ \sum_{k=0}^{t-1} (X(k+1) - X(k)) \right\}_{t \in \mathbb{Z}} = \{X(t) - X(0)\}_{t \in \mathbb{Z}}. \quad (2.5.6)$$

The passage from (2.5.5) to (2.5.6) follows from viewing $k+h$ as time and 1 as the increment. Since Y is stationary, $k+h$ can be replaced by k .

Remark 2.5.4 One often chooses $h = 1$ in (2.5.4) and considers a stationary series

$$Y_n = X(n+1) - X(n), \quad n \in \mathbb{Z}.$$

Definition 2.5.5 An H -self-similar process $\{X(t)\}_{t \in \mathbb{R}}$ will be referred to as an SS or H -SS process. An H -self-similar process $\{X(t)\}_{t \in \mathbb{R}}$ with stationary increments will be referred to as SSSI or H -SSSI process.

The next result establishes several basic properties of SSSI processes. We always suppose $H > 0$.

Proposition 2.5.6 Let $X = \{X(t)\}_{t \in \mathbb{R}}$ be an H -SSSI process. Then, the following properties hold:

(a) $X(0) = 0$ a.s.

- (b) If $H \neq 1$ and $\mathbb{E}|X(1)| < \infty$, then $\mathbb{E}X(t) = 0$ for all $t \in \mathbb{R}$.
 (c) If $\mathbb{E}|X(1)|^\gamma < \infty$ for $0 < \gamma \leq 1$, then $H \leq 1/\gamma$.
 (d) If $\mathbb{E}|X(1)|^2 < \infty$, then $\mathbb{E}(X(t))^2 = |t|^{2H} \mathbb{E}(X(1))^2$, $t \in \mathbb{R}$.
 (e) If $\mathbb{E}|X(1)|^2 < \infty$, then, for $s, t \in \mathbb{R}$,

$$\mathbb{E}X(t)X(s) = \frac{\mathbb{E}X(1)^2}{2} \{ |t|^{2H} + |s|^{2H} - |t-s|^{2H} \}. \quad (2.5.7)$$

- (f) If $H = 1$ and $\mathbb{E}|X(1)|^2 < \infty$, then for any $t \in \mathbb{R}$, $X(t) = tX(1)$ a.s.

Proof Property (a) holds since, for $c > 0$,

$$X(0) \stackrel{d}{=} X(c \cdot 0) \stackrel{d}{=} c^H X(0)$$

and $c^H \rightarrow \infty$, as $c \rightarrow \infty$. Property (b) follows from

$$2^H \mathbb{E}X(t) = \mathbb{E}X(2t) = \mathbb{E}(X(2t) - X(t)) + \mathbb{E}X(t) = 2\mathbb{E}X(t),$$

by using self-similarity and stationarity of the increments. Property (c) follows from a similar argument leading to

$$\begin{aligned} 2^{H\gamma} \mathbb{E}|X(1)|^\gamma &= \mathbb{E}|X(2)|^\gamma \leq \mathbb{E}(|X(2) - X(1)| + |X(1)|)^\gamma \\ &\leq \mathbb{E}(|X(2) - X(1)|^\gamma + |X(1)|^\gamma) = 2\mathbb{E}|X(1)|^\gamma, \end{aligned}$$

by using the inequality $(a+b)^\gamma \leq a^\gamma + b^\gamma$, $a, b \geq 0$, $\gamma \in (0, 1]$. If $t > 0$, Property (d) follows from

$$\mathbb{E}(X(t))^2 = \mathbb{E}t^{2H}(X(1))^2 = t^{2H} \mathbb{E}(X(1))^2$$

and, if $t < 0$, from

$$\mathbb{E}(X(t))^2 = \mathbb{E}(X(t) - X(0))^2 = \mathbb{E}(X(0) - X(-t))^2 = \mathbb{E}(X(-t))^2 = (-t)^{2H} \mathbb{E}(X(1))^2.$$

Stationarity of the increments ensures that

$$\mathbb{E}(X(t) - X(s))^2 = \mathbb{E}(X(t-s) - X(0))^2 = \mathbb{E}X(t-s)^2$$

and therefore Property (e) is a consequence of

$$\begin{aligned} \mathbb{E}X(t)X(s) &= \frac{1}{2} \{ \mathbb{E}X(t)^2 + \mathbb{E}X(s)^2 - \mathbb{E}(X(t) - X(s))^2 \} \\ &= \frac{1}{2} \{ \mathbb{E}X(t)^2 + \mathbb{E}X(s)^2 - \mathbb{E}(X(t-s))^2 \}, \end{aligned} \quad (2.5.8)$$

and Property (d). Property (f) follows from

$$\begin{aligned} \mathbb{E}(X(t) - tX(1))^2 &= \mathbb{E}X(t)^2 - 2t\mathbb{E}X(t)X(1) + t^2\mathbb{E}X(1)^2 \\ &= t^2\mathbb{E}X(1)^2 - 2t \cdot t \frac{\mathbb{E}X(1)^2}{2} (t^2 + 1 - (t-1)^2) + t^2\mathbb{E}X(1)^2 = 0. \quad \square \end{aligned}$$

With a bit more effort, one can show that for $0 < \gamma < 1$ in (c), one has $H < 1/\gamma$ and that the conclusion in (f) holds when $\mathbb{E}|X(1)| < \infty$ (Exercise 2.12).

2.6 Examples of Self-Similar Processes

2.6.1 Fractional Brownian Motion

We shall first focus on Gaussian SSSI processes. These are defined through their covariance function and hence the following result shows that the function in Proposition 2.5.6, (e), is a valid covariance function.

Proposition 2.6.1 *Let $0 < H \leq 1$. Then, the function*

$$R_H(t, s) = |t|^{2H} + |s|^{2H} - |t - s|^{2H}, \quad s, t \in \mathbb{R},$$

*is positive semidefinite.*⁸

Proof We have to show that $\sum_{i=1}^n \sum_{j=1}^n R_H(t_i, t_j) u_i u_j \geq 0$ for any real numbers $t_1, \dots, t_n$ and $u_1, \dots, u_n$. View each u_i as a mass at the points t_i , $i = 1, \dots, n$. Now add a mass $u_0 = -\sum_{i=1}^n u_i$ at the origin $t_0 = 0$. Then $\sum_{i=0}^n u_i = 0$ and

$$\sum_{i=1}^n \sum_{j=1}^n R_H(t_i, t_j) u_i u_j = \sum_{i=1}^n \sum_{j=1}^n (|t_i|^{2H} + |t_j|^{2H} - |t_i - t_j|^{2H}) u_i u_j = - \sum_{i=0}^n \sum_{j=0}^n |t_i - t_j|^{2H} u_i u_j.$$

But, for any constant $c > 0$,

$$\sum_{i=0}^n \sum_{j=0}^n e^{-c|t_i - t_j|^{2H}} u_i u_j = \sum_{i=0}^n \sum_{j=0}^n (e^{-c|t_i - t_j|^{2H}} - 1) u_i u_j = -c \sum_{i=0}^n \sum_{j=0}^n |t_i - t_j|^{2H} u_i u_j + o(c),$$

as c tends to 0. It is therefore enough to prove that $\sum_{i=0}^n \sum_{j=0}^n e^{-c|t_i - t_j|^{2H}} u_i u_j \geq 0$. The function $e^{-c|t|^{2H}}$, $0 < H \leq 1$, is the characteristic function of a $2H$ -stable random variable ξ with the scaling exponent $c^{1/2H}$; that is, $\mathbb{E}e^{it\xi} = e^{-c|t|^{2H}}$, for all $t \in \mathbb{R}$ (see Appendix A.4). It follows that

$$\sum_{j=0}^n \sum_{k=0}^n e^{-c|t_j - t_k|^{2H}} u_j u_k = \mathbb{E} \sum_{j=0}^n \sum_{k=0}^n u_j u_k e^{i(t_j - t_k)\xi} = \mathbb{E} \left| \sum_{j=0}^n u_j e^{it_j \xi} \right|^2 \geq 0. \quad \square$$

There is then a zero mean Gaussian process $\{Z(t)\}_{t \in \mathbb{R}}$ with the covariance function

$$\text{Cov}(Z(t), Z(s)) = \Gamma_H(t, s) = \frac{\sigma^2}{2} (|t|^{2H} + |s|^{2H} - |t - s|^{2H}),$$

where $\sigma^2 = \text{Var}(Z(1))$. Since the distribution of a zero mean Gaussian process is entirely determined by its covariance function, Proposition 2.5.6, (e), implies that, for a given H , all H -SSSI zero mean Gaussian processes differ by a multiplicative constant only. This leads to the following definition.

Definition 2.6.2 A Gaussian H -SSSI process $\{B_H(t)\}_{t \in \mathbb{R}}$ with $0 < H \leq 1$ is called *fractional Brownian motion* (FBM). It is called *standard* if $\sigma^2 = \mathbb{E}(B_H(1))^2 = 1$.

⁸ Positive semidefiniteness is also referred to as nonnegative definiteness. See Section 9.1.1 for a definition.

The discussion above leads to the following useful corollary.

Corollary 2.6.3 Suppose that $\{X(t)\}_{t \in \mathbb{R}}$ satisfies the following conditions:

- (i) it is a Gaussian process with zero mean and $X(0) = 0$,
- (ii) $\mathbb{E}(X(t))^2 = \sigma^2 |t|^{2H}$ for all $t \in \mathbb{R}$ and some $\sigma > 0$ and $0 < H \leq 1$, and
- (iii) $\mathbb{E}(X(t) - X(s))^2 = \mathbb{E}(X(t - s))^2$ for all $s, t \in \mathbb{R}$.

Then, $\{X(t)\}_{t \in \mathbb{R}}$ is fractional Brownian motion.

Proof A Gaussian process is determined by its mean function and covariance function. Here the mean is zero. To determine the covariance, observe that one has (2.5.8) by (iii). By (ii), this expression becomes (2.5.7), which is the covariance of fractional Brownian motion. $\square$

Remark 2.6.4 If $H = 1$, then by the property (f) in Proposition 2.5.6, $B_H(t) = tB_H(1)$; that is, it is a straight line with a random slope. We will exclude this trivial case in the future and suppose $0 < H < 1$.

One may get a more physical understanding of FBM through its integral representations. The following result concerns integral representation in time domain. Gaussian random measures and corresponding integrals are defined in Appendix B.1. See, in particular, the integral representations in the time and spectral domains, and their relationship in (B.1.14) (see also Section 1.4.1).

Time Domain Representations of FBM

Proposition 2.6.5 For $H \in (0, 1)$, standard FBM admits the following integral representation:

$$\{B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \frac{1}{c_1(H)} \int_{\mathbb{R}} ((t-u)_+^{H-1/2} - (-u)_+^{H-1/2}) B(du) \right\}_{t \in \mathbb{R}}, \quad (2.6.1)$$

where $x_+ = \max\{x, 0\}$, $c_1(H)^2 = \int_{\mathbb{R}} ((1-u)_+^{H-1/2} - (-u)_+^{H-1/2})^2 du$ is a normalizing constant and $B(du)$ is a Gaussian random measure on $\mathbb{R}$ with the control measure $\mathbb{E}(B(du))^2 = du$.

Proof Denote the stochastic process on the right-hand side of (2.6.1) by $X_H(t)$, without the normalizing constant. We need to show that $X_H(t)$ is FBM. It is enough to prove that X_H satisfies conditions (i)–(iii) of Corollary 2.6.3.

X_H satisfies (i) of Corollary 2.6.3 by construction if it is well-defined. For X_H to be well-defined, we have to check that the deterministic integrand of (2.6.1) is square-integrable. Potential problems may occur only around $u = 0, t$ or $-\infty$. Around $u = t$, we need to consider $((t-u)_+^{H-1/2})^2 = (t-u)_+^{2H-1}$, which is integrable since $(2H-1)+1 = 2H > 0$. The argument for the behavior around $u = 0$ is analogous. Around $u = -\infty$, $((t-u)_+^{H-1/2} - (-u)_+^{H-1/2})^2 \sim C((-u)_+^{H-3/2})^2 = C(-u)_+^{2H-3}$, which is integrable since $(2H-3)+1 = 2(H-1) < 0$.

To see that X_H satisfies (ii) of Corollary 2.6.3, observe that

$$\begin{aligned}\mathbb{E}(X_H(t))^2 &= \int_{\mathbb{R}} \left((t-u)_+^{H-1/2} - (-u)_+^{H-1/2} \right)^2 du \\ &= |t|^{2H} \int_{\mathbb{R}} \left((\text{sign}(t) - v)_+^{H-1/2} - (-v)_+^{H-1/2} \right)^2 dv = |t|^{2H} \int_{\mathbb{R}} \left((1-s)_+^{H-1/2} - (-s)_+^{H-1/2} \right)^2 ds,\end{aligned}$$

where we made the changes of variables $u = |t|v$ and, if $t < 0$, $-1 - v = -s$. Similarly, X_H satisfies (iii) of Corollary 2.6.3 since

$$\begin{aligned}\mathbb{E}(X_H(t) - X_H(s))^2 &= \int_{\mathbb{R}} \left((t-u)_+^{H-1/2} - (s-u)_+^{H-1/2} \right)^2 du \\ &= \int_{\mathbb{R}} \left((t-s-v)_+^{H-1/2} - (-v)_+^{H-1/2} \right)^2 dv = \mathbb{E}(X_H(t-s))^2,\end{aligned}$$

where we made the change of variables $s - u = -v$. $\square$

Remark 2.6.6 If $1/2 < H < 1$, then we can express (2.6.1) as

$$\{B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \frac{H-1/2}{c_1(H)} \int_{\mathbb{R}} \left(\int_0^t (s-u)_+^{H-3/2} ds \right) B(du) \right\}_{t \in \mathbb{R}}. \quad (2.6.2)$$

Remark 2.6.7 When $H = 1/2$, the integrand $((t-u)_+)^0 - ((-u)_+)^0$ is interpreted by convention as $1_{[0,t)}$ if $t \geq 0$ and $1_{(t,0]}$ if $t < 0$. Note also that with this convention, $c_1(1/2) = 1$. In this case, one gets (for $t > 0$)

$$B_{1/2}(t) = \int_0^t B(du) = B(t)$$

so that FBM reduces to Brownian motion.

Remark 2.6.8 The time domain representation (2.6.1) is not unique. For example, for any $a, b \in \mathbb{R}$,

$$\int_{\mathbb{R}} \left(a((t-u)_+^{H-1/2} - (-u)_+^{H-1/2}) + b((t-u)_-^{H-1/2} - (-u)_-^{H-1/2}) \right) B(du), \quad (2.6.3)$$

where $x_- = \max\{0, -x\}$, is another representation of FBM. If $a \neq 0$, $b = 0$ as in (2.6.1), the representation is called *causal* (non-anticipative) because integration over $u \leq t$ is used for $t > 0$. If $a = b$, then it is called *well-balanced*. In this case, $x_+ + x_- = |x|$, and thus

$$\{B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ a \int_{\mathbb{R}} (|t-u|^{H-1/2} - |u|^{H-1/2}) B(du) \right\}_{t \in \mathbb{R}}.$$

Remark 2.6.9 The two integrands in (2.6.1) cannot be separated; that is, the process $Y_H(t) = \int_{\mathbb{R}} (t-u)_+^{H-1/2} B(du)$ is not well-defined. Indeed, if it were, it would be stationary and self-similar (see the discussion preceding Definition 2.5.5). However, $Y_H(t) - Y_H(0)$ once rewritten as (2.6.1) is well-defined and yields $B_H(t)$. The subtraction of $(-u)_+^{H-1/2}$ is called an *infrared correction*.

Spectral Domain Representations of FBM

A spectral domain representation of FBM can be obtained through (B.1.14) in Appendix B.1 by computing the $L^2(\mathbb{R})$ -Fourier transform of the deterministic kernel in (2.6.1) or (2.6.3). We denote the Fourier transform of a function $f \in L^2(\mathbb{R})$ as $\widehat{f}(x) = \int_{\mathbb{R}} e^{ixu} f(u) du$ (see also Appendix A.1.2).

Proposition 2.6.10 *Let $-1/2 < p < 1/2$. The $L^2(\mathbb{R})$ -Fourier transform of the function*

$$f_{t,\pm}(u) = (t-u)_{\pm}^p - (-u)_{\pm}^p, \quad u \in \mathbb{R},$$

is given by

$$\widehat{f}_{t,\pm}(x) = \Gamma(p+1) \frac{e^{itx} - 1}{ix} (\mp ix)^{-p} = \Gamma(p+1) \frac{e^{itx} - 1}{ix} |x|^{-p} e^{\mp i \operatorname{sign}(x) p \pi / 2} \quad (2.6.4)$$

$$= \pm \Gamma(p+1) (e^{itx} - 1) |x|^{-p-1} e^{\mp i \operatorname{sign}(x) (p+1) \pi / 2}. \quad (2.6.5)$$

Proof We start with two observations. Firstly, the second equality in (2.6.4) results from $\mp i = e^{\mp i \pi / 2}$, so that

$$(\mp ix)^{-p} = |x|^{-p} (\mp i \operatorname{sign}(x))^{-p} = |x|^{-p} e^{\mp i \operatorname{sign}(x) p \pi / 2}.$$

Secondly, it is known (Gradshteyn and Ryzhik [421], pp. 436–437, Formulae 3.761.4 and 3.761.9) that

$$\int_0^\infty e^{\pm iv} v^{\alpha-1} dv = \Gamma(\alpha) e^{\pm i \alpha \pi / 2} \quad (2.6.6)$$

for $\alpha \in (0, 1)$, where the integral is understood as $\lim_{u \rightarrow \infty} \int_0^u e^{\pm iv} v^{\alpha-1} dv$ and the convergence can be verified by applying integration by parts. Indeed, note that

$$\int_1^u e^{\pm iv} v^{\alpha-1} dv = (\mp i) \int_1^u v^{\alpha-1} d e^{\pm iv} = (\mp i) \left(u^{\alpha-1} e^{\pm iu} - e^{\pm i} - (\alpha-1) \int_1^u e^{\pm iv} v^{\alpha-2} dv \right)$$

and the last expression has a limit as $u \rightarrow \infty$ since $\alpha < 1$ and $\int_1^\infty v^{\alpha-2} dv < \infty$.

Now consider, for example,

$$f_t(u) := f_{t,+}(u) = (t-u)_+^p - (-u)_+^p. \quad (2.6.7)$$

The cases $p < 0$ and $p > 0$ have to be considered separately. (The case $p = 0$ is trivial since it is interpreted by convention as $f_t(u) = 1_{[0,t)}(u)$.)

When $p < 0$, define a sequence of functions $f_{t,n}(u) = (t-u)_+^p 1_{\{t-u < n\}} - (-u)_+^p 1_{\{-u < n\}}$. Since $f_{t,n} \rightarrow f_t$ in $L^2(\mathbb{R})$, it is enough to show that $\widehat{f}_{t,n} \rightarrow \widehat{f}_t$ in $L^2(\mathbb{R})$, where $\widehat{f}_t = \widehat{f}_{t,+}$ is given in (2.6.4)–(2.6.5). Observe that

$$\begin{aligned} \widehat{f}_{t,n}(x) &= \int_{\mathbb{R}} e^{ixu} (t-u)_+^p 1_{\{t-u < n\}} du - \int_{\mathbb{R}} e^{ixu} (-u)_+^p 1_{\{-u < n\}} du \\ &= (e^{itx} - 1) \int_{\mathbb{R}} e^{-ixs} s_+^p 1_{\{s < n\}} ds = (e^{itx} - 1) \int_{\mathbb{R}} e^{-i \operatorname{sign}(x)|x|s} s_+^p 1_{\{s < n\}} ds \\ &= (e^{itx} - 1) |x|^{-p-1} \int_{\mathbb{R}} e^{-i \operatorname{sign}(x)v} v_+^p 1_{\{v < n|x|\}} dv = (e^{itx} - 1) |x|^{-p-1} \int_0^{n|x|} e^{-i \operatorname{sign}(x)v} v^p dv. \end{aligned}$$

On the other hand, using the expression for $\widehat{f}_t(x)$ in (2.6.5) and the relation (2.6.6) with $\alpha = p + 1$, we have

$$\widehat{f}_t(x) = \widehat{f}_{t,+}(x) = (e^{itx} - 1)|x|^{-p-1} \int_0^\infty e^{-i\operatorname{sign}(x)v} v^p dv,$$

so that

$$\widehat{f}_t(x) - \widehat{f}_{t,n}(x) = (e^{itx} - 1)|x|^{-p-1} \int_{n|x|}^\infty e^{-i\operatorname{sign}(x)v} v^p dv.$$

By integration by parts, $|\int_a^\infty e^{\pm iv} v^p dv| = |a^p e^{\pm iv} + p \int_a^\infty e^{\pm iv} v^{p-1} dv| \leq a^p + |p| \int_a^\infty v^{p-1} dv = 2a^p$. Hence,

$$|\widehat{f}_t(x) - \widehat{f}_{t,n}(x)| \leq C |e^{itx} - 1| |x|^{-1} n^p \quad (2.6.8)$$

and it follows that $\widehat{f}_{t,n} \rightarrow \widehat{f}_t$ in $L^2(\mathbb{R})$ as $n \rightarrow \infty$ since $p < 0$.

When $p > 0$, rewrite (2.6.7) as $f_t(u) = p \int_{\mathbb{R}} 1_{[0,t)}(v)(v-u)_+^{p-1} dv$ and consider

$$f_{t,n}(u) = p \int_{\mathbb{R}} 1_{[0,t)}(v)(v-u)_+^{p-1} 1_{\{v-u < n\}} dv = p(g_t * h_n)(u),$$

where $g_t(v) = 1_{[0,t)}(v)$, $h_n(v) = (-v)_+^{p-1} 1_{\{-v < n\}}$ and $*$ denotes the convolution (see Appendix A.1.2). As above, it is enough to show that $\widehat{f}_{t,n} \rightarrow \widehat{f}_t$ in $L^2(\mathbb{R})$. Observe that $\widehat{g}_t(x) = (e^{itx} - 1)/ix$ and

$$\widehat{h}_n(x) = \int_{\mathbb{R}} e^{ixv} (-v)_+^{p-1} 1_{\{-v < n\}} dv = \int_0^n e^{-ixv} v^{p-1} dv = |x|^{-p} \int_0^{n|x|} e^{-i\operatorname{sign}(x)u} u^{p-1} du$$

so that

$$\widehat{f}_{t,n}(x) = p \frac{e^{itx} - 1}{ix} |x|^{-p} \int_0^{n|x|} e^{-i\operatorname{sign}(x)u} u^{p-1} du.$$

By using the second expression for $\widehat{f}_t(x)$ in (2.6.4) together with (2.6.6),

$$\widehat{f}_t(x) - \widehat{f}_{t,n}(x) = p \frac{e^{itx} - 1}{ix} |x|^{-p} \int_{n|x|}^\infty e^{-i\operatorname{sign}(x)u} u^{p-1} du.$$

It follows that $\widehat{f}_{t,n} \rightarrow \widehat{f}_t$ in $L^2(\mathbb{R})$ as $n \rightarrow \infty$, as in the case $p < 0$ above (by replacing p by $p - 1 < 0$). $\square$

The following result provides an integral representation of FBM in the spectral domain. Hermitian Gaussian random measures are defined in Appendix B.1.

Proposition 2.6.11 For $H \in (0, 1)$, standard FBM admits the following integral representation:

$$\{B_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \frac{1}{c_2(H)} \int_{\mathbb{R}} \frac{e^{itx} - 1}{ix} |x|^{1/2-H} \widehat{B}(dx) \right\}_{t \in \mathbb{R}}, \quad (2.6.9)$$

where $c_2(H)^2 = \int_{\mathbb{R}} |e^{ix} - 1|^2 |x|^{-2H-1} dx$ is a normalizing constant and $\widehat{B}(dx)$ is an Hermitian Gaussian random measure with the control measure $\mathbb{E}|\widehat{B}(dx)|^2 = dx$.

Proof We include two proofs. The first is based on Proposition 2.6.10. By Remark 2.6.8 with $a = b = 1$, the kernel $f_{t,+} + f_{t,-}$ provides a representation for FBM. By Proposition 2.6.10, its Fourier transform is

$$\widehat{f}_{t,+}(x) + \widehat{f}_{t,-}(x) = 2\Gamma(p+1) \cos(p\pi/2) \frac{e^{itx} - 1}{ix} |x|^{-p},$$

since $i^p + i^{-p} = e^{ip\pi/2} + e^{-ip\pi/2} = 2\cos(p\pi/2)$. By setting $p = H - 1/2$, one obtains the representation (2.6.9) of FBM in the spectral domain (see (B.1.14) in Appendix B.1).

The second proof is based on Corollary 2.6.3. Denote the process on the right-hand side of (2.6.9) by $Y_H(t)$, without the normalizing constant. It is enough to check that Y_H satisfies conditions (i)–(iii) of Corollary 2.6.3. Condition (i) holds as long as Y_H is well-defined and real-valued. Y_H is real-valued since the deterministic integrand $g_t(x) = (e^{itx} - 1)|x|^{1/2-H}/(ix)$ satisfies $\overline{g_t(x)} = g_t(-x)$. It is well-defined since $g_t \in L^2(\mathbb{R})$.

Y_H satisfies (ii) of Corollary 2.6.3 since

$$\begin{aligned} \mathbb{E}(Y_H(t))^2 &= \int_{\mathbb{R}} \left| \frac{e^{itx} - 1}{ix} \right|^2 |x|^{1-2H} dx \\ &= \int_{\mathbb{R}} |e^{i|t|x} - 1|^2 |x|^{-1-2H} dx = |t|^{2H} \int_{\mathbb{R}} |e^{iy} - 1|^2 |y|^{-1-2H} dy, \end{aligned}$$

where we made the changes of variables $y = |t|x$. Similarly, Y_H satisfies (iii) of Corollary 2.6.3 since

$$\begin{aligned} \mathbb{E}(Y_H(t) - Y_H(s))^2 &= \int_{\mathbb{R}} \left| \frac{e^{itx} - e^{isx}}{ix} \right|^2 |x|^{1-2H} dx \\ &= \int_{\mathbb{R}} \left| \frac{e^{i(t-s)x} - 1}{ix} \right|^2 |x|^{1-2H} |e^{isx}|^2 dx = \mathbb{E}(Y_H(t-s))^2. \quad \square \end{aligned}$$

Note that the normalizing constant $c_2(H)^2$ in the proposition is there to guarantee that $\mathbb{E}B_H(1)^2 = 1$ and consequently

$$c_2(H)^2 = \int_{\mathbb{R}} \frac{|e^{ix} - 1|^2}{x^2} |x|^{1-2H} dx = \int_{\mathbb{R}} |e^{ix} - 1|^2 |x|^{-2H-1} dx.$$

It can be expressed as

$$\begin{aligned} c_2(H)^2 &= 4 \int_0^\infty (1 - \cos x) x^{-2H-1} dx = \frac{2}{H} \int_0^\infty \sin(x) x^{-2H} dx = \frac{2}{H} \Gamma(1-2H) \sin\left(\frac{(1-2H)\pi}{2}\right) \\ &= \frac{2}{H} \Gamma(1-2H) \cos(H\pi) = \frac{\pi}{H\Gamma(2H) \sin(H\pi)} = \frac{2\pi}{\Gamma(2H+1) \sin(H\pi)} \end{aligned} \quad (2.6.10)$$

by using (2.6.6) and, for the equality before last, (2.2.14) and $\sin(2x) = 2\sin(x)\cos(x)$.

Note also that the normalizing constants $c_1(H)$ and $c_2(H)$ can be related as follows. Let $f_t(u) = (t-u)_+^{H-1/2} - (-u)_+^{H-1/2}$ and $g_t(x) = (e^{itx} - 1)|x|^{1/2-H}/(ix)$ be the kernel functions in (2.6.1) and (2.6.9). Observe that by Proposition 2.6.10 with $p = H - 1/2$, $\widehat{f}_1(x) = \Gamma(H+1/2)g_1(x)e^{-i\text{sign}(x)(H-1/2)\pi/2}$. Then, by using Parseval's identity $\|f\|_{L^2(\mathbb{R})}^2 = \|\widehat{f}\|_{L^2(\mathbb{R})}^2/2\pi$ in (A.1.11) of Appendix A.1.2, we have

$$c_1(H)^2 = \|f_1\|_{L^2(\mathbb{R})}^2 = \frac{\|\widehat{f_1}\|_{L^2(\mathbb{R})}^2}{2\pi} = \frac{\Gamma(H+1/2)^2}{2\pi} \|g_1\|_{L^2(\mathbb{R})}^2 = \frac{\Gamma(H+1/2)^2}{2\pi} c_2(H)^2. \quad (2.6.11)$$

In particular,

$$c_1(H)^2 = \frac{\Gamma(H+1/2)^2}{\Gamma(2H+1)\sin(H\pi)} = \frac{B(H+1/2, H+1/2)}{\sin(H\pi)} = \frac{(H-1/2)B(H-1/2, 2-2H)}{2H}, \quad (2.6.12)$$

which is not obvious from the definition of the normalizing constant in the time domain. The first equality follows from (2.6.11) and (2.6.10), and the second from the definition (2.2.12) of the beta function. The last equality in (2.6.12) follows from

$$\begin{aligned} \frac{\Gamma(H+1/2)\Gamma(H+1/2)}{\Gamma(2H+1)\sin(H\pi)} &= \frac{(H-1/2)^2\Gamma(H-1/2)\Gamma(H-1/2)}{2H\Gamma(2H)\sin(H\pi)} \\ &= \frac{(H-1/2)^2\Gamma(H-1/2)\Gamma(H-1/2)\Gamma(1-2H)}{2H\Gamma(2H)\Gamma(1-2H)\sin(H\pi)} \\ &= -\frac{(H-1/2)\Gamma(H-1/2)\Gamma(2-2H)\Gamma(H-1/2)\sin(2H\pi)}{4H\pi\sin(H\pi)} \\ &= -\frac{(H-1/2)B(H-1/2, 2-2H)\Gamma(H-1/2)\Gamma(1-(H-1/2))\cos(H\pi)}{2H\pi} \\ &= -\frac{(H-1/2)B(H-1/2, 2-2H)\cos(H\pi)}{2H\sin((H-1/2)\pi)} = \frac{(H-1/2)B(H-1/2, 2-2H)}{2H}. \end{aligned}$$

Relation (2.2.14) was used for the third and fifth equalities, and the identity $\sin(2x) = 2\sin(x)\cos(x)$ was used for the fourth equality.

Figure 2.3 provides several realizations of standard FBM B_H on the interval $[0, 1]$ (in fact, on the dyadic grid $k/2^J$, $k = 0, 1, \dots, 2^J$, with $J = 10$) which are obtained by using the circulant matrix embedding method discussed in Section 2.11.2 for generating the stationary increments of FBM. As H becomes larger, the sample paths of FBM become smoother. In fact, as noted in Section 7.1.1 of Chapter 7, the sample paths of FBM are H' -Hölder continuous for any $H' < H$ and hence indeed are smoother for larger H .

2.6.2 Bifractional Brownian Motion

Bifractional Brownian motion provides an interesting and useful extension of fractional Brownian motion.

Definition 2.6.12 *Bifractional Brownian motion* (biFBM) $B_{H,K} = \{B_{H,K}(t)\}_{t \geq 0}$, $H \in (0, 1]$, $K \in (0, 1]$, is a zero mean Gaussian process with covariance function

$$\mathbb{E}B_{H,K}(t)B_{H,K}(s) = \frac{1}{2^K} \left((t^{2H} + s^{2H})^K - |t-s|^{2HK} \right), \quad s, t \geq 0. \quad (2.6.13)$$

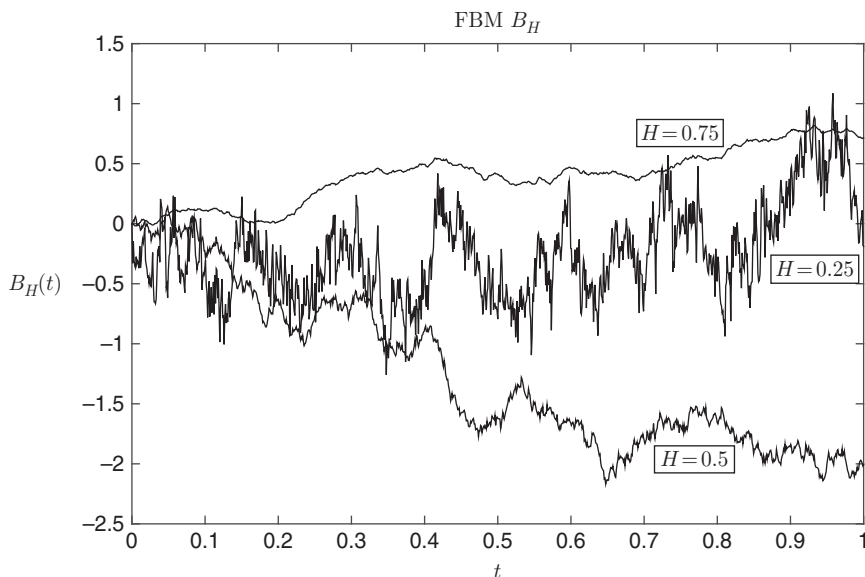


Figure 2.3 Realizations of FBM B_H with $H = 0.25$, $H = 0.5$ (Brownian motion) and $H = 0.75$.

In particular, $\mathbb{E}(B_{H,K}(t))^2 = t^{2HK}$. Note that biFBM is the usual FBM when $K = 1$. When $H = 1/2$, the covariance function (2.6.13) becomes

$$\mathbb{E}B_{1/2,K}(t)B_{1/2,K}(s) = \frac{1}{2^K} \left((t+s)^K - |t-s|^K \right), \quad s, t \geq 0, \quad (2.6.14)$$

and can be checked to be that of $B_{K/2}(t) - B_{K/2}(-t)$, $t \geq 0$, where $B_{K/2}$ is FBM with index $K/2$ and a suitable variance. See Section 3.12 for a physical model involving bifractional Brownian motion.

The fact that the right-hand side of (2.6.13) defines a valid covariance function is proved in Houdré and Villa [489], Proposition 2.1. It can also be shown (Bardina and Es-Sebaiy [100]) that the right-hand side of (2.6.13) is a valid covariance function when $H \in (0, 1)$, $HK \in (0, 1)$ and $K \in (1, 2)$.

The following result provides some basic properties of biFBM.

Proposition 2.6.13 *Let $B_{H,K}$, $H \in (0, 1]$, $K \in (0, 1]$, be biFBM. Then:*

- (i) *It is (HK) -self-similar.*
- (ii) *It has stationary increments if and only if $K = 1$; that is, it is the usual FBM.*
- (iii) *For all $s, t \geq 0$,*

$$2^{-K}|t-s|^{2HK} \leq \mathbb{E}(B_{H,K}(t) - B_{H,K}(s))^2 \leq 2^{1-K}|t-s|^{2HK}. \quad (2.6.15)$$

Proof Part (i) follows from the fact that, for any $c > 0$,

$$\mathbb{E}B_{H,K}(ct)B_{H,K}(cs) = c^{2HK}\mathbb{E}B_{H,K}(t)B_{H,K}(s).$$

Part (ii) follows from the relation

$$\mathbb{E}(B_{H,K}(t) - B_{H,K}(s))^2 = \frac{2}{2^K} |t-s|^{2HK} + \left(t^{2HK} + s^{2HK} - \frac{2}{2^K} (t^{2H} + s^{2H})^K \right), \quad (2.6.16)$$

which depends on $t-s$ only if and only if $K=1$ (see also (2.5.3)). By using (2.6.16), the left-hand side of (2.6.15) in part (iii), for example, is equivalent to

$$|t-s|^{2HK} + 2^K t^{2HK} + 2^K s^{2HK} - 2(t^{2H} + s^{2H})^K \geq 0.$$

This holds for $s=0$ and, taking $u=t/s$ when $s \neq 0$, this is equivalent to

$$f(u) := |u-1|^{2HK} + 2^K u^{2HK} + 2^K - 2(u^{2H} + 1)^K \geq 0.$$

Proving the latter inequality is a calculus exercise (see Houdré and Villa [489], Proposition 3.1, for details). $\square$

Further insight into biFBM can be gained from the following decomposition result due to Lei and Nualart [603]. For $K \in (0, 1)$, set

$$X_K(t) = \int_0^\infty (1 - e^{-ut}) u^{-\frac{1+K}{2}} B(du), \quad t \geq 0, \quad (2.6.17)$$

where $B(du)$ is a Gaussian random measure on $(0, \infty)$ with the control measure du . It can be checked that the covariance function of X_K is given by

$$\mathbb{E}X_K(t)X_K(s) = \frac{\Gamma(1-K)}{K} \left(t^K + s^K - (t+s)^K \right), \quad s, t \geq 0. \quad (2.6.18)$$

Proposition 2.6.14 *Let $B_{H,K}$, $H \in (0, 1]$, $K \in (0, 1)$, be biFBM and let the process X_K be defined by (2.6.17) and independent of $B_{H,K}$. Then,*

$$\left\{ 2^{\frac{K-1}{2}} B_{H,K}(t) + \sqrt{\frac{K}{2\Gamma(1-K)}} X_K(t^{2H}) \right\}_{t \geq 0} \stackrel{d}{=} \{B_{HK}(t)\}_{t \geq 0}, \quad (2.6.19)$$

where B_{HK} is a standard FBM with index HK .

Proof Write the covariance function (2.6.13) of biFBM as

$$\mathbb{E}B_{H,K}(t)B_{H,K}(s) = \frac{1}{2^K} \left((t^{2H} + s^{2H})^K - t^{2HK} - s^{2HK} \right) + \frac{1}{2^K} \left(t^{2HK} + s^{2HK} - |t-s|^{2HK} \right).$$

Then,

$$\begin{aligned} \mathbb{E}B_{HK}(t)B_{HK}(s) &= \frac{1}{2} \left(t^{2HK} + s^{2HK} - |t-s|^{2HK} \right) \\ &= 2^{K-1} \mathbb{E}B_{H,K}(t)B_{H,K}(s) + \frac{1}{2} \left((t^{2H})^K + (s^{2H})^K - (t^{2H} + s^{2H})^K \right) \\ &= 2^{K-1} \mathbb{E}B_{H,K}(t)B_{H,K}(s) + \frac{K}{2\Gamma(1-K)} \mathbb{E}X_K(t^{2H})X_K(s^{2H}). \end{aligned}$$

This yields the decomposition (2.6.19). $\square$

The process X_K can be shown to have a version with paths that are infinitely differentiable (Lei and Nualart [603], Theorem 2). This should not be surprising since informally differentiating under the integral sign in (2.6.17) leads to a well-defined process. Then, Proposition 2.6.14 shows that biFBM with indices H and K can be thought as FBM with index HK up to an infinitely differentiable process with a fractional time change.

2.6.3 The Rosenblatt Process

We shall give next an example of a non-Gaussian SSSI process with finite variance. We shall use the following proposition. The $L^2(\mathbb{R}^2)$ -Fourier transform is discussed in Appendix A.1.2.

Proposition 2.6.15 *Let $1/4 < p < 1/2$. Then, the function*

$$f_t(u_1, u_2) = \int_0^t (s - u_1)_+^{p-1} (s - u_2)_+^{p-1} ds$$

is in $L^2(\mathbb{R}^2)$ with

$$\|f_t\|_{L^2(\mathbb{R}^2)}^2 = \frac{B(p, 1-2p)^2}{(4p-1)2p} |t|^{4p}, \quad (2.6.20)$$

and its $L^2(\mathbb{R}^2)$ -Fourier transform is

$$\widehat{f}_t(x_1, x_2) = \Gamma(p)^2 \frac{e^{it(x_1+x_2)} - 1}{i(x_1+x_2)} |x_1|^{-p} |x_2|^{-p} e^{-i\operatorname{sign}(x_1)\frac{p\pi}{2}} e^{-i\operatorname{sign}(x_2)\frac{p\pi}{2}}. \quad (2.6.21)$$

Proof It is enough to prove the result for $t = 1$. Observe that, using the assumption $1/4 < p < 1/2$,

$$\begin{aligned} \|f_1\|_{L^2(\mathbb{R}^2)}^2 &= \int_{\mathbb{R}} du_1 \int_{\mathbb{R}} du_2 \left(\int_0^1 (s - u_1)_+^{p-1} (s - u_2)_+^{p-1} ds \right)^2 \\ &= \int_{\mathbb{R}} du_1 \int_{\mathbb{R}} du_2 \int_0^1 ds_1 \int_0^1 ds_2 (s_1 - u_1)_+^{p-1} (s_1 - u_2)_+^{p-1} (s_2 - u_1)_+^{p-1} (s_2 - u_2)_+^{p-1} \\ &= \int_0^1 ds_1 \int_0^1 ds_2 \left(\int_{\mathbb{R}} (s_1 - u)_+^{p-1} (s_2 - u)_+^{p-1} du \right)^2 \\ &= 2 \int_0^1 ds_1 \int_{s_1}^1 ds_2 \left(\int_0^\infty z^{p-1} (s_2 - s_1 + z)^{p-1} dz \right)^2 \\ &= 2 \left(\int_0^\infty v^{p-1} (1+v)^{p-1} dv \right)^2 \int_0^1 ds_1 \int_{s_1}^1 ds_2 (s_2 - s_1)^{4p-2} \\ &= 2C_p^2 \int_0^1 ds_1 \int_{s_1}^1 ds_2 (s_2 - s_1)^{4p-2} = \frac{C_p^2}{(4p-1)2p}, \end{aligned}$$

since $4p-1 > 0$. Here $C_p = \int_0^\infty v^{p-1} (1+v)^{p-1} dv = B(p, 1-2p)$ by (2.2.12) and the fact that $1-2p > 0$.

For the Fourier transform, let $f_{1,n}(u_1, u_2) = \int_0^1 (s - u_1)_+^{p-1} 1_{\{s-u_1 < n\}} (s - u_2)_+^{p-1} 1_{\{s-u_2 < n\}} ds$. It converges to $f_1(u_1, u_2)$ in $L^2(\mathbb{R}^2)$ by arguing as above and using the dominated convergence theorem. The Fourier transform of $f_{1,n}(u_1, u_2)$ is

$$\begin{aligned}\widehat{f}_{1,n}(x_1, x_2) &= \int_{\mathbb{R}} \int_{\mathbb{R}} e^{i(x_1 u_1 + x_2 u_2)} f_{1,n}(u_1, u_2) du_1 du_2 \\ &= \int_0^1 e^{i(x_1 + x_2)s} ds \int_0^n e^{-ix_1 v_1} v_1^{p-1} dv_1 \int_0^n e^{-ix_2 v_2} v_2^{p-1} dv_2 \\ &= \frac{e^{i(x_1 + x_2)} - 1}{i(x_1 + x_2)} |x_1|^{-p} |x_2|^{-p} \int_0^{n|x_1|} e^{-i \operatorname{sign}(x_1) u_1} u_1^{p-1} du_1 \int_0^{n|x_2|} e^{-i \operatorname{sign}(x_2) u_2} u_2^{p-1} du_2.\end{aligned}$$

Finally, note that this converges to $\widehat{f}_1(x_1, x_2)$ for each $x_1 \neq 0, x_2 \neq 0$, as $n \rightarrow \infty$, by using the relations (2.6.8) and (2.6.6). $\square$

The following is a corollary of Proposition 2.6.15 (see Appendix B.2 for definitions of the multiple integrals with respect to Gaussian random measures).

Proposition 2.6.16 *Let $H_0 \in (3/4, 1)$ and $H = 2H_0 - 1 \in (1/2, 1)$. Let also $B(du)$ be a real-valued Gaussian random measure on $\mathbb{R}$ with the control measure $\mathbb{E}(B(du))^2 = du$, and $\widehat{B}(dx)$ be an Hermitian Gaussian random measure on $\mathbb{R}$ with the control measure $\mathbb{E}|\widehat{B}(dx)|^2 = dx$. Then, the process*

$$\{Z_H(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \frac{1}{d_1^0(H_0)} \int_{\mathbb{R}^2}' \left(\int_0^t (s - u_1)_+^{H_0-3/2} (s - u_2)_+^{H_0-3/2} ds \right) B(du_1) B(du_2) \right\}_{t \in \mathbb{R}} \quad (2.6.22)$$

$$\stackrel{d}{=} \left\{ \frac{1}{d_2^0(H_0)} \int_{\mathbb{R}^2}'' \frac{e^{it(x_1+x_2)} - 1}{i(x_1 + x_2)} |x_1|^{1/2-H_0} |x_2|^{1/2-H_0} \widehat{B}(dx_1) \widehat{B}(dx_2) \right\}_{t \in \mathbb{R}} \quad (2.6.23)$$

is an H -SSSI process, where the normalizing constants are related as

$$d_1^0(H_0) = \frac{\Gamma(H_0 - 1/2)^2}{2\pi} d_2^0(H_0). \quad (2.6.24)$$

Moreover, $\mathbb{E}Z_H(1)^2 = 1$ if

$$(d_1^0(H_0))^2 = \frac{2B(H_0 - 1/2, 2 - 2H_0)^2}{(2H_0 - 1)(4H_0 - 3)} \quad (2.6.25)$$

or, equivalently, if

$$(d_2^0(H_0))^2 = \frac{2(2\Gamma(2 - 2H_0) \cos(H_0\pi))^2}{(2H_0 - 1)(4H_0 - 3)}. \quad (2.6.26)$$

Proof By setting $p - 1 = H_0 - 3/2$, one gets $p = H_0 - 1/2 \in (1/4, 1/2)$ since $H_0 \in (3/4, 1)$. The representations (2.6.22) and (2.6.23) are well-defined and equal in distribution. The equality in distribution follows from (B.2.16) in Appendix B and Proposition 2.6.15, since the kernels in the right-hand sides of (2.6.22) and (2.6.23) are related through the

Fourier transform. Note also from Proposition 2.6.15 that the Fourier transform of the kernel in (2.6.22) produces the complex exponentials

$$e^{-i \operatorname{sign}(x_1) \frac{(H_0-1/2)\pi}{2}} e^{-i \operatorname{sign}(x_2) \frac{(H_0-1/2)\pi}{2}}$$

which do not appear in the right-hand side of (2.6.23). The exponentials can be omitted by the formula (B.2.15) for the change of variables found in Appendix B.

The process defined by the integral representation (2.6.22) can be shown to have stationary increments by using the fact that, for fixed $h \in \mathbb{R}$, the random measures $B(du)$ and $B(d(u+h))$ have the same properties. Similarly, the process is also H -SS by using the fact that, for fixed $c > 0$, the random measures $B(d(cu))$ and $c^{1/2}B(du)$ have the same properties. (Strictly speaking, it is necessary to use the change of variables formula (B.2.11) found in Appendix B.) The normalizations (2.6.25) and (2.6.26) follow from (2.6.20). $\square$

Definition 2.6.17 The process $\{Z_H(t)\}_{t \in \mathbb{R}}$, $H \in (1/2, 1)$, in (2.6.22)–(2.6.23) is called the *Rosenblatt process*.

Remark 2.6.18 It is important to distinguish between the self-similarity parameter of fractional Brownian motion, which appears as H_0 in Proposition 2.6.16, and the self-similarity parameter H of the Rosenblatt process. Observe that the exponent $H_0 - 3/2$ appears in the representation (2.6.2) of FBM. Similarly, the exponent $1/2 - H_0$ appears in the integral representation (2.6.9) of FBM. As indicated in the proposition, we have

$$H_0 = \frac{H+1}{2} \in \left(\frac{3}{4}, 1\right) \quad \text{and} \quad H = 2H_0 - 1 \in \left(\frac{1}{2}, 1\right). \quad (2.6.27)$$

It is sometimes desirable to express the formulas in Proposition 2.6.16 by using the self-similarity parameter H instead of H_0 . Thus,

$$\begin{aligned} \{Z_H(t)\}_{t \in \mathbb{R}} &\stackrel{d}{=} \left\{ \frac{1}{d_1(H)} \int_{\mathbb{R}^2} \left(\int_0^t (s-u_1)_+^{-\frac{2-H}{2}} (s-u_2)_+^{-\frac{2-H}{2}} ds \right) B(du_1) B(du_2) \right\}_{t \in \mathbb{R}} \\ &\stackrel{d}{=} \left\{ \frac{1}{d_2(H)} \int_{\mathbb{R}^2} \frac{e^{it(x_1+x_2)} - 1}{i(x_1+x_2)} |x_1|^{-\frac{H}{2}} |x_2|^{-\frac{H}{2}} \widehat{B}(dx_1) \widehat{B}(dx_2) \right\}_{t \in \mathbb{R}} \end{aligned} \quad (2.6.28)$$

is H -SSSI process, where the normalizing constants are related as

$$d_1(H) = \frac{\Gamma(H/2)^2}{2\pi} d_2(H). \quad (2.6.29)$$

Moreover, $\mathbb{E}Z_H(1)^2 = 1$ if

$$d_1(H)^2 = \frac{B(H/2, 1-H)^2}{(H/2)(2H-1)} \quad (2.6.30)$$

or, equivalently, if

$$d_2(H)^2 = \frac{(2\Gamma(1-H) \sin(H\pi/2))^2}{(H/2)(2H-1)}. \quad (2.6.31)$$

2.6.4 $S\alpha S$ Lévy Motion

This is an example of a SSSI process with infinite variance. The idea is simple. Brownian motion can be represented as $B(t) = \int_0^t B(du)$, namely as the integral of Gaussian noise. Let us replace the Gaussian noise $B(du)$ by a symmetric α -stable ($S\alpha S$) noise $M_\alpha(du)$ with $0 < \alpha < 2$ (see Appendix B.1 for a definition).

Definition 2.6.19 The $S\alpha S$ process

$$S_\alpha(t) = \int_0^t M_\alpha(du), \quad t \geq 0,$$

where $M_\alpha(du)$ is a $S\alpha S$ random measure with control measure du is called the $S\alpha S$ Lévy motion.

The $S\alpha S$ Lévy motion $S_\alpha(t)$ has independent increments because $M_\alpha(A)$ and $M_\alpha(B)$ are independent random variables if $A \cap B = \emptyset$. It is self-similar with

$$H = \frac{1}{\alpha} \in \left(\frac{1}{2}, \infty\right)$$

because for any $c > 0$, $M_\alpha(cdu) \stackrel{d}{=} c^{1/\alpha} M_\alpha(du)$. A more precise way of showing the self-similarity is by using the characteristic function. For any $t_1, \dots, t_J \geq 0, \theta_1, \dots, \theta_J \in \mathbb{R}$,

$$\begin{aligned} \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j S_\alpha(ct_j) \right\} &= \exp \left\{ - \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j 1_{[0, ct_j]}(u) \right|^\alpha du \right\} = \exp \left\{ - \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j 1_{[0, t_j]}(u) \right|^\alpha cdu \right\} \\ &= \exp \left\{ - \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j c^{1/\alpha} 1_{[0, t_j]}(u) \right|^\alpha du \right\} = \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j c^{1/\alpha} S_\alpha(t_j) \right\}. \end{aligned}$$

Remark 2.6.20 If $\alpha = 2$, the $S\alpha S$ Lévy motion becomes Brownian motion. For $t < 0$, one defines the $S\alpha S$ Lévy motion as $S_\alpha(t) = \int_t^0 M_\alpha(du)$, $t < 0$. One could also replace the $S\alpha S$ measure $M_\alpha(du)$ by a skewed one (see Appendix B.1).

2.6.5 Linear Fractional Stable Motion

Here is another example of SSSI process with infinite variance. The basic idea is to replace the Gaussian noise in the integral representations of FBM by a $S\alpha S$ noise (see Appendix B.1 for a definition).

Proposition 2.6.21 For $\alpha \in (0, 2)$, $H \in (0, 1)$, $H \neq 1/\alpha$ and $a, b \in \mathbb{R}$, the process

$$\{L_{\alpha, H}(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ \int_{\mathbb{R}} f_t(u) M_\alpha(du) \right\}_{t \in \mathbb{R}}, \quad (2.6.32)$$

where $M_\alpha(du)$ is a $S\alpha S$ random measure on $\mathbb{R}$ with the Lebesgue control measure du , and

$$f_t(u) = a((t-u)_+^{H-1/\alpha} - (-u)_+^{H-1/\alpha}) + b((t-u)_-^{H-1/\alpha} - (-u)_-^{H-1/\alpha}), \quad (2.6.33)$$

is $S\alpha S$ and H -SSSI.

Remark 2.6.22 When $H = 1/\alpha$, the exponents in (2.6.33) become $H - 1/\alpha = 0$ and the following convention is used,

$$a_{\pm}^0 = (a_{\pm})^0, \quad a^0 = \begin{cases} 0, & \text{if } a = 0, \\ 1, & \text{if } a \neq 0. \end{cases} \quad (2.6.34)$$

With this convention, for $t > 0$,

$$(t - u)_+^0 - (-u)_+^0 = 1_{[0,t)}(u), \quad (t - u)_-^0 - (-u)_-^0 = -1_{(0,t]}(u). \quad (2.6.35)$$

When $H = 1/\alpha$, the kernel function thus becomes

$$f_t(u) = (a - b)1_{[0,t)}(u), \quad u \neq 0, t.$$

Hence, when $a \neq b$ and $H = 1/\alpha$, the process $L_{\alpha,H}$ is the $S\alpha S$ Lévy motion.

Proof $L_{\alpha,H}$ is a $S\alpha S$ stochastic process as long as it is well-defined or $f_t \in L^\alpha(\mathbb{R})$. Around $u = t$, for example, $|f_t(u)|^\alpha$ behaves as (up to a constant) $(|t - u|^{H-1/\alpha})^\alpha = |t - u|^{H\alpha-1}$, which is integrable since $(H\alpha - 1) + 1 = H\alpha > 0$. Around $u = \infty$, $|f_t(u)|^\alpha$ behaves as (up to a constant) $(|u|^{H-1/\alpha-1})^\alpha = |u|^{H\alpha-1-\alpha}$, which is integrable since $(H\alpha - 1 - \alpha) + 1 = (H - 1)\alpha < 0$. $L_{\alpha,H}$ has stationary increments by using the property $M_\alpha(d(u + h)) \stackrel{d}{=} M_\alpha(du)$. H -SS follows from the property $M_\alpha(d(cu)) \stackrel{d}{=} c^{1/\alpha} M_\alpha(du)$. To understand H -SS in a more rigorous way, observe that

$$\begin{aligned} \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j L_{\alpha,H}(ct_j) \right\} &= \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j \int_{\mathbb{R}} f_{ct_j}(u) M_\alpha(du) \right\} \\ &= \exp \left\{ - \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j f_{ct_j}(u) \right|^\alpha du \right\} = \exp \left\{ - \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j c^H f_{t_j}(v) \right|^\alpha dv \right\} \\ &= \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j c^H L_{\alpha,H}(t_j) \right\}, \end{aligned}$$

since $f_{ct}(cv) = c^{H-1/\alpha} f_t(v)$. □

Definition 2.6.23 The process $L_{\alpha,H} = \{L_{\alpha,H}(t)\}_{t \in \mathbb{R}}$, $\alpha \in (0, 2)$, $H \in (0, 1)$, $H \neq 1/\alpha$, in (2.6.32) is called a *linear fractional stable motion* (LFSM).

We should also note that, in contrast to the Gaussian case of FBM, there are infinitely many different $S\alpha S$ H -SSSI processes (for fixed α and H). More details can be found in Pipiras and Taqqu [820].

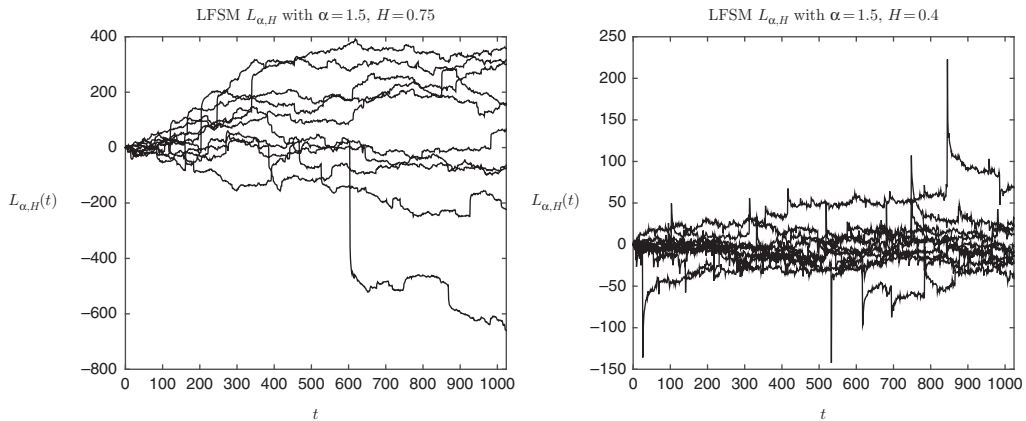


Figure 2.4 Realizations of LFSM $L_{\alpha,H}$. Left: $\alpha = 1.5$, $H = 0.75$. Right: $\alpha = 1.5$, $H = 0.4$.

Remark 2.6.24 One can replace the $S\alpha S$ measure $M_\alpha(du)$ by a skewed α -stable measure (see Appendix B.1). Note also that, since $L_{\alpha,H}(1)$ is a $S\alpha S$ random variable, its characteristic function is $\mathbb{E}e^{i\theta L_{\alpha,H}(1)} = e^{-\sigma^\alpha |\theta|^\alpha}$ for a scale parameter $\sigma > 0$. LFSM $L_{\alpha,H}$ is called *standard* when $\sigma = 1$.

Figure 2.4 depicts sample paths of LFSM when $\alpha = 1.5$, $H = 0.75$ and $\alpha = 1.5$, $H = 0.4$. They are obtained using an approximate method based on FFT developed in Stoev and Taqqu [926]. When $H - 1/\alpha > 0$ (the left plot in the figure), the sample paths of LFSM can be shown to be continuous (e.g., Samorodnitsky and Taqqu [883], Example 12.2.3).

2.6.6 Log-Fractional Stable Motion

We used the convention (2.6.34) to conclude that LFSM is the $S\alpha S$ Lévy motion when $H = 1/\alpha$. Another way to proceed is to let $H \rightarrow 1/\alpha$. This approach leads to another $S\alpha S$ SSSI process. More specifically, let

$$a = b = 1$$

in (2.6.33), and consider

$$\frac{1}{H - 1/\alpha} L_{\alpha,H}(t) = \int_{\mathbb{R}} \left(\frac{|t - u|^{H-1/\alpha} - 1}{H - 1/\alpha} - \frac{|u|^{H-1/\alpha} - 1}{H - 1/\alpha} \right) M_\alpha(du).$$

The fact that, for any $u \neq 0$,

$$\lim_{H \rightarrow 1/\alpha} \frac{|u|^{H-1/\alpha} - 1}{H - 1/\alpha} = \lim_{H \rightarrow 1/\alpha} \frac{e^{(H-1/\alpha) \log |u|} - 1}{H - 1/\alpha} = \log |u|,$$

suggests the following definition.

Definition 2.6.25 Let $1 < \alpha < 2$ and $M_\alpha(du)$ be a $S\alpha S$ random measure on $\mathbb{R}$ with the Lebesgue control measure du . Then,

$$\int_{\mathbb{R}} (\log |t - u| - \log |u|) M_\alpha(du), \quad t \in \mathbb{R}, \quad (2.6.36)$$

is called a *log-fractional stable motion* (log-FSM).

It is easy to show that log-FSM is well-defined, has stationary increments and is self-similar with $H = 1/\alpha$ (see Exercise 2.15, or Samorodnitsky and Taqqu [883], Proposition 7.3.6).

Remark 2.6.26 If we set $\alpha = 2$, then log-FSM becomes Gaussian and is H -SSSI with $H = 1/2$. The resulting process must therefore be Brownian motion. The expression (2.6.36) then provides an alternative representation for Brownian motion.

2.6.7 The Telecom Process

The Telecom process appears as a limit of stochastic processes used to model computer workloads (see Section 3.3). It is a δ -stable process⁹ with $1 < \delta < 2$. Its integral representation involves a δ -stable measure defined not on $\mathbb{R}$ but on $\mathbb{R} \times (0, \infty)$.

Definition 2.6.27 Let $1 < \gamma < \delta < 2$, $c > 0$ and

$$K_t(s, u) = (t - s)_+ \wedge u - (-s)_+ \wedge u, \quad t, s \in \mathbb{R}, \quad u > 0,$$

where $x \wedge y = \min\{x, y\}$. The *Telecom process* is

$$Z_{\gamma, \delta}(t) = c \int_{-\infty}^{\infty} \int_0^{\infty} K_t(s, u) u^{-(1+\gamma)/\delta} M_\delta(ds, du), \quad t \in \mathbb{R}, \quad (2.6.37)$$

where $M_\delta(ds, du)$ is a δ -stable, totally skewed to the right random measure on $\mathbb{R} \times (0, \infty)$ with the control measure¹⁰ $dsdu$.

Properties of the kernel $K_t(s, u)$ are given in Lemma 3.3.1 of Chapter 3. Observe that, for $t > 0$,

$$\begin{aligned} Z_{\gamma, \delta}(t) &= c \int_{-\infty}^0 \int_0^{\infty} (t \wedge (s + u)_+) u^{-(1+\gamma)/\delta} M_\delta(ds, du) \\ &\quad + c \int_0^t \int_0^{\infty} ((t - s) \wedge u) u^{-(1+\gamma)/\delta} M_\delta(ds, du). \end{aligned} \quad (2.6.38)$$

⁹ We use δ instead of α in this context, as is done in Section 3.3.

¹⁰ In Section 3.3, $M_\delta(ds, du)$ is defined with control measure $dsu^{-(1+\gamma)}du$, and therefore the integral representation of the Telecom process does not contain the factor $u^{-(1+\gamma)/\delta}$.

Proposition 2.6.28 *The Telecom process is well-defined, has stationary increments and is H -self-similar with parameter*

$$H = \frac{\delta + 1 - \gamma}{\delta} \in \left(\frac{1}{\delta}, 1\right).$$

Proof The integral (2.6.37) is well-defined when $K_t(s, u)u^{-(1+\gamma)/\delta} \in L^\delta(\mathbb{R} \times \mathbb{R}_0, ds du)$. This can be shown by considering the five regions of s and u in (3.3.11) below. Stationarity of the increments can be shown using the facts that $K_{t+h}(s, u) - K_h(s, u) = K_t(s - h, u)$ and that $M_\delta(d(s + h), du) \stackrel{d}{=} M_\delta(ds, du)$. Similarly, the H -self-similarity is a consequence of the facts that $K_{ct}(s, u)u^{-(1+\gamma)/\delta} = c^H K_t(s/c, u/c)(u/c)^{-(1+\gamma)/\delta} c^{-2/\delta}$ and that $M_\delta(d(cs), d(cu)) \stackrel{d}{=} c^{2/\delta} M_\delta(ds, du)$. $\square$

It can be shown (Pipiras and Taqqu [820]) that LFSM and the Telecom processes, with M_δ being $S\delta S$, have different finite-dimensional distributions (for the same H and $\delta = \alpha$).

Remark 2.6.29 Suppose we set $\delta = 2$ in Definition 2.6.27 of the Telecom process. Then, the stable random measure becomes Gaussian. We can then apply Corollary 2.6.3. We conclude that the resulting process $Z_{\gamma,2}(t)$ is FBM because it is Gaussian, has stationary increments and is self-similar with

$$H = \frac{3 - \gamma}{2} \in \left(\frac{1}{2}, 1\right).$$

Thus, (2.6.37) with $\delta = 2$ provides an alternative representation of FBM.

2.6.8 Linear Fractional Lévy Motion

FBM and linear fractional stable motion have similar linear integral representations (2.6.1) and (2.6.32) with respect to Gaussian and $S\alpha S$ random measures, respectively. We introduce here a process having this type of linear integral representation but with respect to a symmetric Lévy random measure. Symmetric Lévy random measures and integrals with respect to them are defined in Appendix B.1.5. The resulting process is not exactly self-similar but it has the same covariance structure as FBM.

Definition 2.6.30 A linear fractional Lévy motion is defined as the process

$$\{Z_{\rho,H}(t)\}_{t \in \mathbb{R}} \stackrel{d}{=} \left\{ a \int_{\mathbb{R}} \left((t-x)_+^{H-1/2} - (-x)_+^{H-1/2} \right) Z(dx) \right\}_{t \in \mathbb{R}}, \quad (2.6.39)$$

where $H \in (0, 1)$, $a \in \mathbb{R}$ is a constant and Z is a symmetric Lévy random measure on $\mathbb{R}$ with the Lebesgue control measure $m(dx) = dx$ and a symmetric Lévy measure ρ satisfying $\int_{\mathbb{R}} u^2 \rho(du) < \infty$.

By (B.1.45), a linear fractional Lévy motion has the same covariance structure as FBM; that is,

$$\mathbb{E} Z_{\rho,H}(t) Z_{\rho,H}(s) = \frac{a^2 \int_{\mathbb{R}} u^2 \rho(du)}{2} \left(|t|^{2H} + |s|^{2H} - |s-t|^{2H} \right), \quad s, t \in \mathbb{R}, \quad (2.6.40)$$

in addition to having zero mean $\mathbb{E}Z_{\rho,H}(t) = 0$. The process $Z_{\rho,H}$ is not expected to be exactly self-similar. Indeed, note from (B.1.43) that the characteristic function of $Z_{\rho,H}(t)$ is given by: for $\theta \in \mathbb{R}$,

$$\mathbb{E}e^{i\theta Z_{\rho,H}(t)} = \exp \left\{ - \int_{\mathbb{R}} \int_{\mathbb{R}} \left(1 - \cos \left(\theta u a((t-x)_+^{H-1/2} - (-x)_+^{H-1/2}) \right) \right) dx \rho(du) \right\}. \quad (2.6.41)$$

Since $\rho(du)$ does not scale – that is, it does not satisfy $\rho(d(cu)) = c^p \rho(du)$ for some $p \in \mathbb{R}$ – there is no reason a priori for $Z_{\rho,H}(t)$ and $t^H Z_{\rho,H}(1)$ to have the same characteristic function (or the same distribution).

Though $Z_{\rho,H}$ is not self-similar, it is *asymptotically self-similar at infinity* in the sense of (2.6.42) in the next proposition.

Proposition 2.6.31 *With the notation and assumptions above,¹¹*

$$\frac{Z_{\rho,H}(Rt)}{R^H} \xrightarrow{fdd} \sigma_H B_H(t), \quad t \in \mathbb{R}, \quad (2.6.42)$$

as $R \rightarrow \infty$, where $\sigma_H = |a|(\int_{\mathbb{R}} u^2 \rho(du))^{1/2} c_1(H)$ with $c_1(H)$ given in (2.6.1) and (2.6.12), and B_H is a standard FBM with index H .

Proof Let $f_t(x) = a((t-x)_+^{H-1/2} - (-x)_+^{H-1/2})$ be the deterministic kernel function of a linear fractional Lévy motion appearing in (2.6.39). The characteristic function of a linear combination $\sum_{j=1}^J \theta_j Z_{\rho,H}(Rt_j)/R^H$, $\theta_j, t_j \in \mathbb{R}$, is

$$\begin{aligned} \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j Z_{\rho,H}(Rt_j)/R^H \right\} &= \mathbb{E} \exp \left\{ i \int_{\mathbb{R}} \left(\sum_{j=1}^J \theta_j f_{Rt_j}(x)/R^H \right) Z(dx) \right\} \\ &= \exp \left\{ - \int_{\mathbb{R}} \int_{\mathbb{R}} \left(1 - \cos \left(u \sum_{j=1}^J \theta_j f_{Rt_j}(x)/R^H \right) \right) dx \rho(du) \right\} \\ &= \exp \left\{ - \int_{\mathbb{R}} \int_{\mathbb{R}} \left(1 - \cos \left(u \sum_{j=1}^J \theta_j f_{t_j}(y)/R^{1/2} \right) \right) R dy \rho(du) \right\} \\ &=: \exp \left\{ - \int_{\mathbb{R}} \int_{\mathbb{R}} g_R(u, y) dy \rho(du) \right\}, \end{aligned}$$

after the change of variables $x = Ry$. As $R \rightarrow \infty$, $g_R(u, y) \rightarrow \frac{u^2}{2} (\sum_{j=1}^J \theta_j f_{t_j}(y))^2$ and $|g_R(u, y)| \leq \frac{u^2}{2} (\sum_{j=1}^J \theta_j f_{t_j}(y))^2$. The dominated convergence theorem implies that the characteristic function converges to

$$\exp \left\{ - \frac{1}{2} \int_{\mathbb{R}} u^2 \rho(du) \int_{\mathbb{R}} \left(\sum_{j=1}^J \theta_j f_{t_j}(y) \right)^2 dy \right\},$$

which yields (2.6.42). □

¹¹ “fdd” stands for “finite-dimensional distributions”.

Under additional assumptions on ρ , $Z_{\rho,H}$ is also *locally asymptotically self-similar* in the sense of (2.6.43) in the following proposition.

Proposition 2.6.32 *With the notation and assumptions above, assume in addition that the symmetric Lévy measure ρ is given by*

$$\rho(du) = 1_{\{|u| \leq 1\}} \frac{du}{|u|^{1+\alpha}},$$

where $\alpha \in (0, 2)$, and that $\tilde{H}$ satisfying $\tilde{H} - 1/\alpha = H - 1/2$ is such that $\tilde{H} \in (0, 1)$. Then,

$$\frac{Z_{\rho,H}(\epsilon s)}{\epsilon^{\tilde{H}}} \xrightarrow{fdd} L_{\alpha,\tilde{H}}(s), \quad s \in \mathbb{R}, \quad (2.6.43)$$

where $L_{\alpha,\tilde{H}}$ is LFSM with H , a , b in (2.6.33) replaced by $\tilde{H}$, $aC(\alpha)^{1/\alpha}$ and 0, respectively, with $C(\alpha) = 2 \int_0^\infty (1 - \cos w) w^{-1-\alpha} dw$.

Remark 2.6.33 The same convergence (2.6.43) holds for $Z_{\rho,H}(\epsilon s)$ replaced by $Z_{\rho,H}(t + \epsilon s) - Z_{\rho,H}(t)$, for fixed t (Cohen and Istas [253], Proposition 4.4.5). We also note that the constant $C(\alpha)$ above can be computed as at the end of the proof of Proposition 2.2.24.

Proof As in the proof of Proposition 2.6.31, using its notation, the characteristic function of a linear combination $\sum_{j=1}^J \theta_j Z_{\rho,H}(\epsilon s_j)/\epsilon^{\tilde{H}}$, $\theta_j, s_j \in \mathbb{R}$, is

$$\begin{aligned} \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j Z_{\rho,H}(\epsilon s_j)/\epsilon^{\tilde{H}} \right\} &= \exp \left\{ - \int_{\mathbb{R}} \int_{\mathbb{R}} \left(1 - \cos \left(u \sum_{j=1}^J \theta_j f_{\epsilon s_j}(x)/\epsilon^{\tilde{H}} \right) \right) dx 1_{\{|u| \leq 1\}} \frac{du}{|u|^{1+\alpha}} \right\} \\ &= \exp \left\{ - \int_{\mathbb{R}} \int_{\mathbb{R}} \left(1 - \cos \left(v \sum_{j=1}^J \theta_j f_{s_j}(y) \right) \right) dy 1_{\{|v| \leq \epsilon^{-1/\alpha}\}} \frac{dv}{|v|^{1+\alpha}} \right\}, \end{aligned}$$

after the changes of variables $x = \epsilon y$ and $u = \epsilon^{1/\alpha} v$. As $\epsilon \rightarrow 0$, the characteristic function converges to

$$\begin{aligned} &\exp \left\{ - \int_{\mathbb{R}} \int_{\mathbb{R}} \left(1 - \cos \left(v \sum_{j=1}^J \theta_j f_{s_j}(y) \right) \right) dy \frac{dv}{|v|^{1+\alpha}} \right\} \\ &= \exp \left\{ - \int_{\mathbb{R}} (1 - \cos w) \frac{dw}{|w|^{1+\alpha}} \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j f_{s_j}(y) \right|^\alpha dy \right\}, \end{aligned}$$

after the change of variables $v | \sum_{j=1}^J \theta_j f_{s_j}(y) | = w$. This limit is the characteristic function of the respective linear combination of the limit process in (2.6.43). $\square$

2.7 The Lamperti Transformation

We provided above some examples of SSSI processes. Suppose that we now require only that the processes are self-similar. Then many examples can be constructed using stationary processes as a starting point as the next result shows.

Theorem 2.7.1 (i) If $X = \{X(t)\}_{t \geq 0}$ is H -SS, then $Y(t) = e^{-tH} X(e^t)$, $t \in \mathbb{R}$, is stationary.

(ii) Conversely, if $Y = \{Y(t)\}_{t \in \mathbb{R}}$ is stationary, then $X(t) = t^H Y(\log t)$, $t > 0$, is H -SS.

Proof Suppose (i), namely that X is self-similar. Observe that

$$\begin{aligned} (Y(t_1 + h), \dots, Y(t_n + h))' &= (e^{-(t_1+h)H} X(e^{t_1+h}), \dots, e^{-(t_n+h)H} X(e^{t_n+h}))' \\ &\stackrel{d}{=} (e^{-t_1H} X(e^{t_1}), \dots, e^{-t_nH} X(e^{t_n}))' = (Y(t_1), \dots, Y(t_n))', \end{aligned}$$

since $\{X(e^h s)\} \stackrel{d}{=} \{e^{hH} X(s)\}$ by self-similarity.

Turning to (ii), suppose that Y is stationary. Observe that

$$\begin{aligned} (X(cs_1), \dots, X(cs_n))' &= (c^H s_1^H Y(\log s_1 + \log c), \dots, c^H s_n^H Y(\log s_n + \log c))' \\ &\stackrel{d}{=} (c^H s_1^H Y(\log s_1), \dots, c^H s_n^H Y(\log s_n))' = (c^H X(s_1), \dots, c^H X(s_n))', \end{aligned}$$

since $\{Y(t + \log c)\} \stackrel{d}{=} \{Y(t)\}$ by stationarity. □

The theorem shows in particular that there are at least as many SS processes as there are stationary processes. The transformations in the theorem are called the *Lamperti transformation* after Lamperti [589].

In the following example, we illustrate the Lamperti transformation through a well-known example.

Example 2.7.2 (*Lamperti transformation*) Let the SS process X in Theorem 2.7.1 be a standard Brownian motion, having the covariance function

$$\mathbb{E}X(t)X(s) = \min\{t, s\}, \quad s, t \geq 0.$$

The process X is self-similar with the self-similarity parameter $H = 1/2$. The Lamperti transformation yields a stationary process $Y(t) = e^{-t/2} X(e^t)$, $t \in \mathbb{R}$. The process Y is a zero mean Gaussian process with the covariance function

$$\mathbb{E}Y(t)Y(s) = e^{-t/2} e^{-s/2} \min\{e^t, e^s\} = e^{-|t-s|/2}, \quad s, t \in \mathbb{R}. \quad (2.7.1)$$

It is thus the Ornstein–Uhlenbeck (OU) process in Examples 1.1.2 and 1.4.3.

Example 2.7.3 (*Lamperti transformation*) The covariance of any H -SSSI process X with $H \in (0, 1)$ and finite variance is given in (2.5.7). Therefore, the covariance of the stationary process $Y(t)$, $t \in \mathbb{R}$, in Theorem 2.7.1, resulting from the Lamperti transformation is

$$\begin{aligned} R(t) &= \mathbb{E}Y(t)Y(0) = e^{-tH} \mathbb{E}X(e^t)X(1) = \frac{\sigma^2}{2} e^{-Ht} (e^{2Ht} + 1 - |e^t - 1|^{2H}) \\ &= \frac{\sigma^2}{2} (e^{Ht} + e^{-Ht} - |e^{t/2} - e^{-t/2}|^{2H}) = \sigma^2 (\cosh(Ht) - 2^{2H-1} |\sinh(t/2)|^{2H}), \end{aligned}$$

where $\sigma^2 = \mathbb{E}X(1)^2$. As $t \rightarrow \infty$, we have

$$\begin{aligned} R(t) &= \frac{\sigma^2}{2} e^{Ht} \left(1 + e^{-2Ht} - (1 - e^{-t})^{2H} \right) \\ &= \frac{\sigma^2}{2} e^{Ht} \left(1 + e^{-2Ht} - 1 + 2He^{-t} + O(e^{-2t}) \right) \\ &= \frac{\sigma^2}{2} \left(e^{-Ht} + 2He^{-(1-H)t} + O(e^{-(2-H)t}) \right) \end{aligned}$$

and hence $\{Y(t)\}$ is short-range dependent. If $H = 1/2$, one recovers $\mathbb{E}Y(t)Y(0) = e^{-t/2}$ when $\sigma^2 = 1$.

2.8 Connections Between Long-Range Dependent Series and Self-Similar Processes

There is often a close relationship between LRD series and SSSI processes. SSSI processes lead to LRD series by taking differences and SSSI processes are obtained as normalized limits of partial sum processes of LRD series. We discuss these connections here in some detail, focusing on series and processes with finite variance.

First, consider going from SSSI processes to LRD series. Suppose $Y = \{Y(t)\}_{t \in \mathbb{R}}$ is an H -SSSI process with $0 < H < 1$, and consider a stationary series

$$X_n = Y(n) - Y(n-1), \quad n \in \mathbb{Z}. \quad (2.8.1)$$

The resulting series has the following basic properties.

Proposition 2.8.1 *The series X defined in (2.8.1) has zero mean, $\mathbb{E}X_n^2 = \mathbb{E}Y(1)^2$, autocovariances*

$$\gamma_X(k) = \frac{\mathbb{E}Y(1)^2}{2} \left(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H} \right) \sim \mathbb{E}Y(1)^2 H(2H-1)k^{2H-2}, \quad \text{as } k \rightarrow \infty, \quad (2.8.2)$$

and spectral density

$$f_X(\lambda) = \frac{\mathbb{E}Y(1)^2}{c_2(H)^2} |1 - e^{-i\lambda}|^2 \sum_{n=-\infty}^{\infty} |\lambda + 2\pi n|^{-1-2H} \sim \frac{\mathbb{E}Y(1)^2}{c_2(H)^2} \lambda^{1-2H}, \quad \text{as } \lambda \rightarrow 0, \quad (2.8.3)$$

where $c_2(H)$ is given in Proposition 2.6.11 and in (2.6.10).

Proof The equality in (2.8.2) follows immediately from Proposition 2.5.6, (e). For the asymptotic relation in (2.8.2), observe that, for large k and $f(x) = x^{2H}$,

$$\begin{aligned} |k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H} &= k^{2H-2} \frac{f(1 + \frac{1}{k}) - 2f(1) + f(1 - \frac{1}{k})}{\frac{1}{k^2}} \\ &\sim k^{2H-2} f''(x)|_{x=1} = 2H(2H-1)k^{2H-2}. \end{aligned}$$

For the spectral density, we suppose without loss of generality that $Y(t) = (\mathbb{E}Y(1)^2)^{1/2} B_H(t)$, where B_H is a standard FBM. We may do so because the spectral density depends only on the second-order properties of the process which are the same as those of standard FBM. By using Proposition 2.6.11, we can write

$$X_n = \frac{(\mathbb{E}Y(1)^2)^{1/2}}{c_2(H)} \int_{\mathbb{R}} e^{inx} \frac{(1 - e^{-ix})}{ix} |x|^{\frac{1}{2}-H} \widehat{B}(dx),$$

where $\widehat{B}(dx)$ satisfies $\mathbb{E}|\widehat{B}(dx)|^2 = dx$. Then,

$$\begin{aligned} \gamma_X(k) &= \mathbb{E}X_k X_0 = \frac{\mathbb{E}Y(1)^2}{c_2(H)^2} \int_{\mathbb{R}} e^{ikx} |1 - e^{-ix}|^2 |x|^{-1-2H} dx \\ &= \frac{\mathbb{E}Y(1)^2}{c_2(H)^2} \sum_{n=-\infty}^{\infty} \int_{-\pi+2\pi n}^{\pi+2\pi n} e^{ikx} |1 - e^{-ix}|^2 |x|^{-1-2H} dx \\ &= \frac{\mathbb{E}Y(1)^2}{c_2(H)^2} \int_{-\pi}^{\pi} e^{ik\lambda} |1 - e^{-i\lambda}|^2 \sum_{n=-\infty}^{\infty} |\lambda + 2\pi n|^{-1-2H} d\lambda = \int_{-\pi}^{\pi} e^{ik\lambda} f_X(\lambda) d\lambda, \end{aligned}$$

from which (2.8.3) follows. $\square$

Remark 2.8.2 Proposition 2.8.1 shows that if $1/2 < H < 1$, then the series X in (2.8.1) is LRD in the sense of conditions II and IV with

$$d = H - \frac{1}{2} \in \left(0, \frac{1}{2}\right). \quad (2.8.4)$$

Formula (2.8.4) is a standard and important relation between the LRD parameter d and the SS parameter H .

We should also note that the series (2.8.1) has a special name when Y is FBM.

Definition 2.8.3 If $Y = B_H$ is FBM, the series X in (2.8.1) is called *fractional Gaussian noise*¹² (FGN).

We now explain how SSSI processes Y arise from LRD series X . It is more instructive in this context to start with a general result concerning SS processes. This is the so-called fundamental limit theorem of Lamperti [589]. It uses the convergence to types theorem which we state next without proof. See, for example, Resnick [846], Theorem 8.7.1, p. 275, for a proof.

Theorem 2.8.4 (Convergence to types theorem.) Suppose that $\{X_n\}_{n \geq 0}$ is a collection of random variables, and U, V are two non-degenerate¹³ random variables. Let $a_n > 0$, $\alpha_n > 0$, $b_n \in \mathbb{R}$, $\beta_n \in \mathbb{R}$.

¹² Typically, the word “motion” is associated with a process having stationary increments, e.g., Brownian

motion, and the word “noise” is associated with a (possibly second-order) stationary process, e.g., white noise.

¹³ A random variable is non-degenerate (proper) if it is not concentrated at a fixed point.

(a) If, as $n \rightarrow \infty$,

$$\frac{X_n - b_n}{a_n} \xrightarrow{d} U, \quad \frac{X_n - \beta_n}{\alpha_n} \xrightarrow{d} V, \quad (2.8.5)$$

then there are $A > 0$, $B \in \mathbb{R}$ such that, as $n \rightarrow \infty$,

$$\frac{\alpha_n}{a_n} \rightarrow A > 0, \quad \frac{\beta_n - b_n}{a_n} \rightarrow B, \quad (2.8.6)$$

$$V \stackrel{d}{=} \frac{U - B}{A}. \quad (2.8.7)$$

(b) Conversely, if (2.8.6) holds, then either relation of (2.8.5) implies the other, and (2.8.7) holds.

The following is the fundamental limit theorem of Lamperti [589].

Theorem 2.8.5 (i) If a stochastic process Y satisfies

$$\frac{Y(\xi t) + g(\xi)}{f(\xi)} \xrightarrow{fdd} X(t), \quad t \geq 0, \text{ as } \xi \rightarrow \infty, \quad (2.8.8)$$

for some proper¹⁴ process $X(t)$, deterministic function¹⁵ $g(\xi) \in \mathbb{R}$ and some positive deterministic function $f(\xi) \rightarrow \infty$, as $\xi \rightarrow \infty$, then there is $H \geq 0$ such that

$$f(\xi) = \xi^H L(\xi), \quad g(\xi) = w(\xi) \xi^H L(\xi), \quad (2.8.9)$$

where $L(\xi)$ is a slowly varying function at infinity and $w(\xi)$ is a function which converges to a constant w as $\xi \rightarrow \infty$. Moreover, the process $\{X(t) - w\}_{t \geq 0}$ is H -SS.

(ii) If $\{X(t) - w\}_{t \geq 0}$ is H -SS, then there are functions $f(\xi)$, $g(\xi)$ and a process Y such that the above convergence (2.8.8) holds.

Proof The proof of (ii) is easy. With $Y(t) = X(t) - w$, $g(\xi) = w\xi^H$, $f(\xi) = \xi^H$, we have, by self-similarity,

$$\left\{ \frac{Y(\xi t) + g(\xi)}{f(\xi)} \right\} = \left\{ \frac{X(\xi t) - w}{\xi^H} + w \right\} \stackrel{d}{=} \{(X(t) - w) + w\} = \{X(t)\}.$$

Consider now the case (i). First, we will derive (2.8.9). If $t = 1$, then

$$\frac{Y(\xi) + g(\xi)}{f(\xi)} \xrightarrow{d} X(1).$$

Similarly, with ξ replaced by $a\xi$ and $t = 1/a$, we have

$$\frac{Y(\xi) + g(a\xi)}{f(a\xi)} \xrightarrow{d} X(1/a).$$

¹⁴ A process $\{X(t)\}$ is called *proper* if each random variable $X(t)$, $t > 0$, is proper (non-degenerate); that is, it is not concentrated at a fixed point.

¹⁵ Note that unlike $Y(\xi t)$, the function $g(\xi)$ does not depend on t .

By Theorem 2.8.4, for any $a \geq 0$,

$$\frac{f(a\xi)}{f(\xi)} \rightarrow F(a) \in (0, \infty), \quad \text{as } \xi \rightarrow \infty.$$

(Note that 0 and ∞ are excluded as possible values of $F(a)$ because the limits $X(1)$ and $X(1/a)$ are proper (non-degenerate).)

Observe next that

$$\frac{f(ab\xi)}{f(\xi)} \rightarrow F(ab), \quad \frac{f(ab\xi)}{f(\xi)} = \frac{f(ab\xi)}{f(b\xi)} \frac{f(b\xi)}{f(\xi)} \rightarrow F(a)F(b).$$

This implies $F(ab) = F(a)F(b)$, or $k(x+y) = k(x) + k(y)$, $x, y \in \mathbb{R}$, with $k(x) = \log F(e^x)$. The function k is (Lebesgue) measurable as the limit of measurable functions. Then, by Theorem 1.1.8 in Bingham et al. [157], $k(x) = cx$ for a constant c . This implies that $F(a) = a^H$. Hence,

$$\frac{f(a\xi)}{f(\xi)} \rightarrow a^H \in (0, \infty), \quad \text{as } \xi \rightarrow \infty,$$

or

$$\frac{(a\xi)^{-H} f(a\xi)}{\xi^{-H} f(\xi)} \rightarrow 1, \quad \text{as } \xi \rightarrow \infty,$$

that is, $\xi^{-H} f(\xi) = L(\xi)$ is a slowly varying function. Hence, $f(\xi) = \xi^H L(\xi)$ where $H \geq 0$. (The case $H < 0$ is excluded because $f(\xi) \rightarrow \infty$.) Moreover, since

$$\frac{Y(0) + g(\xi)}{f(\xi)} \xrightarrow{fdd} X(0),$$

there is a function $w(\xi)$ such that $g(\xi) = w(\xi)\xi^H L(\xi)$ with $w(\xi) \rightarrow w \in \mathbb{R}$. Note also that (2.8.8) now yields

$$\frac{Y(\xi t)}{f(\xi)} \xrightarrow{fdd} X(t) - w =: \tilde{X}(t).$$

Finally, to establish H -self-similarity, observe that

$$\frac{Y(a\xi t)}{\xi^H L(\xi)} \xrightarrow{fdd} \tilde{X}(at)$$

and

$$\frac{Y(a\xi t)}{\xi^H L(\xi)} = a^H \frac{L(a\xi)}{L(\xi)} \frac{Y(a\xi t)}{(a\xi)^H L(a\xi)} \xrightarrow{fdd} a^H \tilde{X}(t)$$

imply that $\tilde{X}(at)$ and $a^H \tilde{X}(t)$ have the same finite-dimensional distributions. $\square$

Theorem 2.8.5 shows that the only processes that arise as large scale limits of processes Y are self-similar. A common example of Y and also the choice leading to SS processes with stationary increments in the limit is given by

$$Y(t) = \sum_{k=1}^{[t]} X_k, \quad (2.8.10)$$

where $\{X_k\}_{k \geq 1}$ is a stationary series, as indicated in the following corollary.

Corollary 2.8.6 (i) Let $\{X_k\}_{k \geq 1}$ be a stationary series and $\{Y(t)\}_{t \geq 0}$ be the process defined by (2.8.10). If (i) of Theorem 2.8.5 holds, then the limit $\{X(t) - w\}_{t \geq 0}$ is H -SSSI.

(ii) If $\{X(t) - w\}_{t \geq 0}$ is H -SSSI and continuous in probability,¹⁶ then there are functions $f(\xi)$, $g(\xi)$ and a stationary series $\{X_k\}_{k \geq 1}$ such that the process Y defined by (2.8.10) satisfies (2.8.8).

Proof (i) If (i) of Theorem 2.8.5 holds, then the limit $\{X(t) - w\}_{t \geq 0}$ is H -SS. This limit has also stationary increments because $\{X_k\}$ is stationary. (ii) For the series $\{X_k\}$, take the increments of the SSSI process, $X_k = X(k) - X(k-1)$. Moreover, take $f(\xi) = \xi^H$ and $g(\xi) = \xi^H w - w$. Then, by using Proposition 2.5.6, (a), and H -self-similarity,

$$\begin{aligned} \left\{ \frac{Y(\xi t) + g(\xi)}{f(\xi)} \right\} &= \left\{ \frac{\sum_{k=1}^{[\xi t]} X_k + g(\xi)}{f(\xi)} \right\} = \left\{ \frac{X([\xi t]) - w + \xi^H w}{\xi^H} \right\} \\ &\stackrel{d}{=} \left\{ \frac{\xi^H (X([\xi t]/\xi) - w) + \xi^H w}{\xi^H} \right\} = \left\{ X\left(\frac{[\xi t]}{\xi}\right) \right\}. \end{aligned}$$

Since X is continuous in probability,

$$X\left(\frac{[\xi t]}{\xi}\right) \xrightarrow{p} X(t),$$

as $\xi \rightarrow \infty$. Since the convergence in probability implies that of distributions, we obtain (2.8.8). $\square$

To conclude this section, we shall illustrate Corollary 2.8.6 in three special cases: the first two involving FBM and the third involving the Rosenblatt process as the limiting SSSI process. We shall also replace the parameter $\xi \rightarrow \infty$ by an integer $N \rightarrow \infty$. Another special case involving LFSM is considered in Section 2.9.1.

The next result concerns FBM in the LRD regime.

Proposition 2.8.7 Let $\{X_n\}$ be a Gaussian LRD series in the sense (2.1.8) in condition V with parameter $0 < d < 1/2$. Then, as $N \rightarrow \infty$,

$$\frac{1}{L_5(N)^{1/2} N^{d+1/2}} \sum_{n=1}^{[Nt]} (X_n - \mathbb{E}X_n) \xrightarrow{fdd} B_H(t), \quad t \geq 0, \quad (2.8.11)$$

where B_H is a standard FBM with SS parameter

$$H = d + \frac{1}{2} \in \left(\frac{1}{2}, 1\right).$$

Proof Suppose without loss of generality that $\mathbb{E}X_n = 0$ and denote the normalized partial sum process in (2.8.11) by $X^N(t)$. Because of Gaussianity, it is enough to check that $\mathbb{E}X^N(t)X^N(s) \rightarrow \mathbb{E}B_H(t)B_H(s)$, for $t > s > 0$, as $N \rightarrow \infty$. This follows from

¹⁶ The process $\{Z(t)\}$ is continuous in probability if for any t , $Z(t+h) \xrightarrow{p} Z(t)$, as $h \rightarrow 0$

$$\begin{aligned}
\mathbb{E}X^N(t)X^N(s) &= \frac{1}{L_5(N)N^{2d+1}} \mathbb{E} \sum_{n=1}^{[Nt]} X_n \sum_{n=1}^{[Ns]} X_n \\
&= \frac{1}{2L_5(N)N^{2d+1}} \left\{ \mathbb{E} \left(\sum_{n=1}^{[Nt]} X_n \right)^2 + \mathbb{E} \left(\sum_{n=1}^{[Ns]} X_n \right)^2 - \mathbb{E} \left(\sum_{n=[Ns]+1}^{[Nt]} X_n \right)^2 \right\} \\
&= \frac{1}{2L_5(N)N^{2d+1}} \left\{ \mathbb{E} \left(\sum_{n=1}^{[Nt]} X_n \right)^2 + \mathbb{E} \left(\sum_{n=1}^{[Ns]} X_n \right)^2 - \mathbb{E} \left(\sum_{n=1}^{[Nt]-[Ns]} X_n \right)^2 \right\} \\
&\sim \frac{1}{2} \{ t^{2d+1} + s^{2d+1} - (t-s)^{2d+1} \} = \mathbb{E}B_{d+\frac{1}{2}}(t)B_{d+\frac{1}{2}}(s),
\end{aligned}$$

as $N \rightarrow \infty$, since, for example,

$$\frac{\mathbb{E}(\sum_{n=1}^{[Nt]} X_n)^2}{L_5(N)N^{2d+1}} = \frac{\mathbb{E}(\sum_{n=1}^{[Nt]} X_n)^2}{L_5([Nt])[Nt]^{2d+1}} \frac{L_5([Nt])}{L_5(N)} \left(\frac{[Nt]}{N} \right)^{2d+1} \sim t^{2d+1}$$

by using (2.1.8). $\square$

Proposition 2.8.7 continues to hold for $d = 0$ (and BM with $H = 1/2$ in the limit), as well as for antipersistent series (with $-1/2 < d < 0$ and $0 < H < 1/2$).

The assumption of Gaussianity in Proposition 2.8.7 is not necessary to obtain FBM in the limit. The following result extends Proposition 2.8.7 to the case of linear series.

Proposition 2.8.8 *Let $\{X_n\}_{n \in \mathbb{Z}}$ be a linear series*

$$X_n = \sum_{k=-\infty}^{\infty} c_k \epsilon_{n-k}, \quad n \in \mathbb{Z}, \quad (2.8.12)$$

with i.i.d. random variables ϵ_n , $n \in \mathbb{Z}$, with zero mean and unit variance. Suppose also that the series $\{X_n\}_{n \in \mathbb{Z}}$ is LRD in the sense (2.1.8) in condition V with parameter $0 < d < 1/2$. Then, as $N \rightarrow \infty$,

$$\frac{1}{L_5(N)^{1/2} N^{d+1/2}} \sum_{n=1}^{[Nt]} X_n \xrightarrow{fdd} B_H(t), \quad t \geq 0, \quad (2.8.13)$$

where B_H is a standard FBM with SS parameter

$$H = d + \frac{1}{2} \in \left(\frac{1}{2}, 1 \right).$$

If the sequence $\{c_k\}$ in (2.8.12) satisfies (2.1.4) in condition I, that is, $c_k = 0$, $k \leq -1$, and $c_k = L_1(k)k^{d-1}$, as $k \rightarrow \infty$, with $d \in (0, 1/2)$ and a slowly varying function $L_1(k)$, then the series $\{X_n\}_{n \in \mathbb{Z}}$ is LRD in the sense (2.1.8) (see Corollaries 2.2.10 and 2.2.7).

Proof Denote by $X^N(t)$ the normalized partial sum process on the left-hand side of (2.8.13). For $t_1, \dots, t_J \geq 0$ and $\theta_1, \dots, \theta_J \in \mathbb{R}$, consider

$$Z_N = \sum_{j=1}^J \theta_j X^N(t_j).$$

It is enough to show that

$$Z_N \xrightarrow{d} \mathcal{N}(0, \sigma^2), \quad (2.8.14)$$

with $\sigma^2 = \lim_{N \rightarrow \infty} \mathbb{E} Z_N^2$, and that

$$\mathbb{E} Z_N^2 \rightarrow \mathbb{E} \left(\sum_{j=1}^J \theta_j B_H(t_j) \right)^2. \quad (2.8.15)$$

The last condition can be replaced by

$$\mathbb{E} X^N(t_j) X^N(t_k) \rightarrow \mathbb{E} B_H(t_j) B_H(t_k). \quad (2.8.16)$$

The convergence (2.8.16) can be proved as in the proof of Proposition 2.8.7. We shall therefore prove only that the convergence (2.8.14) holds.

Let $\sigma_N = L_5(N)^{1/2} N^{d+1/2}$. Observe that

$$\begin{aligned} Z_N &= \frac{1}{\sigma_N} \sum_{j=1}^J \theta_j \sum_{n=1}^{[Nt_j]} X_n = \sum_{m=-\infty}^{\infty} \frac{1}{\sigma_N} \left(\sum_{j=1}^J \theta_j \sum_{n=1}^{[Nt_j]} c_{n-m} \right) \epsilon_m \\ &=: \sum_{m=-\infty}^{\infty} \left(\sum_{j=1}^J \theta_j a_{j,m}^{(N)} \right) \epsilon_m =: \sum_{m=-\infty}^{\infty} d_{N,m} \epsilon_m. \end{aligned}$$

For an integer $M = M_N \rightarrow \infty$ to be chosen below, consider independent random variables $\eta_{N,1}, \dots, \eta_{N,l_N}$ defined by

$$\eta_{N,1} = \sum_{|m| > M} d_{N,m} \epsilon_m,$$

$$\eta_{N,2} = d_{N,-M} \epsilon_{-M}, \quad \eta_{N,3} = d_{N,-M+1} \epsilon_{-M+1}, \dots, \eta_{N,l_N} = d_{N,M} \epsilon_M.$$

(We have $l_N = 2M + 1 = 2M_N + 1$.) Note that

$$Z_N = \eta_{N,1} + \dots + \eta_{N,l_N}.$$

To show (2.8.14), it is enough to show that the independent, non-identically distributed random variables $\eta_{N,1}, \dots, \eta_{N,l_N}$ satisfy the Lindeberg condition, namely, for any $\delta > 0$,

$$I_N := \sum_{k=1}^{l_N} \mathbb{E} |\eta_{N,k}|^2 1_{\{|\eta_{N,k}| > \delta\}} \rightarrow 0. \quad (2.8.17)$$

Note that

$$\sup_N \sum_{m=-\infty}^{\infty} d_{N,m}^2 = \sup_N \mathbb{E} Z_N^2 < \infty$$

by the convergence (2.8.15). Then, one can choose $M = M_N \rightarrow \infty$ such that

$$\mathbb{E} |\eta_{N,1}|^2 = \sum_{|m| > M} d_{N,m}^2 \rightarrow 0. \quad (2.8.18)$$

To deal with the remaining variables $\eta_{N,2}, \dots, \eta_{N,l_N}$, observe that

$$\begin{aligned}(a_{j,m}^{(N)})^2 &= \frac{1}{\sigma_N^2} (c_{1-m} + \dots + c_{[Nt_j]-m})^2 = \frac{1}{\sigma_N^2} (\sigma_N a_{j,m-1}^{(N)} + c_{[Nt_j]-m})^2 \\ &= (a_{j,m-1}^{(N)})^2 + \frac{2}{\sigma_N} a_{j,m-1}^{(N)} c_{[Nt_j]-m} + \frac{c_{[Nt_j]-m}^2}{\sigma_N^2}.\end{aligned}$$

Summing over $m = k-l, \dots, k$, we get after simplification, that

$$(a_{j,k}^{(N)})^2 = (a_{j,k-l-1}^{(N)})^2 + \frac{2}{\sigma_N} \sum_{m=k-l}^k a_{j,m-1}^{(N)} c_{[Nt_j]-m} + \frac{1}{\sigma_N^2} \sum_{m=k-l}^k c_{[Nt_j]-m}^2.$$

Hence,

$$(a_{j,k}^{(N)})^2 \leq (a_{j,k-l-1}^{(N)})^2 + \frac{2}{\sigma_N} \left(\sum_{m=-\infty}^{\infty} (a_{j,m}^{(N)})^2 \right)^{1/2} \left(\sum_{m=-\infty}^{\infty} c_m^2 \right)^{1/2} + \frac{1}{\sigma_N^2} \sum_{m=-\infty}^{\infty} c_m^2.$$

Letting $l \rightarrow -\infty$, and using the facts that $\sum_{m=-\infty}^{\infty} c_m^2 < \infty$ and $\sup_N \sum_{m=-\infty}^{\infty} (a_{j,m}^{(N)})^2 = \sup_N \mathbb{E} X^N(t_j)^2 < \infty$, we get that, for all k ,

$$|a_{j,k}^{(N)}| \leq \frac{C_j}{\sigma_N^{1/2}}.$$

This also yields that, for all k ,

$$|d_{N,k}| \leq \frac{C}{\sigma_N^{1/2}}. \quad (2.8.19)$$

Now, by using (2.8.19),

$$\begin{aligned}\sum_{k=2}^{l_N} \mathbb{E} |\eta_{N,k}|^2 1_{\{|\eta_{N,k}| > \delta\}} &\leq \left(\sum_{m=-\infty}^{\infty} d_{N,m}^2 \right) \mathbb{E} |\epsilon_0|^2 1_{\{|\epsilon_0| > \delta \sigma_N^{1/2}/C\}} \\ &= (\mathbb{E} Z_N^2) \mathbb{E} |\epsilon_0|^2 1_{\{|\epsilon_0| > \delta \sigma_N^{1/2}/C\}} \rightarrow 0,\end{aligned} \quad (2.8.20)$$

since $\sigma_N \rightarrow \infty$. Finally, the Lindeberg condition (2.8.17) follows from (2.8.18) and (2.8.20). $\square$

The third illustration involves the Rosenblatt process. The presented result is a special case of Theorem 5.3.1 in Chapter 5, whose rigorous proof is given in that chapter. We shall only explain here why the Rosenblatt process is a natural limiting process, without providing a rigorous proof.

Proposition 2.8.9 *Let $\{X_n\}_{n \in \mathbb{Z}}$ be a Gaussian series which is LRD in the sense (2.1.5) in condition II with parameter $1/4 < d < 1/2$. Then, as $N \rightarrow \infty$,*

$$\frac{1}{L_2(N)N^{2d}} \sum_{n=1}^{[Nt]} (X_n^2 - \mathbb{E} X_n^2) \xrightarrow{fdd} \beta_{2,H} Z_H(t), \quad t \geq 0, \quad (2.8.21)$$

where Z_H is a standard Rosenblatt process with SS parameter

$$H = 2d \in \left(\frac{1}{2}, 1\right)$$

and the constant $\beta_{2,H}$ is such that $\beta_{2,H}^2 = 2/(H(2H - 1))$.

The Rosenblatt process is a natural limiting process in (2.8.21). Indeed, this fact is easiest to see through the spectral domain. Assuming that the series $\{X_n\}$ is LRD in the sense (2.1.7) in condition IV expressed in the spectral domain. Let $L(\lambda) = L_4(\lambda)^{1/2}$ and $g(\lambda) = f_X(\lambda)^{1/2} = L(|\lambda|)|\lambda|^{-d}$, where $f_X(\lambda)$ is the spectral density function of the series $\{X_n\}$. Note that $L(\lambda)$ is a slowly varying function at zero. Now write the series $\{X_n\}$ in its spectral representation as $X_n = \int_{(-\pi, \pi]} e^{in\lambda} g(\lambda) \widehat{B}(d\lambda)$, where $\widehat{B}(d\lambda)$ is a Gaussian Hermitian random measure (see Section 1.3.4 and Appendix B.1). It is natural to expect that X_n^2 is related to a double integral with respect to the random measure $\widehat{B}(d\lambda)$. In fact, the exact relation is

$$X_n^2 - \mathbb{E}X_n^2 = \int_{(-\pi, \pi]^2}'' e^{in(\lambda_1 + \lambda_2)} g(\lambda_1) g(\lambda_2) \widehat{B}(d\lambda_1) \widehat{B}(d\lambda_2).$$

This is a special case of Proposition 4.1.2, (ii), in Chapter 4, and also of the relation (B.2.23) with $p = 1$ in Appendix B.2. Denoting the normalized partial sum process in (2.8.21) by $X^N(t)$, we deduce that

$$\begin{aligned} X^N(t) &\stackrel{d}{=} \frac{L_4(1/N)}{L_2(N)} \frac{1}{L(1/N)^2 N^{2d}} \int_{(-\pi, \pi]^2}'' \frac{e^{i[Nt](\lambda_1 + \lambda_2)} - 1}{e^{i(\lambda_1 + \lambda_2)} - 1} g(\lambda_1) g(\lambda_2) \widehat{B}(d\lambda_1) \widehat{B}(d\lambda_2) \\ &= \frac{L_4(1/N)}{L_2(N)} \frac{1}{L(1/N)^2 N^{2d}} \int_{\mathbb{R}^2}'' \frac{e^{i\frac{[Nt]}{N}(x_1 + x_2)} - 1}{e^{i\frac{x_1 + x_2}{N}} - 1} g\left(\frac{x_1}{N}\right) g\left(\frac{x_2}{N}\right) 1_{(-\pi N, \pi N]^2}(x_1, x_2) \\ &\quad \times \widehat{B}\left(d\frac{x_1}{N}\right) \widehat{B}\left(d\frac{x_2}{N}\right) \\ &\stackrel{d}{=} \frac{L_4(1/N)}{L_2(N)} \int_{\mathbb{R}^2}'' f_t^N(x_1, x_2) \widehat{B}(dx_1) \widehat{B}(dx_2), \end{aligned}$$

where $\stackrel{d}{=}$ denotes equality of the finite-dimensional distributions,

$$f_t^N(x_1, x_2) = \frac{e^{i\frac{[Nt]}{N}(x_1 + x_2)} - 1}{N(e^{i\frac{x_1 + x_2}{N}} - 1)} \frac{g(\frac{x_1}{N})}{L(1/N)N^d} \frac{g(\frac{x_2}{N})}{L(1/N)N^d} 1_{(-\pi N, \pi N]^2}(x_1, x_2).$$

Observe that

$$f_t^N(x_1, x_2) \rightarrow \frac{e^{it(x_1 + x_2)} - 1}{i(x_1 + x_2)} |x_1|^{-d} |x_2|^{-d}$$

and that, by Corollary 2.2.15,

$$\frac{L_4(1/N)}{L_2(N)} \rightarrow \frac{1}{\pi} \Gamma(2d) \cos(d\pi) = \frac{1}{\pi} \Gamma(H) \cos(H\pi/2).$$

This suggests that the limit of $X_N(t)$ in the sense of finite-dimensional distributions is the process

$$\frac{\Gamma(H) \cos(H\pi/2)}{\pi} \int_{\mathbb{R}^2}'' \frac{e^{it(x_1 + x_2)} - 1}{i(x_1 + x_2)} |x_1|^{-d} |x_2|^{-d} \widehat{B}(dx_1) \widehat{B}(dx_2) = \frac{\Gamma(H) \cos(H\pi/2) d_2(H)}{\pi} Z_H(t),$$

where the constant $d_2(H)$ appears in (2.6.31). The multiplicative constant in the last expression reduces to $\beta_{2,H}$ by using the relation (2.2.14).

Note that the limit process in Proposition 2.8.9 is non-Gaussian and has finite variance. The result is quite striking in view of the classical central limit theorem for SRD series, and again points to fundamental differences between SRD and LRD series.

2.9 Long- and Short-Range Dependent Series with Infinite Variance

While there is a relative consensus on what SRD and LRD mean when the time series has finite variance, the same cannot be said when dealing with series having infinite variance. This is partly because there is presently no widely accepted and well developed theory for modeling with these series. More importantly, the class of infinite variance time series cannot be described through simple measures of dependence such as the covariance. In this section, we shall discuss several attempts to distinguish between SRD and LRD in the infinite variance case. We focus on α -heavy-tailed series with parameter $\alpha \in (0, 2)$ as discussed in Appendix A.4. We shall now provide a number of possible alternative (non-equivalent) definitions of LRD in the case of heavy-tailed series.

2.9.1 First Definition of LRD Under Heavy Tails: Condition A

One way to proceed is by analogy to the case of finite variance. Since second moments are infinite for heavy-tailed random variables, covariances and spectral densities cannot be used. One possible approach is to define LRD through the behavior of the partial sum process of the series. Recall from (A.4.11) and (A.4.8) in Appendix A.4 that if $\{X_n\}$ is a series of i.i.d. α -heavy-tailed variables, $\alpha \in (0, 2)$, then

$$\frac{1}{L_0(N)N^{1/\alpha}} \left(\sum_{n=1}^{[Nt]} X_n - A([Nt]) \right) \xrightarrow{fdd} S_\alpha(t), \quad t \geq 0,$$

where $A(\cdot)$ is a deterministic function, L_0 is a slowly varying function and S_α is an α -stable Lévy motion. The case $\alpha = 2$ is associated with i.i.d. variables X_n having finite variance and Brownian motion in the limit. By analogy to the finite variance case (cf. Propositions 2.8.7 and 2.8.8), this suggests the following definition for LRD.

Condition A An α -heavy-tailed, $\alpha \in (0, 2)$, stationary time series $\{X_n\}_{n \in \mathbb{Z}}$ is LRD if it satisfies

$$\frac{1}{L(N)N^{d+1/\alpha}} \left(\sum_{n=1}^{[Nt]} X_n - A([Nt]) \right) \xrightarrow{fdd} X_\alpha(t), \quad t \geq 0, \quad (2.9.1)$$

with

$$d > 0,$$

where L is a slowly varying function and X_α is a (non-degenerate) stochastic process.

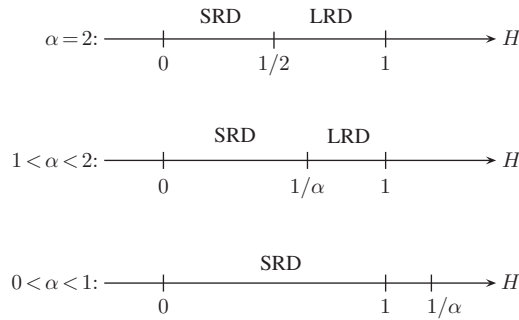


Figure 2.5 SRD and LRD regimes for various ranges of α .

When a time series satisfies (2.9.1) with $d \leq 0$, it could be called SRD. By Corollary 2.8.6 or by arguing directly, the limiting process X_α in condition A is H -SSSI with

$$H = \frac{1}{\alpha} + d,$$

so that

$$d = H - \frac{1}{\alpha}.$$

Example 2.9.1 (*Linear fractional stable noise*) Consider the LFSM process $\{L_{\alpha,H}(t)\}_{t \in \mathbb{R}}$ defined in Proposition 2.6.21, with $H \in (0, 1)$. The series $X_n = L_{\alpha,H}(n) - L_{\alpha,H}(n-1)$, $n \in \mathbb{Z}$, of its increments is called *linear fractional stable noise* (LFSN). It satisfies condition A with $d = H - 1/\alpha$. The series is LRD in the sense of this condition when $H > 1/\alpha$ which is possible only when $\alpha \in (1, 2)$ because $H \in (0, 1)$. This is illustrated in Figure 2.5 which includes the case $\alpha = 2$ where $L_{2,H}(t)$ is FBM.

Why is it not possible for LFSN to be LRD when $\alpha < 1$? A possible (heuristic) explanation is that $\alpha < 1$ corresponds to very heavy tails and thus X_n may take large values with high probability. If this were combined with LRD, there would be long strings of high values and the integral would explode.

Note that, in general, one does not expect LRD in the sense of condition A when $\alpha \in (0, 1)$. For example, if $\mathbb{E}|X_\alpha(t)|^\gamma < \infty$ with $\gamma \leq 1$, Proposition 2.5.6, (c), implies that $H \leq 1/\gamma$. Then, if X_α is α -stable, we can take $\gamma < \alpha$. By taking $\gamma < \alpha$ arbitrarily close to α , we get that $H \leq 1/\alpha$. This precludes having $H = 1/\alpha + d$ with $d > 0$. The process, therefore, cannot be LRD in the sense of condition A. A somewhat stronger statement of this fact can be found in Samorodnitsky [879], Proposition 8.1.

Here is another example of a series satisfying condition A.

Example 2.9.2 (*Heavy-tailed linear series*) Consider a linear series

$$X_n = \sum_{k=0}^{\infty} a_k \epsilon_{n-k}, \quad n \in \mathbb{Z}, \quad (2.9.2)$$

where $\epsilon_k, k \in \mathbb{Z}$, are i.i.d. α -heavy-tailed random variables. Suppose for simplicity that the random variables ϵ_k are symmetric. By (A.4.5) in Appendix A.4, this is equivalent to requiring that the characteristic function of ϵ_k has the form

$$\mathbb{E} \exp\{i\theta\epsilon_k\} = \exp \left\{ - (C_0 + o(1)) |\theta|^\alpha h(|\theta|^{-1}) \right\}, \quad (2.9.3)$$

as $\theta \rightarrow 0$, where $C_0 = \Gamma(1 - \alpha)(c_1 + c_2) \cos(\alpha\pi/2)$ and h is a slowly varying function at infinity. These quantities appear in the tail probabilities

$$\mathbb{P}(\epsilon_k < -x) = \mathbb{P}(\epsilon_k > x) = cx^{-\alpha}h(x). \quad (2.9.4)$$

Because ϵ_k is symmetric, we require $c = c_1 = c_2$. See (A.4.4) and (A.4.5) in Appendix A.4. Now consider the coefficients $\{a_k\}$. As in (2.1.4), suppose that

$$a_k = L_1(k)k^{d-1}, \quad k \geq 1, \quad (2.9.5)$$

with L_1 slowly varying at infinity but d satisfying

$$0 < d < 1 - \frac{1}{\alpha}. \quad (2.9.6)$$

Observe that this implies $\alpha > 1$. One could show using (2.9.3) that under the assumptions above, the linear series (2.9.2) is well-defined and α -heavy-tailed (Exercise 2.16).

Finally, let h_α be a slowly varying function at infinity satisfying

$$h(N^{1/\alpha}h_\alpha(N)^{1/\alpha}) \sim h_\alpha(N), \quad (2.9.7)$$

as $N \rightarrow \infty$, where h is given in (2.9.4). The existence of h_α is shown following (A.4.8) in Appendix A.4. The following result shows that $\{X_n\}_{n \in \mathbb{Z}}$ satisfies condition A.

Proposition 2.9.3 *With the above notation and assumptions on the linear series $\{X_n\}_{n \in \mathbb{Z}}$, one has*

$$\frac{1}{L(N)N^{d+1/\alpha}} \sum_{n=1}^{[Nt]} X_n \xrightarrow{fdd} L_{\alpha,H}(t), \quad t \geq 0, \quad (2.9.8)$$

where $L_{\alpha,H}$ is LFSM with $H = d + 1/\alpha$, $a = 1$, $b = 0$ in (2.6.33), and where

$$L(N) = d^{-1}C_0^{1/\alpha}L_1(N)h_\alpha(N)^{1/\alpha} \quad (2.9.9)$$

is a slowly varying function at infinity.

Proof We suppose for simplicity that $a_0 = 0$. Set $a_k = 0$ for $k \leq -1$, and

$$a_N(m) = \sum_{n=1}^N a_{n-m}.$$

Then $X_n = \sum_{m=-\infty}^{\infty} a_{n-m}\epsilon_m$ and hence

$$\sum_{n=1}^{[Nt]} X_n = \sum_{m=-\infty}^{\infty} \left(\sum_{n=1}^{[Nt]} a_{n-m} \right) \epsilon_m = \sum_{m=-\infty}^{\infty} a_{[Nt]}(m) \epsilon_m.$$

Denote by $Y_N(t)$ the process on the left-hand side of (2.9.8) and let $A_N = L(N)N^{1/\alpha+d}$ be the normalization. Since we want to prove convergence of finite-dimensional distributions, we will consider an arbitrary linear combination of $Y_N(t_1), \dots, Y_N(t_J)$ with $J > 0$. By using (2.9.3), for $\theta_j \in \mathbb{R}, t_j \geq 0, j = 1, \dots, J$,

$$\begin{aligned} I_N &:= -\log \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j Y_N(t_j) \right\} \\ &= A_N^{-\alpha} \sum_{m=-\infty}^{\infty} (C_0 + o(1)) \left| \sum_{j=1}^J \theta_j a_{[Nt_j]}(m) \right|^\alpha h \left(A_N \left| \sum_{j=1}^J \theta_j a_{[Nt_j]}(m) \right|^{-1} \right). \end{aligned} \quad (2.9.10)$$

The terms $o(1)$ in (2.9.10) depend, in particular, on m . We want to show first that these terms can be taken out of the sum $\sum_{m=-\infty}^{\infty}$. To accomplish this, note that

$$A_N^{-1} a_N(m) = d C_0^{-1/\alpha} h_\alpha(N)^{-1/\alpha} L_1(N)^{-1} N^{-1/\alpha-d} \sum_{n=1}^N L_1(n-m)(n-m)_+^{d-1}.$$

Since h_α and L_1 are slowly varying, by using Potter's bound (2.2.2), we get that, for arbitrarily small fixed $\epsilon_1, \epsilon_2 > 0$,

$$\begin{aligned} |A_N^{-1} a_N(m)| &\leq C N^{\epsilon_1} N^{-1/\alpha-d} \sum_{n=1}^N (n-m)_+^{d+\epsilon_2-1} \\ &\leq C N^{-1/\alpha-d+\epsilon_1} \sum_{n=1}^N n^{d+\epsilon_2-1} \leq C' N^{-1/\alpha+\epsilon_1+\epsilon_2}. \end{aligned}$$

Hence, by taking ϵ_1, ϵ_2 so that $-1/\alpha + \epsilon_1 + \epsilon_2 < 0$, we have

$$\sup_{m \in \mathbb{Z}} |A_N^{-1} a_N(m)| \rightarrow 0. \quad (2.9.11)$$

This implies, as desired, that

$$I_N = (C_0 + o(1)) A_N^{-\alpha} \sum_{m=-\infty}^{\infty} \left| \sum_{j=1}^J \theta_j a_{[Nt_j]}(m) \right|^\alpha h \left(A_N \left| \sum_{j=1}^J \theta_j a_{[Nt_j]}(m) \right|^{-1} \right). \quad (2.9.12)$$

Using the expressions for A_N and $L(N)$, we have

$$\begin{aligned} I_N &= (1 + o(1)) \frac{1}{N} \sum_{m=-\infty}^{\infty} \left| d \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}(m)}{L_1(N)N^d} \right|^\alpha \\ &\quad \times h_\alpha(N)^{-1} h \left(h_\alpha(N)^{1/\alpha} N^{1/\alpha} \left| C_0^{-1/\alpha} d \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}(m)}{L_1(N)N^d} \right|^{-1} \right) \\ &=: (1 + o(1)) \int_{\mathbb{R}} f_N(u) du, \end{aligned} \quad (2.9.13)$$

where

$$f_N(u) = \sum_{m=-\infty}^{\infty} \left| d \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}(m)}{L_1(N)N^d} \right|^\alpha \\ \times h_\alpha(N)^{-1} h \left(h_\alpha(N)^{1/\alpha} N^{1/\alpha} \left| C_0^{-1/\alpha} d \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}(m)}{L_1(N)N^d} \right|^{-1} \right) 1_{[\frac{m}{N}, \frac{m+1}{N})}(u).$$

Note that

$$\frac{a_{[Nt]}(m)}{L_1(N)N^d} = \sum_{n=1}^{[Nt]} \frac{L_1(\frac{n-m}{N}N)}{L_1(N)} \left(\frac{n}{N} - \frac{m}{N} \right)_+^{d-1} \frac{1}{N} \rightarrow \int_0^t (s-u)_+^{d-1} ds,$$

as $N \rightarrow \infty$ and $m/N \rightarrow u$. Then, by using the fact that h is a slowly varying function and the relation (2.9.7), we get that

$$f_N(u) \rightarrow \left| d \sum_{j=1}^J \theta_j \int_0^{t_j} (s-u)_+^{d-1} ds \right|^\alpha =: f(u).$$

We want to conclude that

$$\int_{\mathbb{R}} f_N(u) du \rightarrow \int_{\mathbb{R}} f(u) du. \quad (2.9.14)$$

By the generalized dominated convergence theorem, it is enough to find functions g_N such that $|f_N(u)| \leq C g_N(u)$, and

$$g_N(u) \rightarrow g(u) \quad \text{a.e. } du \quad (2.9.15)$$

and

$$\int_{\mathbb{R}} g_N(u) du \rightarrow \int_{\mathbb{R}} g(u) du. \quad (2.9.16)$$

By using (2.9.11), Potter's bound (2.2.1) and the relation (2.9.7), observe that $|f_N(u)| \leq C g_N(u)$, where

$$g_N(u) = \sum_{m=-\infty}^{\infty} \left| \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}(m)}{L_1(N)N^d} \right|^{\alpha+\epsilon} 1_{[\frac{m}{N}, \frac{m+1}{N})}(u)$$

and $\epsilon > 0$ is arbitrarily small and fixed. As in the case of f_N above, we have

$$g_N(u) \rightarrow \left| \sum_{j=1}^J \theta_j \int_0^{t_j} (s-u)_+^{d-1} ds \right|^{\alpha+\epsilon} =: g(u), \quad (2.9.17)$$

so that (2.9.15) holds. To show (2.9.16), note first that

$$\int_{\mathbb{R}} g_N(u) du = \sum_{m=-\infty}^{\infty} \left| \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}(m)}{L_1(N)N^d} \right|^{\alpha+\epsilon} \frac{1}{N}.$$

We will now approximate the sequence $a_k = L_1(k)k_+^{d-1}$ by the sequence $a_k^0 = k_+^{d-1}$. Set $a_N^0(m) = \sum_{n=1}^N a_{n-m}^0$. Consider

$$T_N := \sum_{m=-\infty}^{\infty} \left| \frac{a_N(m)}{L_1(N)N^d} - \frac{a_N^0(m)}{N^d} \right|^{\alpha+\epsilon} \frac{1}{N}.$$

Note that, by using Potter's bound (2.2.1),

$$\begin{aligned} T_N &\leq \sum_{m=-\infty}^{\infty} \left(\sum_{n=1}^N \left| \frac{L_1(n-m)}{L_1(N)} - 1 \right| \left(\frac{n}{N} - \frac{m}{N} \right)_+^{d-1} \frac{1}{N} \right)^{\alpha+\epsilon} \frac{1}{N} \\ &\leq C \sum_{m=-\infty}^{\infty} \left(\sum_{n=1}^N \left(1_{\{\frac{n}{N} - \frac{m}{N} > A\}} \left(\frac{n}{N} - \frac{m}{N} \right)^{d+\tilde{\epsilon}-1} + 1_{\{\frac{n}{N} - \frac{m}{N} < B\}} \left(\frac{n}{N} - \frac{m}{N} \right)_+^{d-\tilde{\epsilon}-1} \right) \frac{1}{N} \right)^{\alpha+\epsilon} \frac{1}{N} \\ &\quad + \sum_{m=-\infty}^{\infty} \left(\sum_{n=1}^N \left| \frac{L_1(n-m)}{L_1(N)} - 1 \right| 1_{\{B \leq \frac{n}{N} - \frac{m}{N} \leq A\}} \left(\frac{n}{N} - \frac{m}{N} \right)_+^{d-1} \frac{1}{N} \right)^{\alpha+\epsilon} \frac{1}{N} =: T_{N,1} + T_{N,2}, \end{aligned}$$

where $\tilde{\epsilon} > 0$ is arbitrarily small. The first sum $T_{N,1}$ in the bound is arbitrarily small for large enough A and small enough B since $\int_0^\infty (\int_0^1 ((u-v)_+^{d+\tilde{\epsilon}-1} + (u-v)_+^{d-\tilde{\epsilon}-1}) du)^{\alpha+\epsilon} dv < \infty$ for small enough $\epsilon, \tilde{\epsilon} > 0$. The second term $T_{N,2}$ in the bound converges to zero for fixed A, B by using the uniform convergence theorem (2.2.4) for the slowly varying function L_1 , and the fact that $\int_0^\infty (\int_0^1 (u-v)_+^{d-1} du)^{\alpha+\epsilon} dv < \infty$ for small enough $\epsilon > 0$. This shows that $T_N \rightarrow 0$. Hence,

$$\int_{\mathbb{R}} g_N(u) du = (1 + o(1)) \sum_{m=-\infty}^{\infty} \left| \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}^0(m)}{N^d} \right|^{\alpha+\epsilon} \frac{1}{N}.$$

Finally, note that

$$\begin{aligned} \sum_{m=-\infty}^{\infty} \left| \sum_{j=1}^J \theta_j \frac{a_{[Nt_j]}^0(m)}{N^d} \right|^{\alpha+\epsilon} \frac{1}{N} &= \sum_{m=-\infty}^{\infty} \left| \sum_{j=1}^J \theta_j \sum_{n=1}^{[Nt_j]} \left(\frac{n}{N} - \frac{m}{N} \right)_+^{d-1} \frac{1}{N} \right|^{\alpha+\epsilon} \frac{1}{N} \\ &= (1 + o(1)) d^\alpha \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j \int_0^{t_j} (u-v)_+^{d-1} du \right|^{\alpha+\epsilon} dv = (1 + o(1)) \int_{\mathbb{R}} g(u) du, \end{aligned}$$

where g is defined in (2.9.17). This proves (2.9.16) and establishes (2.9.14).

In view of (2.9.10), (2.9.12), (2.9.13) and (2.9.14), it follows that

$$\begin{aligned} I_N &\rightarrow \int_{\mathbb{R}} \left| d \sum_{j=1}^J \theta_j \int_0^{t_j} (s-v)_+^{d-1} ds \right|^\alpha dv = \int_{\mathbb{R}} \left| \sum_{j=1}^J \theta_j \left((t_j - v)_+^d - (-v)_+^d \right) \right|^\alpha dv \\ &= -\log \mathbb{E} \exp \left\{ i \sum_{j=1}^J \theta_j L_{\alpha,H}(t_j) \right\}, \end{aligned}$$

where $L_{\alpha,H}$ is LFSM as in the statement of the proposition. This establishes the convergence of the finite-dimensional distributions. $\square$

2.9.2 Second Definition of LRD Under Heavy Tails: Condition B

Example 2.9.2 also suggests the following alternative definition of LRD.

Condition B Consider a linear series $X_n = \sum_{k=0}^{\infty} a_k \epsilon_{n-k}$, $n \in \mathbb{Z}$, with i.i.d. α -heavy-tailed variables ϵ_k , $k \in \mathbb{Z}$, $\alpha \in (1, 2)$, and a sequence $\{a_k\}$ satisfying

$$a_k = L_1(k)k^{d-1}, \quad (2.9.18)$$

where L_1 is a slowly varying function at infinity. This series is LRD if

$$0 < d < 1 - \frac{1}{\alpha}. \quad (2.9.19)$$

A drawback of condition B is that linear series form only a small sub-class of heavy-tailed series, even in the case of α -stable series. When $\{X_n\}$ is a linear series with absolutely summable coefficients $\{a_k\}$ such that $|\sum_k a_k| > 0$, one can show that Proposition 2.9.3 holds for $\alpha \in (0, 2)$ with $d = 0$ and as $N \rightarrow \infty$, the limit is stable Lévy motion (Astrauskas [53]). Such series could then be called SRD.

Example 2.9.4 (Heavy-tailed FARIMA series) An example is a heavy-tailed FARIMA $(0, d, 0)$ series defined by $X_n = (I - B)^{-d} \epsilon_n$ with a series $\{\epsilon_n\}$ as in condition B. When the innovations ϵ_k are α -stable, the series is referred to as stable FARIMA $(0, d, 0)$. Several realizations of $S\alpha S$ FARIMA $(0, d, 0)$ series, for d s both satisfying (2.9.19) and violating (2.9.19), are given in Figure 2.6, and are interesting to compare to the Gaussian FARIMA $(0, d, 0)$ series in Figure 2.2. They are obtained using an approximate method based on FFT developed in Stoev and Taqqu [926] (cf. Figure 2.4). Extensions to heavy-tailed FARIMA (p, d, q) series are also possible. In the stable case, see Kokoszka and Taqqu [571, 572, 573], Stoev and Taqqu [926].

2.9.3 Third Definition of LRD Under Heavy Tails: Codifference

Another way to study serial dependence for heavy-tailed series is to replace covariances by other measures of dependence. This is typically done for α -stable series. One common measure of dependence is given in the following definition.

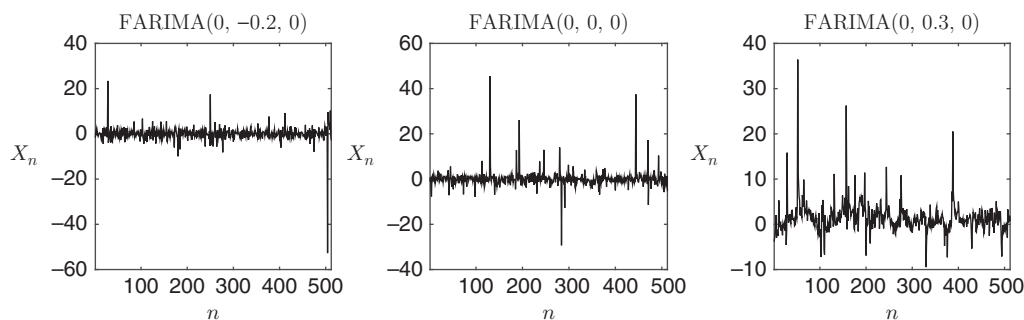


Figure 2.6 Several realizations of $S\alpha S$ FARIMA $(0, d, 0)$ series for $\alpha = 1.5$. Left: $d = -0.2$. Middle: $d = 0$. Right: $d = 0.3$.

Definition 2.9.5 Let $(X, Y)'$ be an α -stable vector¹⁷ with $\alpha \in (0, 2)$. The *codifference* of X and Y is

$$\tau_{X,Y} = \|X\|_\alpha^\alpha + \|Y\|_\alpha^\alpha - \|X - Y\|_\alpha^\alpha,$$

where $\|\cdot\|_\alpha$ denotes a scale parameter. For α -stable series $\{X_n\}_{n \in \mathbb{Z}}$,¹⁸ set

$$\tau(k) = \tau_{X_k, X_0}.$$

Note that when $\alpha = 2$, the codifference becomes the covariance.

The asymptotics of codifference $\tau(k)$ at large lags has been obtained for a number of candidate heavy-tailed LRD series. Let us focus on the case $\alpha \in (1, 2)$ only. For example, in the case of LFSN, it has been shown (Astrauskas, Levy, and Taqqu [54]) that

$$\tau(k) \sim C_1 k^{\alpha H - \alpha}, \quad (2.9.20)$$

if $\overline{H}(\alpha) < H < 1$, $H \neq 1/\alpha$, and

$$\tau(k) \sim C_2 k^{H - \frac{1}{\alpha} - 1}, \quad (2.9.21)$$

if $0 < H < \overline{H}(\alpha)$, where

$$\overline{H}(\alpha) = 1 - (\alpha(\alpha - 1))^{-1}.$$

In the case of FARIMA(0, d , 0) series with α -stable innovations, the same results hold by taking $H = d + 1/\alpha$ (Kokoszka and Taqqu [571]).

Despite interesting asymptotics, it is not too clear how to define LRD using the codifference. One possibility would be as follows:

Condition C An α -stable stationary series $\{X_n\}_{n \in \mathbb{Z}}$, $\alpha \in (0, 2)$, is LRD if

$$\sum_{k=-\infty}^{\infty} |\tau(k)| = \infty. \quad (2.9.22)$$

For LFSN, by using (2.9.20) and when $\overline{H}(\alpha) < H < 1$, $H \neq 1/\alpha$, condition C holds when $\alpha H - \alpha + 1 > 0$ or

$$H > 1 - \frac{1}{\alpha}.$$

Note that since $\overline{H}(\alpha) < 1 - 1/\alpha < 1/\alpha$, the LRD range under condition C is not the same as the LRD range $H > 1/\alpha$ under conditions A and B (in the latter case, only for FARIMA series as increments of LFSM do not have a linear representation). Another drawback of condition C is that the codifference of many other fractional stable noises (that is, increments of stable SSSI processes) does not have such nice asymptotic behavior. For example, there are fractional noises for which the codifference does not even converge to zero at large lags. See Samorodnitsky and Taqqu [883] for more details.

¹⁷ One can think of an α -stable vector as having an integral representation $(\int_S f(x) M_\alpha(dx), \int_S g(x) M_\alpha(dx))'$ where $M_\alpha(dx)$ is an α -stable random measure and f, g are deterministic functions.

¹⁸ Similarly, think of X_n as $\int_S f_n(x) M_\alpha(dx)$.

We should also mention that there are dependence measures akin to the codifference, for example, the “covariation” (Samorodnitsky and Taqqu [883]). As with the codifference, these measures of dependence are interesting but of limited practical use for defining LRD.

2.10 Heuristic Methods of Estimation

We consider here several preliminary estimation methods to detect LRD and to estimate related parameters. The methods are heuristic in the sense that a rigorous statistical theory for these estimators will not be developed in this section. The theory behind some methods can be developed as discussed in the notes in Section 2.13. For other methods, a rigorous theory presents mathematical difficulties or is not sought because of inferior performance. Many of the methods described here are widely used.

We should also note that we focus here on series with finite variance. As is common in statistical applications, all slowly varying functions appearing in definitions of LRD will be assumed constant. The focus throughout is on estimation of LRD parameter d , though estimation of the associated constants will also be mentioned whenever appropriate.

2.10.1 The R/S Method

One of the first methods which was used to detect and to estimate LRD is based on the so-called R/S statistic. The R/S statistic originates with Hurst [501] and is related to dam construction. Suppose that one would like to construct a dam which does not overflow and also has enough storage capacity for future use. Let X_n be the inflow to the dam in period n , $n = 1, \dots, k$, and suppose that the outflow from the dam equals to a constant a which is the amount withdrawn in every period:

period	1	2	3	...	$k-1$	k
input	X_1	X_2	X_3	...	X_{k-1}	X_k
output	a	a	a	...	a	a

It is reasonable to consider a k -horizon policy where

$$a = \frac{X_1 + \dots + X_k}{k}$$

so that the water level by period k is the same as in period 0. Then, the difference between inflow and outflow in any period $1 \leq j \leq k$ is¹⁹

$$S_j = \sum_{n=1}^j X_n - aj = \sum_{n=1}^j X_n - \frac{j}{k} \sum_{n=1}^k X_n, \quad j = 1, \dots, k. \quad (2.10.1)$$

For convenience set $S_0 = 0$ and observe that $S_k = 0$.

The dam should also have a sufficient initial amount so that it never ends lacking water under this withdrawal policy. Hence, in order to avoid both lack of water and overflow, the dam should be at least of size

$$R(k) = \max_{0 \leq j \leq k} S_j^+ + \max_{0 \leq j \leq k} S_j^- = \max_{0 \leq j \leq k} S_j - \min_{0 \leq j \leq k} S_j, \quad (2.10.2)$$

¹⁹ Do not confuse S_n defined in (2.10.1) with the standard deviation $S(k)$ in (2.10.5).

where the second equality follows from the convention $S_0 = 0$ and the “ R ” refers to “Range”. Observe that

$$R(k) = \max_{0 \leq j \leq k} \left\{ \sum_{n=1}^j X_n - \frac{j}{k} \sum_{n=1}^k X_n \right\} - \min_{0 \leq j \leq k} \left\{ \sum_{n=1}^j X_n - \frac{j}{k} \sum_{n=1}^k X_n \right\}.$$

Here are two toy examples of the situation:

period	1	2	3	4
input	100	50	100	50
output	75	75	75	75
cumulative input	100	150	250	300
cumulative output	75	150	225	300
difference	25	0	25	0
$R(k)$				25

in which case $R(4) = 25$, and

period	1	2	3	4
input	100	100	50	50
output	75	75	75	75
cumulative input	100	200	250	300
cumulative output	75	150	225	300
difference	25	50	25	0
$R(k)$				50

in which case $R(4) = 50$. In the latter case, the input appears more persistent leading to a higher minimum dam size. This is generally the case with LRD time series.

In the R/S method, one is interested in the behavior of

$$Q(k) = \frac{R(k)}{S(k)}, \quad (2.10.3)$$

as $k \rightarrow \infty$, where

$$S(k)^2 = \frac{1}{k} \sum_{j=1}^k (X_j - \bar{X}_k)^2$$

with $\bar{X}_k = \frac{1}{k} \sum_{j=1}^k X_j$. One divides by $S(k)$ in order to reduce the fluctuations of $R(k)$. The ratio $Q(k)$ in (2.10.3) is called the R/S (*rescaled adjusted range*) statistic.

In practice, given observations $X_1, \dots, X_N$, $(N-k+1)$ replicates of $Q(k)$ are considered. These are defined as follows. Given a time series $\{X_n\}$, let $Y_0 = 0$, $Y_n = \sum_{j=1}^n X_j$ and

$$R(l, k) = \max_{0 \leq j \leq k} \left\{ Y_{l+j} - Y_l - \frac{j}{k} (Y_{l+k} - Y_l) \right\} - \min_{0 \leq j \leq k} \left\{ Y_{l+j} - Y_l - \frac{j}{k} (Y_{l+k} - Y_l) \right\},$$

$$S(l, k)^2 = \frac{1}{k} \sum_{j=l+1}^{l+k} (X_j - \bar{X}_{l,k})^2,$$

where $\bar{X}_{l,k} = \frac{1}{k} \sum_{j=l+1}^{l+k} X_j$. Then set

$$Q(l, k) = \frac{R(l, k)}{S(l, k)}. \quad (2.10.4)$$

Observe that

$$R(k) = R(0, k) = \max_{0 \leq j \leq k} \left\{ Y_j - \frac{j}{k} Y_k \right\} - \min_{0 \leq j \leq k} \left\{ Y_j - \frac{j}{k} Y_k \right\},$$

$S(k) = S(0, k)$ and

$$Q(k) = \frac{R(k)}{S(k)} = Q(0, k). \quad (2.10.5)$$

The asymptotics of $Q(k)$, as $k \rightarrow \infty$, can informally be derived as follows. Suppose without loss of generality that $\mathbb{E}X_n = 0$. Observe that

$$\begin{aligned} R(k) &= \max_{0 \leq j \leq k} - \min_{n=1}^j \left\{ \sum_{n=1}^j X_n - \frac{j}{k} \sum_{n=1}^k X_n \right\} \\ &= \max_{0 \leq j/k \leq 1} - \min_{n=1}^{\lfloor \frac{j}{k} \rfloor} \left\{ \sum_{n=1}^{\lfloor \frac{j}{k} \rfloor} X_n - \frac{j}{k} \sum_{n=1}^k X_n \right\} = \max_{0 \leq j/k \leq 1} - \min \left\{ Y^k\left(\frac{j}{k}\right) - \frac{j}{k} Y^k(1) \right\}, \end{aligned}$$

where

$$Y^k(t) = \sum_{n=1}^{\lfloor kt \rfloor} X_n.$$

For many time series $\{X_n\}$, one expects that there is $\rho > 0$ such that

$$k^{-\rho} Y^k(t) \xrightarrow{d} Y(t), \quad (2.10.6)$$

where the convergence in distribution happens in a function space (weak convergence). Then,

$$k^{-\rho} R(k) \xrightarrow{d} \max_{0 \leq t \leq 1} - \min \{ Y(t) - tY(1) \}.$$

The process $\tilde{Y}(t) = Y(t) - tY(1)$, $t \in [0, 1]$, is a bridge process associated with Y , and $\max - \min$ is its range. One similarly expects that by the weak law of large numbers,

$$S(k) \xrightarrow{P} \sqrt{\text{Var}(X_1)},$$

where $\xrightarrow{P}$ denotes convergence in probability.

Now, for SRD stationary series $\{X_n\}$, one expects that (2.10.6) holds with $\rho = 1/2$ and the limit $Y(t) = cB(t)$, where B is a standard Brownian motion and

$$c^2 = \lim_{k \rightarrow \infty} \mathbb{E} \left(\frac{1}{k^{1/2}} \sum_{n=1}^k X_n \right)^2 = \lim_{k \rightarrow \infty} \frac{1}{k} \sum_{n=1}^k \sum_{m=1}^k \mathbb{E} X_n X_m = \sum_{h=-\infty}^{\infty} \gamma_X(h)$$

with $\gamma_X(h) = \mathbb{E}X_h X_0$, is the long-run variance (see the discussion following (2.3.3)). Hence, under SRD, one expects that

$$k^{-1/2} Q(k) \xrightarrow{d} \frac{c}{\sqrt{\text{Var}(X_1)}} \max_{0 \leq u \leq 1} - \min \{B(u) - uB(1)\}.$$

The limit involves the range of a Brownian bridge $\{B(u) - uB(1)\}_{0 \leq u \leq 1}$ which has been well studied (its distribution can be found in, for example, Feller [359] or Resnick [845], pp. 526–539).

On the other hand, under LRD, one expects that for many stationary time series, (2.10.6) holds with

$$\rho = H = d + 1/2$$

and the limit $Y(t) = B_H(t)$, where B_H is FBM. Hence,

$$k^{-H} Q(k) \xrightarrow{d} \frac{1}{\sqrt{\text{Var}(X_1)}} \max_{0 \leq u \leq 1} - \min \{B_H(u) - uB_H(1)\}.$$

Thus, for large k , $Q(k) \approx k^H \xi$ or

$$\log Q(k) \approx H \log k + \log \xi,$$

for some random variable ξ , and $H = 1/2$ under SRD, and $H > 1/2$ under LRD.

In practice, one plots $(N - k + 1)$ replicates $\log Q(0, k), \dots, \log Q(N - k, k)$ of $\log Q(k)$ versus $\log k$ and performs ordinary least squares to estimate the parameter H . The resulting plot is known as the *R/S plot* or the *poxplot*. See Figure 2.7.

Though *R/S* statistic is widely used, there is no satisfactory statistical theory for the resulting estimator of H . For example, confidence intervals are not known for the estimator. One difficulty is that $\log Q(k)$ (or $Q(k)$) is non-Gaussian, non-symmetric and dependent over k in a nontrivial way.

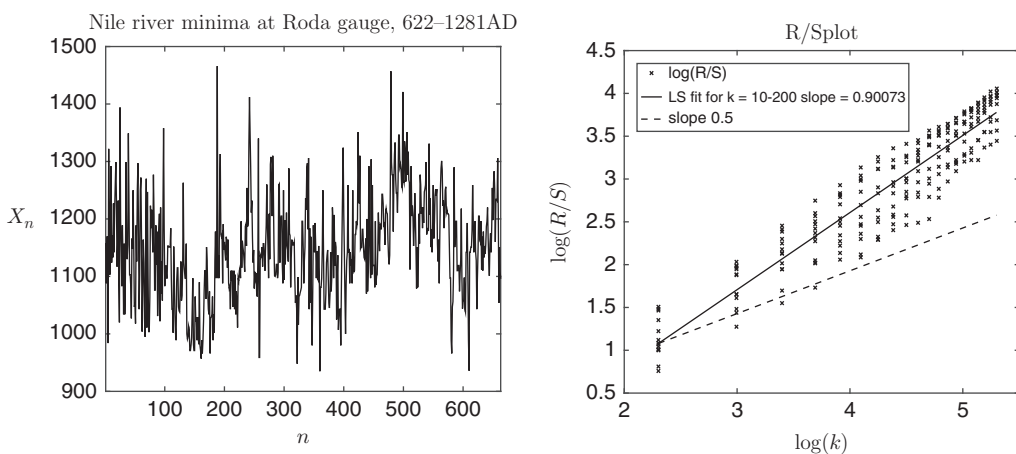


Figure 2.7 Left: Time series of the Nile river minima. Right: The corresponding *R/S* plot.

2.10.2 Aggregated Variance Method

This estimation method is based on the relation

$$\text{Var}(\bar{X}_N) = \text{Var}\left(\frac{1}{N} \sum_{k=1}^N X_k\right) \approx c_5 N^{2d-1}, \quad \text{as } N \rightarrow \infty, \quad (2.10.7)$$

where $d = 0$ for SRD series, and $0 < d < 1/2$ for LRD series. See Proposition 2.2.5. The LRD parameter d can then be estimated through the following steps:

1. For $k = 2, \dots, N/2$, compute sample means $\bar{X}_1(k), \bar{X}_2(k), \dots, \bar{X}_{m_k}(k)$ over non-overlapping windows of size k .
2. Calculate

$$s^2(k) = \frac{1}{m_k - 1} \sum_{j=1}^{m_k} (\bar{X}_j(k) - \bar{X}(k))^2,$$

where

$$\bar{X}(k) = \frac{1}{m_k} \sum_{j=1}^{m_k} \bar{X}_j(k) = \frac{1}{km_k} \sum_{n=1}^{km_k} X_n.$$

3. Plot $\log s^2(k)$ versus $\log k$, and estimate d and c_5 by ordinary least squares.

In view of (2.10.7), one indeed expects that, for large k , $s^2(k) \approx c_5 k^{2d-1}$ and thus

$$\log s^2(k) \approx (2d - 1) \log k + \log c_5.$$

The method has the same drawbacks as the R/S method.

2.10.3 Regression in the Spectral Domain

This estimation method is based on a relation for the spectral density f_X of a series $\{X_n\}$ around zero,

$$f_X(\lambda) \approx c_4 \lambda^{-2d}, \quad \text{as } \lambda \rightarrow 0, \quad (2.10.8)$$

where $d = 0$ corresponds to SRD series, and $0 < d < 1/2$ to LRD series. With

$$I_X(\lambda_k) = \frac{1}{N} \left| \sum_{j=1}^N X_j e^{-ij\lambda_k} \right|^2 \quad (2.10.9)$$

denoting the periodogram of $X_1, \dots, X_N$ at Fourier frequencies

$$\lambda_k = \frac{2\pi k}{N} \quad (2.10.10)$$

(see Section 1.3.3), this suggests that, for $k/N \approx 0$,

$$\log \frac{I_X(\lambda_k)}{2\pi} \approx \log c_4 - 2d \log \lambda_k, \quad (2.10.11)$$

and hence that d and c_4 could be estimated by least squares regression of $\log I_X(\lambda_k)$ on $\log \lambda_k$ for $k = 1, \dots, m$. Here, m is the number of lower frequencies used in the estimation

with $m/N \approx 0$. The resulting estimator of d is called the *GPH estimator* after Geweke and Porter-Hudak [387].

We shall describe next the expected asymptotics of GPH estimator, and the resulting confidence intervals. The argument below is partly informal and, in a sense, not completely correct. It nevertheless provides good intuition and points to advantages of using methods in the spectral domain. See also Section 10.6 for another related and popular estimator in the spectral domain. The analogous estimation of the constant c_4 (or $\log c_4$), and the joint estimation of d , c_4 are left as an exercise (see Exercise 2.18).

Consider the quantity

$$w(\lambda) = \frac{1}{\sqrt{2\pi N}} \sum_{j=1}^N X_j e^{-ij\lambda}, \quad (2.10.12)$$

so that $|w(\lambda)|^2 = I_X(\lambda)$. The next result establishes some of its basic properties for LRD series.

Proposition 2.10.1 *Let $\{X_n\}_{n \in \mathbb{Z}}$ be a LRD series in the sense (2.1.5) in condition II with a quasi-monotone slowly varying function L_2 and let $w(\lambda)$ be defined by (2.10.12). Then, for fixed $\lambda_k = 2\pi k/N$ ($\neq 0$),*

$$\mathbb{E}(\Re w(\lambda_k))^2 \rightarrow \frac{f_X(\lambda_k)}{2}, \quad \mathbb{E}(\Im w(\lambda_k))^2 \rightarrow \frac{f_X(\lambda_k)}{2}, \quad \mathbb{E}\Re w(\lambda_k)\Im w(\lambda_k) \rightarrow 0, \quad (2.10.13)$$

where f_X is the spectral density of the series, and, for fixed $\lambda_k \neq \lambda_j$ ($\neq 0$),

$$\mathbb{E}\Re w(\lambda_k)\Re w(\lambda_j) \rightarrow 0, \quad \mathbb{E}\Im w(\lambda_k)\Im w(\lambda_j) \rightarrow 0, \quad \mathbb{E}\Re w(\lambda_k)\Im w(\lambda_j) \rightarrow 0. \quad (2.10.14)$$

Proof Using

$$\sum_{j=1}^N e^{-ij\lambda_k} = \frac{e^{-i\lambda_k}(e^{-iN\lambda_k} - 1)}{e^{-i\lambda_k} - 1} = \frac{e^{-i\lambda_k}(e^{-i2\pi k} - 1)}{e^{-i2\pi k/N} - 1} = 0,$$

for $\lambda_k = 2\pi k/N \neq 0$, allows writing $\sum_{j=1}^N X_j e^{-ij\lambda_k} = \sum_{j=1}^N (X_j - \mathbb{E}X_j) e^{-ij\lambda_k}$. Hence,

$$\mathbb{E}(\Re w(\lambda_k))^2 = \frac{1}{2\pi N} \sum_{m=1}^N \sum_{n=1}^N \gamma_X(m-n) \cos(m\lambda_k) \cos(n\lambda_k) =: T_1 + T_2,$$

where

$$T_1 = \frac{1}{4\pi N} \sum_{m=1}^N \sum_{n=1}^N \gamma_X(m-n) \cos((m-n)\lambda_k),$$

$$T_2 = \frac{1}{4\pi N} \sum_{m=1}^N \sum_{n=1}^N \gamma_X(m-n) \cos((m+n)\lambda_k).$$

For T_1 , note that

$$T_1 = \frac{1}{4\pi} \left(\gamma_X(0) + \frac{2}{N} \sum_{h=1}^{N-1} (N-h) \gamma_X(h) \cos(h\lambda_k) \right) =: \frac{1}{2} (T_{1,1} - T_{1,2}),$$

where

$$T_{1,1} = \frac{1}{2\pi} \left(\gamma_X(0) + 2 \sum_{h=1}^{N-1} \gamma_X(h) \cos(h\lambda_k) \right), \quad T_{1,2} = \frac{1}{\pi N} \sum_{h=1}^{N-1} h \gamma_X(h) \cos(h\lambda_k).$$

Now, by Proposition 2.2.14, $T_{1,1} \rightarrow f_X(\lambda_k)$. As in (A.2.6) in the proof of Proposition A.2.1 in Appendix A.2,

$$N|T_{1,2}| \leq CN^{2d}L_2(N),$$

which implies $T_{1,2} \rightarrow 0$ since $d < 1/2$. For T_2 , note that

$$T_2 = \frac{1}{4\pi N} \sum_{h=-(N-1)}^{N-1} \gamma_X(h) \sum_{q=|h|+1}^N \cos((2q-|h|)\lambda_k).$$

Indeed, the possible values of $m-n=h$ in $\gamma_X(m-n)$ are $-(N-1), \dots, N-1$. For example, the value $m-n=h \geq 0$ is achieved by $(m, n) = (h+1, 1), \dots, (N, N-h)$, for which $m+n=2m-(m-n)$ takes values $2q-h$ with $q=h+1, \dots, N$.

By arguing as in (A.2.2) of Appendix A.2,

$$\begin{aligned} \left| \sum_{q=|h|+1}^N \cos((2q-|h|)\lambda_k) \right| &= \left| \sum_{r=1}^{N-|h|} \cos((|h|+2r)\lambda_k) \right| = \left| \Re \left(\sum_{r=1}^{N-|h|} e^{i(|h|+2r)\lambda_k} \right) \right| \\ &\leq \left| \sum_{r=1}^{N-|h|} e^{i(|h|+2r)\lambda_k} \right| = \left| \sum_{r=1}^{N-|h|} e^{i2r\lambda_k} \right| = \left| \frac{e^{i2\lambda_k} - e^{i2(N-|h|)\lambda_k}}{1 - e^{i2\lambda_k}} \right| \leq \frac{2}{|1 - e^{i2\lambda_k}|} = \frac{1}{\sin \lambda_k} \leq C. \end{aligned}$$

Hence, $|T_2| \leq C' \sum_{h=0}^{N-1} |\gamma_X(h)|/N$ and $T_2 \rightarrow 0$ in view of Proposition 2.2.1. Gathering the results above, we get

$$\mathbb{E}(\Re w(\lambda_k))^2 \rightarrow \frac{f_X(\lambda_k)}{2}.$$

One can show similarly that $\mathbb{E}(\Im w(\lambda_k))^2 \rightarrow \frac{f_X(\lambda_k)}{2}$. For $\mathbb{E}\Re w(\lambda_k)\Im w(\lambda_k)$, note that

$$\begin{aligned} \mathbb{E}\Re w(\lambda_k)\Im w(\lambda_k) &= \frac{1}{2\pi N} \sum_{m=1}^N \sum_{n=1}^N \gamma_X(m-n) \cos(m\lambda_k) \sin(n\lambda_k) \\ &= \frac{1}{4\pi N} \sum_{m=1}^N \sum_{n=1}^N \gamma_X(m-n) (\sin((m+n)\lambda_k) - \sin((m-n)\lambda_k)) \\ &= \frac{1}{4\pi N} \sum_{m=1}^N \sum_{n=1}^N \gamma_X(m-n) \sin((m+n)\lambda_k), \end{aligned}$$

since the terms with $\sin((m - n)\lambda_k)$ cancel out by symmetry. The remaining expression converges to zero by arguing as for T_2 above. The rest of the statements (2.10.14) can also be proved similarly by arguing as for T_2 above (Exercise 2.17). $\square$

Though Proposition 2.10.1 is stated for LRD series and hence $d > 0$, the proof above works also for SRD series and hence the case $d = 0$.

Suppose now for simplicity that $\{X_n\}$ is a Gaussian series. Then, the vector

$$(\Re w(\lambda_{k_1}), \Im w(\lambda_{k_1}), \dots, \Re w(\lambda_{k_n}), \Im w(\lambda_{k_n}))'$$

is Gaussian. By Proposition 2.10.1, for fixed distinct $\lambda_{k_1}, \dots, \lambda_{k_n}$, it converges in distribution to a Gaussian vector

$$\left(\sqrt{\frac{f_X(\lambda_{k_1})}{2}} \eta_{k_1}, \sqrt{\frac{f_X(\lambda_{k_1})}{2}} \zeta_{k_1}, \dots, \sqrt{\frac{f_X(\lambda_{k_n})}{2}} \eta_{k_n}, \sqrt{\frac{f_X(\lambda_{k_n})}{2}} \zeta_{k_n} \right)',$$

where $\eta_{k_1}, \zeta_{k_1}, \dots, \eta_{k_n}, \zeta_{k_n}$ are i.i.d. $\mathcal{N}(0, 1)$ random variables since their covariances vanish. On the other hand, consider the periodograms $I_X(\lambda_k)$. One has

$$\frac{I_X(\lambda_k)}{2\pi} = |w(\lambda_k)|^2 = (\Re w(\lambda_k))^2 + (\Im w(\lambda_k))^2 \xrightarrow{d} \frac{f_X(\lambda_k)}{2} (\eta_k^2 + \zeta_k^2).$$

This implies that, for fixed distinct $\lambda_{k_1}, \dots, \lambda_{k_n}$,

$$\left(\frac{I_X(\lambda_{k_1})}{2\pi}, \dots, \frac{I_X(\lambda_{k_n})}{2\pi} \right)' \xrightarrow{d} \left(f_X(\lambda_{k_1}) \xi_{k_1}, \dots, f_X(\lambda_{k_n}) \xi_{k_n} \right)', \quad (2.10.15)$$

where $\xi_{k_1}, \dots, \xi_{k_n}$ are independent $\chi^2_2/2$ (or standard exponential) random variables.

We shall now focus on the m Fourier frequencies closest to the origin, namely, $\lambda_j = 2\pi j/N$ for $j = 1, \dots, m$. In view of (2.10.11), one has

$$\log \frac{I_X(\lambda_k)}{2\pi} \approx \log c_4 + d(-2 \log \lambda_k).$$

Hence, we can set $y_j = \log(I_X(\lambda_j)/2\pi)$ and $z_j = -2 \log \lambda_j$, let $\bar{y} = \frac{1}{m} \sum_{j=1}^m y_j$ and $\bar{z} = \frac{1}{m} \sum_{j=1}^m z_j$ and focus on the linear regression

$$y_j = \log c_4 + d z_j. \quad (2.10.16)$$

Set

$$a_j = \frac{z_j - \bar{z}}{\sum_{j=1}^m (z_j - \bar{z})^2}$$

and observe that $\sum_{j=1}^m a_j = 0$. The GPH estimator of d is the linear regression estimator of the slope in (2.10.16), namely,

$$\hat{d}_{gph} = \frac{\sum_{j=1}^m (z_j - \bar{z})(y_j - \bar{y})}{\sum_{j=1}^m (z_j - \bar{z})^2} = \sum_{j=1}^m a_j (y_j - \bar{y}) = \sum_{j=1}^m a_j y_j = \sum_{j=1}^m a_j \log I_X(\lambda_j).$$

The next result concerns the asymptotics of the numerator and denominator of the regression weights a_j .

Proposition 2.10.2 For $z_j = -2 \log \frac{2\pi j}{N}$, $j = 1, \dots, m$, as $m \rightarrow \infty$,

$$z_j - \bar{z} = -2 \left(\log \left(\frac{j}{m} \right) + 1 \right) + o(1), \quad \sum_{j=1}^m (z_j - \bar{z})^2 = 4m(1 + o(1)),$$

so that

$$a_j = \frac{z_j - \bar{z}}{\sum_{j=1}^m (z_j - \bar{z})^2} = -\frac{1}{2m} \left(\log \left(\frac{j}{m} \right) + 1 \right) + o(1), \quad (2.10.17)$$

where $o(1)$ is uniform over $j = 1, \dots, m$.

Proof For the first relation, note that

$$\begin{aligned} -\frac{1}{2}(z_j - \bar{z}) &= \log j - \frac{1}{m} \sum_{j=1}^m \log j = \log \left(\frac{j}{m} \right) - \frac{1}{m} \sum_{j=1}^m \log \left(\frac{j}{m} \right) \\ &= \log \left(\frac{j}{m} \right) - \int_0^1 \log x \, dx + o(1) = \log \left(\frac{j}{m} \right) + 1 + o(1). \end{aligned}$$

For the second relation, by using the first relation, note that

$$\frac{1}{m} \sum_{j=1}^m (z_j - \bar{z})^2 = \frac{4}{m} \sum_{j=1}^m \left(\log \left(\frac{j}{m} \right) + 1 \right)^2 + o(1) = 4 \int_0^1 (\log x + 1)^2 dx + o(1) = 4 + o(1).$$

□

In view of (2.10.15) and (2.10.8), one may expect that

$$\begin{aligned} \widehat{d}_{gph} &= \sum_{j=1}^m a_j \log \frac{I_X(\lambda_j)}{2\pi} \approx \sum_{j=1}^m a_j \log f_X(\lambda_j) + \sum_{j=1}^m a_j \log \xi_j \approx \sum_{j=1}^m a_j \log c \lambda_j^{-2d} + \sum_{j=1}^m a_j \log \xi_j \\ &= -2d \left(\sum_{j=1}^m a_j \log \frac{2\pi j}{N} \right) + \sum_{j=1}^m a_j \log \xi_j = -2d \left(\sum_{j=1}^m a_j \log \frac{j}{m} \right) + \sum_{j=1}^m a_j \left(\log \xi_j - \mathbb{E} \log \xi_j \right), \end{aligned}$$

since $\sum_{j=1}^m a_j = 0$ and where ξ_j s are i.i.d. exponential random variables. By Proposition 2.10.2,

$$\sum_{j=1}^m a_j \log \left(\frac{j}{m} \right) \approx -\frac{1}{2} \sum_{j=1}^m \left(\log \left(\frac{j}{m} \right) + 1 \right) \left(\log \frac{j}{m} \right) \frac{1}{m} \approx -\frac{1}{2} \int_0^1 (\log x + 1) \log x \, dx = -\frac{1}{2}.$$

Hence

$$\sqrt{m}(\widehat{d}_{gph} - d) \approx \sqrt{m} \sum_{j=1}^m a_j \left(\log \xi_j - \mathbb{E} \log \xi_j \right).$$

By the Lindeberg–Feller central limit theorem for independent, non-identically distributed random variables (Feller [360], Chapter VIII, Section 4), one expects that

$$\sqrt{m} \sum_{j=1}^m a_j \left(\log \xi_j - \mathbb{E} \log \xi_j \right) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

where, in view of (2.10.17), the asymptotic variance is given by

$$\begin{aligned}\sigma^2 &= \lim_{m \rightarrow \infty} m \sum_{j=1}^m a_j^2 \operatorname{Var}(\log \xi_1) = \lim_{m \rightarrow \infty} \frac{1}{4m} \sum_{j=1}^m \left(\log \left(\frac{j}{m} \right) + 1 \right)^2 \operatorname{Var}(\log \xi_1) \\ &= \frac{1}{4} \int_0^1 (\log x + 1)^2 dx \operatorname{Var}(\log \xi_1) = \frac{\operatorname{Var}(\log \xi_1)}{4}.\end{aligned}$$

The exponential variable ξ_1 has the density e^{-x} , $x > 0$, and hence

$$\operatorname{Var}(\log \xi_1) = \int_0^\infty (\log x)^2 e^{-x} dx - \left(\int_0^\infty (\log x) e^{-x} dx \right)^2 = \frac{\pi^2}{6} + C^2 - C^2 = \frac{\pi^2}{6},$$

where C is the Euler constant, by using Gradshteyn and Ryzhik [421], p. 571, Formula 4.331.1 and p. 572, Formula 4.335.1. Gathering the above observations, one expects that

$$\sqrt{m}(\hat{d}_{gph} - d) \xrightarrow{d} \mathcal{N}(0, \frac{\pi^2}{24}). \quad (2.10.18)$$

Observe that the variance does not depend on d .

The result (2.10.18) is proved rigorously in Robinson [852]. We note that the asymptotics of m in Robinson [852] is different from above. It is assumed that $m = m(N)$ is such that $m/N \rightarrow 0$. In contrast, in Proposition 2.10.1, it was assumed that λ_k and, in particular, $\lambda_m = 2\pi m/N$ are fixed as $N \rightarrow \infty$. If λ_m is fixed, then $m/N \rightarrow \text{const} > 0$. Assuming $m/N \rightarrow 0$ is more realistic since (2.10.8) is an asymptotic relation as $\lambda \rightarrow 0$. Figure 2.8 illustrates the GPH method. For further discussion, see the notes to this section in Section 2.13.

2.10.4 Wavelet-Based Estimation

We begin by describing briefly what wavelets are. More details can be found in Section 8.2 below. A wavelet $\psi : \mathbb{R} \rightarrow \mathbb{R}$ is a function with special properties. For the purpose of this

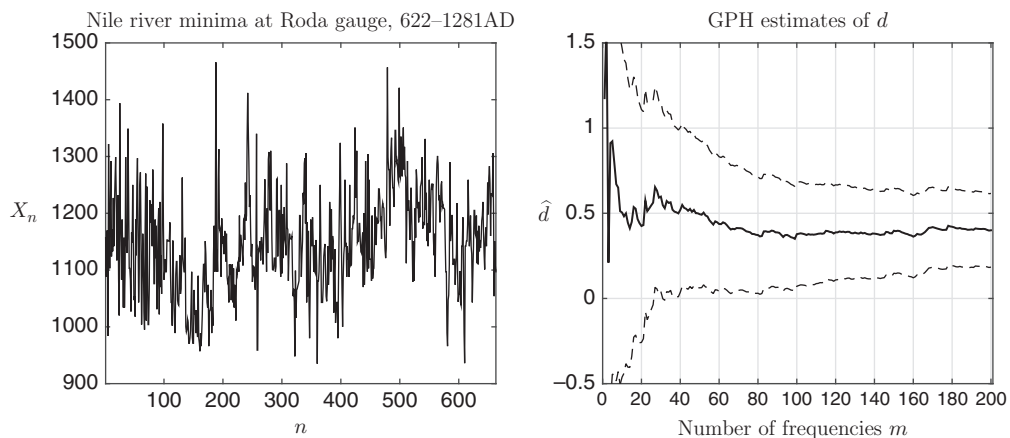


Figure 2.8 Left: Time series of the Nile river minima. Right: The corresponding GPH plot (with confidence intervals in dashed lines).

section, suppose that ψ has a compact support, is bounded, and has $Q \geq 1$ zero moments, that is,

$$\int_{\mathbb{R}} \psi(t) t^q dt = 0, \quad q = 0, \dots, Q-1. \quad (2.10.19)$$

Let

$$\psi_{j,k}(t) = 2^{-j/2} \psi(2^{-j}t - k), \quad j, k \in \mathbb{Z}, \quad (2.10.20)$$

be scaled and translated copies of the wavelet ψ . Here, large j corresponds to large scales (low frequencies) and small or large negative j corresponds to small scales (high frequencies).

It is more convenient to focus here on continuous-time second-order stationary processes $X = \{X(t)\}_{t \in \mathbb{R}}$. We shall use the same notation and terminology which was used earlier. See Remark 2.1.6. For a process X , let

$$d(j, k) = \int_{\mathbb{R}} X(t) \psi_{j,k}(t) dt \quad (2.10.21)$$

be the so-called *detail (wavelet) coefficients* of X . The next result provides some basic properties of these coefficients, in the context of LRD processes X .

Proposition 2.10.3 *The following holds:*

- (i) *For fixed j , the series $\{d(j, k)\}_{k \in \mathbb{Z}}$ has zero mean and is second-order stationary.*
- (ii) *Detail coefficients are invariant to the addition of polynomial trends up to order $Q-1$.*
- (iii) *Let X be LRD in the sense (2.1.5) in condition II. Then, as $j \rightarrow \infty$,*

$$\mathbb{E}d(j, 0)^2 \sim C_d(2^j)^{2d} L_2(2^j), \quad (2.10.22)$$

where $C_d = \int_{\mathbb{R}^2} |s|^{2d-1} \psi(s) \psi(s+t) ds dt$.

- (iv) *Let X be LRD in the sense (2.1.5) in condition II with $L_2(h) \sim c_2$, as $h \rightarrow \infty$. Suppose also that the function $\gamma_X(1/x)x^{2d-1}$ is $2Q$ times continuously differentiable at $x = 0$. Then, for fixed j ,*

$$|\mathbb{E}d(j, 0)d(j, k)| \leq C|k|^{2d-2Q-1},$$

where the constant $C = C_{d,j}$ depends, in particular, on j .

Proof Part (i) follows from $\mathbb{E}d(j, k) = \mu_X \int_{\mathbb{R}} \psi_{j,k}(t) dt = 0$ by setting $\mu_X = \mathbb{E}X(t)$ and using (2.10.19). The second-order stationarity follows from: for $k, h \in \mathbb{Z}$,

$$\begin{aligned} \mathbb{E}d(j, k+h)d(j, k) &= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{E}X(t)X(s) 2^{-j/2} \psi(2^{-j}t - (k+h)) 2^{-j/2} \psi(2^{-j}s - k) dt ds \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{E}X(u + 2^j k)X(v + 2^j k) 2^{-j/2} \psi(2^{-j}u - h) 2^{-j/2} \psi(2^{-j}v) du dv \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{E}X(u)X(v) 2^{-j/2} \psi(2^{-j}u - h) 2^{-j/2} \psi(2^{-j}v) du dv = \mathbb{E}d(j, h)d(j, 0), \end{aligned}$$

where we used the change of variables $t = u + 2^j k$, $s = v + 2^j k$, and the second-order stationarity of the stationarity of the process $\{X(u)\}$.

For part (ii), suppose that the process is $Y(t) = \sum_{j=0}^{Q-1} a_j t^j + X(t)$ instead of $X(t)$. Then, the detail coefficients for $X(t)$ and $Y(t)$ are the same because of (2.10.19).

For part (iii), observe that

$$\mathbb{E}d(j, 0)^2 = \int_{\mathbb{R}^2} \gamma_X(u-v) \psi_{j,0}(u) \psi_{j,0}(v) du dv = 2^{-j} \int_{\mathbb{R}^2} \gamma_X(u-v) \psi(2^{-j}u) \psi(2^{-j}v) du dv.$$

The change of variables $x = 2^{-j}(u-v)$ and $y = 2^{-j}v$ yields

$$\mathbb{E}d(j, 0)^2 = 2^j \int_{\mathbb{R}} \gamma_X(2^j x) \Psi(x) dx,$$

where $\Psi(x) = \int_{\mathbb{R}} \psi(y) \psi(x+y) dy$. The function Ψ also has a compact support and is bounded. With $\gamma_X(h) = h^{2d-1} L_2(h)$, $h > 0$,

$$\begin{aligned} \mathbb{E}d(j, 0)^2 &= (2^j)^{2d} \int_{\mathbb{R}} |x|^{2d-1} L_2(2^j |x|) \Psi(x) dx \\ &= (2^j)^{2d} L_2(2^j) \int_{\mathbb{R}} |x|^{2d-1} \frac{L_2(2^j |x|)}{L_2(2^j)} \Psi(x) dx. \end{aligned}$$

Express the integral as $\int_{\epsilon}^{\infty} + \int_0^{\epsilon}$ for $\epsilon > 0$. For the first integral, by (2.2.4), $L_2(2^j |x|)/L_2(2^j) \rightarrow 1$, uniformly for $|x| \in [\epsilon, A]$, where A is arbitrary and fixed. Since Ψ has compact support, the integral is asymptotic to the right-hand side of (2.10.22), as $j \rightarrow \infty$. On the other hand, the second integral $\int_0^{\epsilon}$ is negligible. Indeed,

$$(2^j)^{2d} \int_0^{\epsilon} x^{2d-1} L_2(2^j x) |\Psi(x)| dx \leq C \int_0^{2^j \epsilon} y^{2d-1} L_2(y) dy \leq \tilde{C} \epsilon^{2d} (2^j)^{2d} L_2(2^j \epsilon)$$

by Proposition 2.2.2. For small ϵ , this is arbitrarily small with respect to $(2^j)^{2d} L_2(2^j)$. This establishes the result (iii).

For part (iv), note that $d(j, k) = \int_{\mathbb{R}} X(t) 2^{-j/2} \psi(2^{-j}t - k) dt = 2^{j/2} \int_{\mathbb{R}} X(2^j t) \psi(t - k) dt$ and that the process $\{X(2^j t)\}_{t \in \mathbb{R}}$ satisfies the same assumptions as $\{X(t)\}_{t \in \mathbb{R}}$. We may thus suppose without loss of generality that $j = 0$. By the assumption on γ_X , $\gamma_X(1/x) = (1/x)^{2d-1} L_2(1/x)$ where $L_2(1/x) = \gamma_X(1/x) x^{2d-1}$ is $2Q$ times continuously differentiable. Then,

$$\gamma_X(1/x) x^{2d-1} = L_2(1/x) = b_2 + b_3 x + \cdots + b_{2Q+2} x^{2Q} + o(x^{2Q}),$$

as $x \rightarrow 0$, or

$$\gamma_X(h) = b_2 h^{2d-1} + \cdots + b_{2Q+2} h^{2d-2Q-1} + o(h^{2d-2Q-1}), \quad (2.10.23)$$

as $h \rightarrow \infty$. With Ψ defined above, note that, for large enough k ,

$$\begin{aligned} \mathbb{E}d(0, 0)d(0, k) &= \int_{\mathbb{R}} \gamma_X(x+k) \Psi(x) dx \\ &= \int_{\mathbb{R}} \sum_{p=1}^{2Q+1} b_{p+1} (x+k)^{2d-p} \Psi(x) dx \\ &\quad + \int_{\mathbb{R}} \left(\gamma_X(x+k) - \sum_{p=1}^{2Q+1} b_{p+1} (x+k)^{2d-p} \right) \Psi(x) dx. \end{aligned} \quad (2.10.24)$$

Consider now the last integral in (2.10.24). Since Ψ has a bounded support $\text{supp}\{\Psi\}$, we may suppose that $x + k$ is of the order k and hence large enough as $k \rightarrow \infty$, uniformly in $x \in \text{supp}\{\Psi\}$. In particular, using (2.10.23), we have

$$\left| \gamma_X(x + k) - \sum_{p=1}^{2Q+1} b_{p+1}(x + k)^{2d-p} \right| = o(k^{2d-2Q-1}), \quad \text{as } k \rightarrow \infty,$$

uniformly in $x \in \text{supp}\{\Psi\}$. Applying this relation and using the fact that Ψ is bounded, we obtain that the last integral in (2.10.24) is of the order $o(k^{2d-2Q-1})$, as $k \rightarrow \infty$.

For the first integral in (2.10.24), we first observe that ψ having Q zero moments implies that Ψ has $2Q$ zero moments. Indeed, for $q < 2Q$,

$$\begin{aligned} \int_{\mathbb{R}} x^q \Psi(x) dx &= \int_{\mathbb{R}^2} (t - s)^q \psi(t) \psi(s) ds dt = \int_{\mathbb{R}^2} \sum_{m=0}^q \binom{q}{m} t^m (-s)^{q-m} \psi(t) \psi(s) ds dt \\ &= \sum_{m=0}^q \binom{q}{m} \int_{\mathbb{R}} t^m \psi(t) dt \int_{\mathbb{R}} (-s)^{q-m} \psi(s) ds = 0, \end{aligned}$$

since ψ has Q zero moments. Then,

$$\begin{aligned} \int_{\mathbb{R}} (x + k)^{2d-p} \Psi(x) dx &= k^{2d-p} \int_{\mathbb{R}} \left(1 + \frac{x}{k}\right)^{2d-p} \Psi(x) dx = k^{2d-p} \int_{\mathbb{R}} \sum_{q=0}^{\infty} a_q \left(\frac{x}{k}\right)^q \Psi(x) dx \\ &= k^{2d-p} \int_{\mathbb{R}} \sum_{q=2Q}^{\infty} a_q \left(\frac{x}{k}\right)^q \Psi(x) dx = O(k^{2d-p-2Q}), \end{aligned}$$

as $k \rightarrow \infty$. Since the term with $p = 1$ dominates, one obtains $O(k^{2p-1-2Q})$ as a bound. $\square$

Proposition 2.10.3 shows that there are several advantages in using wavelets:

- invariance to polynomial trends in (ii),
- recovery of the LRD parameter d in (iii),
- strong decorrelation in (iv).

Another property, namely, that of computational efficiency related to Fast Wavelet Transform is indicated below.

Proposition 2.10.3, (iii) (assuming $L_2(h) \sim c_2$ as $h \rightarrow \infty$), suggests a straightforward way to estimate LRD parameter d . Consider

$$\bar{\mu}_j = \frac{1}{n_j} \sum_{k=1}^{n_j} |d_{j,k}|^2, \quad (2.10.25)$$

where $d_{j,k}$, $k = 1, \dots, n_j$, are detail coefficients available at scale 2^j . They are typically approximations of the detail coefficients $d(j, k)$ defined in (2.10.21). This is because one never observes the path $X(t)$ for all $t \in \mathbb{R}$. In practice, only a sample of the discrete-time series is available. In this case, either the detail coefficients can be viewed as approximations when the discrete time series is considered a sample of a continuous-time process, or they can be viewed as the exact detail coefficients of a suitable continuous-time process

constructed from discrete-time series. There is, in either case, a very fast algorithm, the so-called *Fast Wavelet Transform* (FWT), for computing these coefficients in practice (see Section 8.2).

Suppose here, in addition, that the underlying process X is Gaussian. In this case, the detail coefficients are also Gaussian. Motivated by Proposition 2.10.3, (iv), which indicates that the detail coefficients $d(j, k)$ are weakly correlated, assume also that *all* the detail coefficients $d_{j,k}$ are uncorrelated. In this case, note that, by Proposition 2.10.3, (iii), and for large scales j , the $|d_{j,k}|^2$, $1 \leq k \leq n_j$, are distributed as $C_d c_2 2^{2dj} |X_{j,k}|^2$ where $X_{j,k}$ are independent $\mathcal{N}(0, 1)$ random variables. Then

$$\begin{aligned} \log_2 \bar{\mu}_j &= \log_2 \left(\frac{1}{n_j} \sum_{k=1}^{n_j} |d_{j,k}|^2 \right) \approx \log_2 \left(\frac{2^{2dj} C_d c_2}{n_j} \sum_{k=1}^{n_j} |X_{j,k}|^2 \right) \\ &= \log_2 C_d c_2 + 2dj - \log_2 n_j + \log_2 \sum_{k=1}^{n_j} |X_{j,k}|^2. \end{aligned} \quad (2.10.26)$$

We will compute $\log_2 \bar{\mu}_j$ from the data and use the relation (2.10.26) to estimate d and c_2 through a regression on j . To make this precise, set

$$y_j = \log_2 \bar{\mu}_j - g(j),$$

where

$$g(j) = \mathbb{E} \left(-\log_2 n_j + \log_2 \sum_{k=1}^{n_j} |X_{j,k}|^2 \right).$$

While $\mathbb{E} \sum_{k=1}^{n_j} |X_{j,k}|^2 = \sum_{k=1}^{n_j} \mathbb{E} |X_{j,k}|^2 = n_j$, it is not true that $\mathbb{E} \log_2 \sum_{k=1}^{n_j} |X_{j,k}|^2$ equals $\log_2 n_j$ because $\mathbb{E} \log_2 \neq \log_2 \mathbb{E}$. The term $g(j)$ will serve as a bias correction. Then

$$y_j \approx \log_2 C_d c_2 + 2dj + \epsilon_j, \quad (2.10.27)$$

with

$$\epsilon_j = -\log_2 n_j + \log_2 \sum_{k=1}^{n_j} |X_{j,k}|^2 - g(j)$$

satisfying $\mathbb{E} \epsilon_j = 0$.

The next result sheds light on $g(j)$ and $\text{Var}(\epsilon_j)$.

Proposition 2.10.4 *We have*

$$\begin{aligned} g(j) &= \frac{1}{\log 2} \left(\psi_0 \left(\frac{n_j}{2} \right) - \log \frac{n_j}{2} \right) \sim -\frac{1}{n_j \log 2}, \\ \sigma_j^2 &:= \text{Var}(\epsilon_j) = \frac{\zeta \left(2, \frac{n_j}{2} \right)}{(\log 2)^2} \sim \frac{2}{n_j (\log 2)^2}, \end{aligned}$$

as $n_j \rightarrow \infty$, where $\psi_0(z) = \Gamma'(z)/\Gamma(z)$ is the digamma function and $\zeta(2, z) = \sum_{n=0}^{\infty} (z+n)^{-2}$ is a special case of the generalized Riemann zeta function.

Proof We need the relations: for $\Re \nu > 0$, $\Re \mu > 0$,

$$\int_0^\infty x^{\nu-1} e^{-\mu x} \log x dx = \frac{\Gamma(\nu)}{\mu^\nu} (\psi_0(\nu) - \log \mu),$$

$$\int_0^\infty x^{\nu-1} e^{-\mu x} (\log x)^2 dx = \frac{\Gamma(\nu)}{\mu^\nu} \left((\psi_0(\nu) - \log \mu)^2 + \zeta(2, \nu) \right)$$

(Gradshteyn and Ryzhik [421], p. 573, Formula 4.352.1 and p. 576, Formula 4.358.2). Since $\sum_{k=1}^{n_j} |X_{j,k}|^2$ is distributed like a chi-squared variable with n_j degrees of freedom, it has the density $x^{n_j/2-1} e^{-x/2} / (2^{n_j/2} \Gamma(n_j/2))$, $x > 0$. Then, by using $\log_2 x = \log x / \log 2$,

$$\begin{aligned} g(j) &= -\log_2 n_j + \frac{1}{2^{n_j/2} \Gamma(n_j/2)} \int_0^\infty (\log_2 x) x^{n_j/2-1} e^{-x/2} dx \\ &= -\log_2 n_j + \frac{1}{(\log 2) 2^{n_j/2} \Gamma(n_j/2)} \int_0^\infty (\log x) x^{n_j/2-1} e^{-x/2} dx \\ &= -\log_2 n_j + \frac{1}{\log 2} (\psi_0(n_j/2) + \log 2) = \frac{1}{\log 2} \psi_0\left(\frac{n_j}{2}\right) - \frac{\log n_j - \log 2}{\log 2} \\ &= \frac{1}{\log 2} \left(\psi_0\left(\frac{n_j}{2}\right) - \log \frac{n_j}{2} \right) \sim -\frac{1}{n_j \log 2}, \end{aligned}$$

where the last asymptotic relation follows from $\psi_0(z) = \log z - (1/2z) + O(1/z^2)$, as $z \rightarrow \infty$ (Gradshteyn and Ryzhik [421], p. 902, Formula 8.361.3).

For $\text{Var}(\epsilon_j)$, observe that

$$\begin{aligned} (\log 2)^2 \text{Var}(\epsilon_j) &= (\log 2)^2 \text{Var} \left(\log_2 \sum_{k=1}^{n_j} |X_{j,k}|^2 \right) \\ &= \frac{1}{2^{n_j/2} \Gamma(n_j/2)} \int_0^\infty (\log x)^2 x^{n_j/2-1} e^{-x/2} dx \\ &\quad - \left(\frac{1}{2^{n_j/2} \Gamma(n_j/2)} \int_0^\infty (\log x) x^{n_j/2-1} e^{-x/2} dx \right)^2, \end{aligned}$$

which is $\zeta(2, n_j/2)$ by using the formulas above. The asymptotic relation follows directly from the definition of $\zeta(2, z)$ since for $z > 1$, $z^{-1} = \int_0^\infty (z+u)^{-2} du \leq \zeta(2, z) \leq \int_{-1}^\infty (z+u)^{-2} du = (z-1)^{-1}$ and hence $\zeta(2, z) \sim z^{-1}$ as $z \rightarrow \infty$. $\square$

These results suggest that d and c_2 can be estimated by a weighted least squares regression of y_j against j with weights $1/\sigma_j^2$, for some range $j = J_1, \dots, J_2$ of large j . Since the slope in (2.10.27) is $2d$, the resulting estimator of d is

$$\hat{d}_w = \frac{1}{2} \frac{\sum_{j=J_1}^{J_2} (1/\sigma_j^2) (z_j - \bar{z})(y_j - \bar{y})}{\sum_{j=J_1}^{J_2} (1/\sigma_j^2) (z_j - \bar{z})^2} = \frac{1}{2} \sum_{j=J_1}^{J_2} w_j y_j,$$

where

$$z_j = j, \quad \bar{z} = \frac{\sum_{j=J_1}^{J_2} (1/\sigma_j^2) z_j}{\sum_{j=J_1}^{J_2} (1/\sigma_j^2)}$$

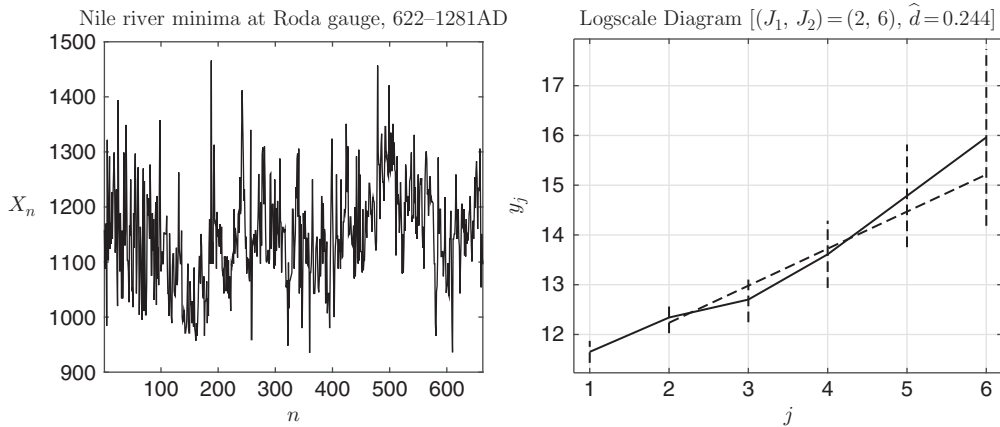


Figure 2.9 Left: Time series of the Nile river minima. Right: The corresponding log-scale diagram. The solid line connects the values of y_j , with confidence intervals indicated in vertical lines. The dashed line is the regression fit over $(J_1, J_2) = (2, 6)$.

and

$$w_j = \frac{(1/\sigma_j^2)(z_j - \bar{z})}{\sum_{j=J_1}^{J_2} (1/\sigma_j^2)(z_j - \bar{z})^2},$$

so that $\sum_{j=J_1}^{J_2} w_j = 0$. The main difference with the case of simple regression considered in Section 2.10.3 is that here, the sums are now weighted sums. One can express the regression weights in a convenient way as

$$w_j = \frac{S_0 j - S_1}{\sigma_j^2 (S_0 S_2 - S_1^2)}$$

with

$$S_k = \sum_{j=J_1}^{J_2} \frac{j^k}{\sigma_j^2}, \quad k = 0, 1, 2.$$

The plot of y_j versus j is known as the log-scale diagram. See Figure 2.9. The constant c_2 can be estimated similarly.

The variance of $\hat{d}_w$ is approximately

$$\text{Var}(\hat{d}_w) \approx \frac{1}{4} \frac{\sum_{j=J_1}^{J_2} ((1/\sigma_j^4)(z_j - \bar{z})^2 \sigma_j^2)}{(\sum_{j=J_1}^{J_2} (1/\sigma_j^2)(z_j - \bar{z})^2)^2} = \frac{1}{4} \frac{1}{\sum_{j=J_1}^{J_2} (1/\sigma_j^2)(z_j - \bar{z})^2} = \frac{S_0}{4(S_0 S_2 - S_1^2)}$$

and $\hat{d}_w$ is expected to be asymptotically normal since such y_j (or $\bar{\mu}_j$) is expected to satisfy a central limit theorem if the number of observations is large.

2.11 Generation of Gaussian Long- and Short-Range Dependent Series

Suppose that one would like to generate N samples $X = (X_0, \dots, X_{N-1})'$ of a stationary, zero mean Gaussian series $\{X_n\}$ having a known covariance function $\gamma_X(n)$, $n \in \mathbb{Z}$. For

notational simplicity, we write γ instead of γ_X throughout the section. The (column) vector X has the covariance structure

$$\Sigma = \begin{pmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(N-1) \\ \gamma(1) & \gamma(0) & \dots & \gamma(N-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(N-1) & \gamma(N-2) & \dots & \gamma(0) \end{pmatrix}. \quad (2.11.1)$$

We discuss below two methods for generation of X .

2.11.1 Using Cholesky Decomposition

According to the Cholesky decomposition, we can write

$$\Sigma = AA', \quad (2.11.2)$$

where A is a lower triangular matrix. Then, let Z be a $N \times 1$ vector of independent $\mathcal{N}(0, 1)$ random variables. Set

$$X = AZ. \quad (2.11.3)$$

The vector X is Gaussian, zero mean and has the desired covariance structure

$$\mathbb{E}XX' = A(\mathbb{E}ZZ')A' = AA' = \Sigma, \quad (2.11.4)$$

by using (2.11.2).

This method is of complexity $O(N^3)$ and hence relatively slow. It can be used to generate the series X of moderate sizes N only.

2.11.2 Using Circulant Matrix Embedding

This is an exact, fast and most popular synthesis method for Gaussian stationary series, with a given covariance structure. Since the method involves circulant matrices, we first recall some of their basic properties.

Recall that a matrix A is called *circulant* if it has a form

$$A = \begin{pmatrix} a_0 & a_1 & \dots & a_{n-2} & a_{n-1} \\ a_{n-1} & a_0 & \dots & a_{n-3} & a_{n-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_1 & a_2 & \dots & a_{n-1} & a_0 \end{pmatrix} = \left(a_{(k-j) \bmod n} \right)_{1 \leq j, k \leq n},$$

that is, the first row of A is $(a_0, a_1, \dots, a_{n-1})$ and every subsequent row is a circular shift by one element of the preceding row. The next result describes the eigenvector and eigenvalue structures of circulant matrices.

Proposition 2.11.1 *A circulant matrix A has orthonormal eigenvectors*

$$f_m = \frac{1}{\sqrt{n}} \left(1, e^{-i \frac{2\pi m}{n}}, e^{-i \frac{2\pi 2m}{n}}, \dots, e^{-i \frac{2\pi (n-1)m}{n}} \right)',$$

$m = 0, \dots, n-1$, and the corresponding eigenvalues

$$\lambda_m = \sum_{j=0}^{n-1} a_j e^{-i \frac{2\pi j m}{n}}. \quad (2.11.5)$$

Proof This is easy to verify. Denoting the k th element of Af_m and f_m by $(Af_m)_k$ and $(f_m)_k$, respectively, $k = 1, \dots, n$, observe that

$$\begin{aligned} (Af_m)_k &= \frac{1}{\sqrt{n}} \sum_{l=1}^n a_{(l-k) \bmod n} e^{-i \frac{2\pi(l-1)m}{n}} \\ &= \frac{1}{\sqrt{n}} \sum_{l=1}^{k-1} a_{n-k+l} e^{-i \frac{2\pi(l-1)m}{n}} + \frac{1}{\sqrt{n}} \sum_{l=k}^n a_{l-k} e^{-i \frac{2\pi(l-1)m}{n}} \\ &= \frac{1}{\sqrt{n}} \sum_{j=n-k+1}^{n-1} a_j e^{-i \frac{2\pi(j-n+k-1)m}{n}} + \frac{1}{\sqrt{n}} \sum_{j=0}^{n-k} a_j e^{-i \frac{2\pi(j+k-1)m}{n}} \\ &= \left(\sum_{j=0}^{n-1} a_j e^{-i \frac{2\pi j m}{n}} \right) \frac{1}{\sqrt{n}} e^{-i \frac{2\pi(k-1)m}{n}} = \lambda_m (f_m)_k. \end{aligned}$$

Orthonormality is easy to prove. □

Note that the eigenvectors f_m above form a discrete Fourier transform (DFT) basis. In practice, the eigenvalues λ_m can be computed using the fast Fourier transform (FFT). We have $Af_m = \lambda_m f_m$ and hence

$$AF = FG,$$

where $F = (f_0 \dots f_{n-1})$ is the DFT matrix, $G = \text{diag}(\lambda_0, \dots, \lambda_{n-1})$ and $*$ denotes Hermitian transpose. Recall that if F is a complex-valued matrix with entries $f_{j,k} = u_{j,k} + i v_{j,k}$, then the Hermitian transpose matrix F^* has entries $f_{j,k}^* = \bar{f}_{k,j} = u_{k,j} - i v_{k,j}$. Since $FF^* = I$, we get therefore

$$A = FGF^*.$$

We now turn to the generation of a Gaussian zero mean vector $X = (X_0, \dots, X_{N-1})'$ having the covariance structure (2.11.1). Set $M = N-1$ and define the following $2M \times 2M$ circulant matrix

$$\tilde{\Sigma} = \begin{pmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(N-2) & \gamma(N-1) & \gamma(N-2) & \dots & \gamma(1) \\ \gamma(1) & \gamma(0) & \dots & \gamma(N-3) & \gamma(N-2) & \gamma(N-1) & \dots & \gamma(2) \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma(N-1) & \gamma(N-2) & \dots & \gamma(1) & \gamma(0) & \gamma(1) & \dots & \gamma(N-2) \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \gamma(1) & \gamma(2) & \dots & \gamma(N-1) & \gamma(N-2) & \gamma(N-3) & \dots & \gamma(0) \end{pmatrix}. \quad (2.11.6)$$

Note that $\tilde{\Sigma}$ contains Σ in its upper-left corner. By the results for circulant matrices noted above,

$$\tilde{\Sigma} = FGF^*, \quad (2.11.7)$$

where, in view of (2.11.5), the diagonal matrix $G = \text{diag}(g_0, \dots, g_{2M-1})$ has diagonal entries

$$\begin{aligned} g_m &= \sum_{j=0}^M \gamma(j) e^{-i \frac{2\pi jm}{2M}} + \sum_{j=M+1}^{2M-1} \gamma(2M-j) e^{-i \frac{2\pi jm}{2M}} \\ &= \sum_{j=0}^M \gamma(j) e^{-i \frac{2\pi jm}{2M}} + \sum_{j=1}^{M-1} \gamma(j) e^{-i \frac{2\pi(2M-j)m}{2M}} \\ &= \gamma(0) + \gamma(M)(-1)^m + \sum_{j=1}^{M-1} \gamma(j) \left(\cos\left(\frac{-\pi jm}{M}\right) + \cos\left(\frac{\pi jm}{M}\right) \right) \\ &= \gamma(0) + \gamma(M)(-1)^m + 2 \sum_{j=1}^{M-1} \gamma(j) \cos\left(\frac{\pi jm}{M}\right). \end{aligned} \quad (2.11.8)$$

We shall now make the following assumption which will be discussed below.

Assumption ND Suppose that the matrix $\tilde{\Sigma}$ above is positive semidefinite. Equivalently, suppose that the eigenvalues $g_m, m = 0, \dots, 2M-1$, are nonnegative.

Under Assumption ND there is a Gaussian vector $\tilde{X}$ with the covariance structure $\tilde{\Sigma}$. We will show below that such $\tilde{X}$ can be constructed easily by using the factorization (2.11.7) of $\tilde{\Sigma}$ above. Since $\tilde{\Sigma}$ contains Σ in its upper-left corner, the first N elements X of $\tilde{X}$ will then have the desired covariance structure Σ .

To construct $\tilde{X}$ above, it is more convenient to work with complex-valued variables. Let

$$\tilde{Y} = FG^{1/2}Z,$$

where $F = (f_0 \dots f_{2M-1})$ is defined in Proposition 2.11.1, $G^{1/2} = \text{diag}(g_0^{1/2}, \dots, g_{2M-1}^{1/2})$ exists by using Assumption ND and $Z = Z^0 + iZ^1$ consists of two independent $\mathcal{N}(0, I_{2M})$ random vectors Z^0 and Z^1 . Note that, for $m = 0, \dots, 2M-1$,

$$\tilde{Y}_m = \frac{1}{\sqrt{2M}} \sum_{j=0}^{2M-1} g_j^{1/2} Z_j e^{-i \frac{2\pi jm}{2M}}, \quad (2.11.9)$$

so that $\tilde{Y}$ can be readily computed by FFT. The following result describes the covariance structure of $\tilde{Y}$.

Proposition 2.11.2 *The Gaussian vectors $\Re(\tilde{Y})$ and $\Im(\tilde{Y})$ are independent, with the covariance structure*

$$\mathbb{E}\Re(\tilde{Y})\Re(\tilde{Y})' = \mathbb{E}\Im(\tilde{Y})\Im(\tilde{Y})' = \tilde{\Sigma}.$$

Proof Note that $\mathbb{E}ZZ' = \mathbb{E}Z^0(Z^0)' - \mathbb{E}Z^1(Z^1)' = 0$, and hence

$$0 = \mathbb{E}\tilde{Y}\tilde{Y}' = \mathbb{E}(\Re(\tilde{Y}) + i\Im(\tilde{Y}))(\Re(\tilde{Y})' + i\Im(\tilde{Y})')$$

or

$$\mathbb{E}\Re(\tilde{Y})\Re(\tilde{Y})' = \mathbb{E}\Im(\tilde{Y})\Im(\tilde{Y})', \quad \mathbb{E}\Re(\tilde{Y})\Im(\tilde{Y})' + \mathbb{E}\Im(\tilde{Y})\Re(\tilde{Y})' = 0.$$

On the other hand, by (2.11.7),

$$\mathbb{E}\tilde{Y}\tilde{Y}^* = FG^{1/2}\mathbb{E}ZZ^*G^{1/2}F^* = 2FGF^* = 2\tilde{\Sigma},$$

since $\mathbb{E}ZZ^* = \mathbb{E}Z^0(Z^0)' + \mathbb{E}Z^1(Z^1)' = 2I_{2M}$. Hence

$$2\tilde{\Sigma} = \mathbb{E}\tilde{Y}\tilde{Y}^* = \mathbb{E}(\Re(\tilde{Y}) + i\Im(\tilde{Y}))(\Re(\tilde{Y})' - i\Im(\tilde{Y})')$$

or

$$\mathbb{E}\Re(\tilde{Y})\Re(\tilde{Y})' + \mathbb{E}\Im(\tilde{Y})\Im(\tilde{Y})' = 2\tilde{\Sigma}, \quad \mathbb{E}\Re(\tilde{Y})\Im(\tilde{Y})' - \mathbb{E}\Im(\tilde{Y})\Re(\tilde{Y})' = 0.$$

It follows that $\mathbb{E}\Re(\tilde{Y})\Re(\tilde{Y})' = \mathbb{E}\Im(\tilde{Y})\Im(\tilde{Y})' = \tilde{\Sigma}$ and $\mathbb{E}\Re(\tilde{Y})\Im(\tilde{Y})' = 0$. $\square$

In view of Proposition 2.11.2, both $\tilde{X} = \Re(\tilde{Y})$ and $\tilde{X} = \Im(\tilde{Y})$ have the covariance structure $\tilde{\Sigma}$. Finally, the desired vector X with the covariance structure Σ is made up of the first $N = M + 1$ elements of $\tilde{X}$.

We have then proved the following.

Theorem 2.11.3 Suppose that Assumption ND holds and let $\tilde{Y} = (Y_0, \dots, Y_{2M-1})'$ be the complex-valued vector defined in (2.11.9). Set either $\tilde{X} = \Re(\tilde{Y})$ or $\tilde{X} = \Im(\tilde{Y})$, and let the sub-vector X consist of the first $M + 1$ elements of $\tilde{X}$. Then, X is a real-valued Gaussian vector with covariance matrix Σ .

We shall discuss next Assumption ND. The first result shows that this assumption is expected to hold for most SRD time series of interest, at least for large enough N .

Proposition 2.11.4 If $\{X_n\}$ is a SRD series with (strictly) positive density function $f(\lambda)$, then Assumption ND holds for large enough N .

Proof For a SRD series, $2\pi f(\lambda) = \gamma(0) + 2 \sum_{j=1}^{\infty} \gamma(j) \cos(j\lambda)$. Moreover, the truncated series

$$G_M(\lambda) := \gamma(0) + \gamma(M) \cos(M\lambda) + 2 \sum_{j=1}^{M-1} \gamma(j) \cos(j\lambda)$$

converges to $2\pi f(\lambda)$ uniformly on $[0, 2\pi)$ because $\sum_{j=1}^{\infty} |\gamma(j)| < \infty$. Since f is positive, then $\inf_{\lambda \in [0, 2\pi)} |G_M(\lambda)| > 0$ for large enough M . It remains to observe that $G_M(\pi m/M) = g_m$, $m = 0, \dots, 2M - 1$, where g_m is given by (2.11.8). $\square$

Note that Proposition 2.11.4 does not indicate how large N should be for Assumption ND to hold. It rather explains why one expects Assumption ND to hold. From a practical perspective, if Assumption ND does not hold for a given N (that is, some of the eigenvalues g_m are negative), the result also suggests that one should try increasing N . Regarding Proposition 2.11.4, see also Exercise 2.21.

The next result, on the other hand, provides sufficient conditions for Assumption ND to hold for a fixed M .

Proposition 2.11.5 (a) Suppose that the sequence $\gamma(0), \dots, \gamma(M)$ is convex, decreasing and nonnegative. Then, Assumption ND holds.

(b) Suppose $\gamma(k) \leq 0$, $k \in \mathbb{Z} \setminus \{0\}$. Then, Assumption ND holds for any M .

Proof (a) Let $D_0(w) = 1$ and

$$\begin{aligned} D_j(w) &= 1 + 2 \sum_{k=1}^j \cos(\pi k w) = 1 + \frac{2}{\sin \frac{\pi w}{2}} \sum_{k=1}^j \cos(\pi k w) \sin \frac{\pi w}{2} \\ &= 1 + \frac{1}{\sin \frac{\pi w}{2}} \sum_{k=1}^j \left(\sin \left(\pi \left(k + \frac{1}{2} \right) w \right) - \sin \left(\pi \left(k - \frac{1}{2} \right) w \right) \right) = \frac{\sin(\pi(j + \frac{1}{2})w)}{\sin \frac{\pi w}{2}}. \end{aligned}$$

Proceeding as in summation by parts (see Exercise 2.2),

$$\begin{aligned} g_m &= \gamma(0) + \gamma(M)(-1)^m + 2 \sum_{j=1}^{M-1} \gamma(j) \cos \left(\frac{\pi j m}{M} \right) \tag{2.11.10} \\ &= \gamma(0) + \gamma(M)(-1)^m + \sum_{j=1}^{M-1} \gamma(j) \left(D_j \left(\frac{m}{M} \right) - D_{j-1} \left(\frac{m}{M} \right) \right) \\ &= \gamma(0) + \gamma(M)(-1)^m + \sum_{j=1}^{M-1} \gamma(j) D_j \left(\frac{m}{M} \right) - \sum_{j=0}^{M-2} \gamma(j+1) D_j \left(\frac{m}{M} \right) \\ &= \gamma(0) + \gamma(M)(-1)^m + \sum_{j=1}^{M-1} \Delta \gamma(j) D_j \left(\frac{m}{M} \right) + \gamma(M) D_{M-1} \left(\frac{m}{M} \right) - \gamma(1) D_0 \left(\frac{m}{M} \right) \\ &= \gamma(0) + \gamma(M)(-1)^m + \sum_{j=0}^{M-1} \Delta \gamma(j) D_j \left(\frac{m}{M} \right) - \gamma(0) D_0 \left(\frac{m}{M} \right) + \gamma(M) D_{M-1} \left(\frac{m}{M} \right) \\ &= \gamma(M) \left((-1)^m + D_{M-1} \left(\frac{m}{M} \right) \right) + \sum_{j=0}^{M-1} \Delta \gamma(j) D_j \left(\frac{m}{M} \right), \end{aligned}$$

where $\Delta \gamma(j) = \gamma(j) - \gamma(j+1)$ (the second term is indeed $\gamma(j+1)$, and not $\gamma(j-1)$).

Proceeding once again as in summation by parts,

$$\begin{aligned} g_m &= \gamma(M) \left((-1)^m + D_{M-1} \left(\frac{m}{M} \right) \right) + \sum_{j=0}^{M-1} \Delta \gamma(j) \left(K_j \left(\frac{m}{M} \right) - K_{j-1} \left(\frac{m}{M} \right) \right) \\ &= \gamma(M) \left((-1)^m + D_{M-1} \left(\frac{m}{M} \right) \right) + \sum_{j=0}^{M-1} \Delta \gamma(j) K_j \left(\frac{m}{M} \right) - \sum_{j=0}^{M-2} \Delta \gamma(j+1) K_j \left(\frac{m}{M} \right) \\ &= \gamma(M) \left((-1)^m + D_{M-1} \left(\frac{m}{M} \right) \right) + \Delta \gamma(M-1) K_{M-1} \left(\frac{m}{M} \right) + \sum_{j=0}^{M-2} \Delta^2 \gamma(j) K_j \left(\frac{m}{M} \right), \end{aligned}$$

where $\Delta^2\gamma(j) = \Delta\gamma(j) - \Delta\gamma(j+1)$, $K_{-1}(w) = 0$ and

$$\begin{aligned} K_j(w) &= \sum_{k=0}^j D_k(w) = \sum_{k=0}^j \frac{\sin(\pi(k + \frac{1}{2})w)}{\sin \frac{\pi w}{2}} = \frac{1}{(\sin \frac{\pi w}{2})^2} \sum_{k=0}^j \sin\left(\pi(k + \frac{1}{2})w\right) \sin\left(\frac{\pi w}{2}\right) \\ &= \frac{1}{2(\sin \frac{\pi w}{2})^2} \sum_{k=0}^j \left(\cos(\pi k w) - \cos(\pi(k+1)w) \right) = \frac{1 - \cos(\pi(j+1)w)}{2(\sin \frac{\pi w}{2})^2}. \end{aligned}$$

By the assumptions, $\gamma(M) \geq 0$, $\Delta\gamma(M-1) \geq 0$ and $\Delta^2\gamma(j) \geq 0$. Also $K_j(m/M) \geq 0$. Furthermore, setting $\alpha = \pi m/2M$, one gets $D_{M-1}(m/M) = \sin(\pi m - \alpha)/\sin \alpha = -(-1)^m$ if $m \neq 0$, and $D_{M-1}(0) = 1 + 2 \sum_{k=1}^{M-1} 1 = 2M - 1$ if $m = 0$. Hence $g_m \geq 0$ for $m = 0, \dots, 2M - 1$.

(b) Note that, since $\gamma(k) \leq 0$, $k = 1, \dots, M$, we have, in view of (2.11.10),

$$g_m \geq \gamma(0) + \gamma(M) + 2 \sum_{j=1}^{M-1} \gamma(j) = g_0.$$

It is therefore enough to show that $g_0 \geq 0$. As in the proof of Proposition 2.2.5, write

$$0 \leq M \operatorname{Var}(\bar{X}_M) = \gamma(0) + 2 \sum_{h=1}^{M-1} \left(1 - \frac{h}{M}\right) \gamma(h)$$

or

$$2 \sum_{h=1}^{M-1} \left(1 - \frac{h}{M}\right) (-\gamma(h)) \leq \gamma(0).$$

The dominated convergence theorem yields

$$2 \sum_{h=1}^{\infty} (-\gamma(h)) \leq \gamma(0).$$

In particular, since $\gamma(h) \leq 0$ for $h \geq M + 1$, this yields $g_0 \geq 0$. □

Many time series models satisfy conditions in Proposition 2.11.5. In the next result, we verify that this is the case for FARIMA(0, d , 0) series. Other series include fractional Gaussian noise (Exercise 2.20).

Proposition 2.11.6 (a) When $d \in (0, 1/2)$, a FARIMA(0, d , 0) series satisfies the conditions of Proposition 2.11.5, part (a), for any $M \geq 1$.

(b) When $d \in (-1/2, 0)$, a FARIMA(0, d , 0) series satisfies the conditions of Proposition 2.11.5, part (b).

Proof For a FARIMA(0, d , 0) series, by using (2.4.9) and (2.2.13), we have

$$\gamma(k) = \gamma(k-1) \frac{k-1+d}{k-d}. \quad (2.11.11)$$

Note that $\gamma(1) > 0$ if $d \in (0, 1/2)$, and $\gamma(1) < 0$ if $d \in (-1/2, 0)$. This implies that $\gamma(k) > 0$, $k \geq 1$, when $d \in (0, 1/2)$, and $\gamma(k) < 0$, $k \geq 1$, when $d \in (-1/2, 0)$. In particular, this proves part (b).

It remains to show that, when $d \in (0, 1/2)$, $\gamma(k) - \gamma(k+1) \geq 0$ and $\gamma(k) - 2\gamma(k+1) + \gamma(k+2) \geq 0$. For this observe that, by using the recursive relation (2.11.11),

$$\gamma(k) - \gamma(k+1) = \gamma(k) \frac{1-2d}{k+1-d}$$

and similarly

$$\gamma(k) - 2\gamma(k+1) + \gamma(k+2) = \gamma(k) \frac{2(1-d)(1-2d)}{(k+1-d)(k+2-d)} \geq 0. \quad \square$$

In Figure 2.2 of Section 2.4, we provided realizations of FARIMA(0, d , 0) series for several values of d , generated by using the circulant matrix embedding method. The realizations of FBM in Figure 2.3 were obtained through the increments of FBM, that is FGN, also generated by using the circulant matrix embedding method.

2.12 Exercises

The more difficult exercises are indicated by *. Open problems are marked by **, though in some instances the line between * and ** is quite blurred.

Exercise 2.1 Prove the inequality (2.2.2).

Exercise 2.2 Show the summation by parts formulae (A.2.3) and (A.2.5) found in Appendix A.2.

Exercise 2.3 Show the equivalence of (2.1.8) and (2.2.9). *Hint:* With $Y_n = \sum_{k=1}^n X_k$, show and use the relation $2N^2\gamma^{(N)}(k) = \text{Var}(Y_{(k+1)N}) - 2\text{Var}(Y_{kN}) + \text{Var}(Y_{(k-1)N})$. See Gubner [436].

Exercise 2.4 Show the relation (A.2.7) in Appendix A.2 using the results for fractional Gaussian noise. *Hint:* Approximate k^{-p} in (A.2.7) by $((k+1)^{2-p} - 2k^{2-p} + (k-1)^{2-p})/(2-p)(1-p)$.

Exercise 2.5 Show the relation (2.3.3) for SRD series.

Exercise 2.6 (i) Let $0 < \epsilon < \pi/2$ and g be a positive integrable function on $(0, \epsilon)$ satisfying (2.1.7) in condition IV. Consider the spectral density

$$f(\lambda) = g(|\lambda|)1_{(0,\epsilon)}(|x|) + g(|\pi - \lambda|)1_{(\pi-\epsilon,\pi)}(|x|), \quad -\pi < \lambda < \pi,$$

which satisfies (2.1.7) as well. Show that the corresponding autocovariance function γ satisfies $\gamma(n) = (1 + (-1)^n)\tilde{\gamma}(n)$ with $\tilde{\gamma}(n) = \int_{-\pi}^{\pi} \cos(n\lambda)g(\lambda)d\lambda$. Thus, condition IV does not necessarily imply condition II. (*Hint*: See Example 5.2 in Samorodnitsky [879].)

(ii) Let

$$g_1(\lambda) = 2^{2j}, \quad \text{if } 2^{-j} \leq \lambda \leq 2^{-j} + 2^{-2j}, \quad \text{for } j = 0, 1, 2, \dots$$

For $\lambda = 2^{-j}$, note that $g_1(\lambda) = \lambda^{-2}$, so that $\limsup_{\lambda \rightarrow 0} \lambda^2 g_1(\lambda) > 0$. Show that, for $n \geq 1$,

$$\left| \int_0^{\pi} \cos(n\lambda)g_1(\lambda)d\lambda \right| \leq Cn^{-1}(\log_2 n)^2.$$

Take now $f(\lambda) = g(\lambda) + g_1(\lambda)$, where $g(\lambda)$ is the spectral density of a LRD time series satisfying both conditions II and IV (e.g., a FARIMA(0, d , 0) series). Conclude that the time series having f as its spectral density satisfies condition II but not condition IV. (*Hint*: See Example 5.3 in Samorodnitsky [879].)

Exercise 2.7* Compare conditions I, II, IV and V for $d < 0$.

Exercise 2.8* For a FARIMA(0, d , 0), $d < 1/2$, series $X = \{X_n\}_{n \in \mathbb{Z}}$, show by induction and the Durbin–Levinson algorithm that the coefficients in the best linear predictor of X_{n+1} in terms of $X_n, \dots, X_1$ as

$$\hat{X}_{n+1} = \phi_{n1}X_n + \dots + \phi_{nn}X_1$$

are given by

$$\phi_{nj} = -\binom{n}{j} \frac{\Gamma(j-d)\Gamma(n-d-j+1)}{\Gamma(-d)\Gamma(n-d+1)}, \quad j = 1, \dots, n.$$

Deduce that the partial autocorrelation coefficients are given by $\alpha(h) = \phi_{hh} = d/(h-d)$.

Exercise 2.9* For a FARIMA(0, d , 0), $d < 1/2$, series $X = \{X_n\}_{n \in \mathbb{Z}}$, show that the coefficients θ_{nj} in the innovations algorithm

$$\hat{X}_n = \sum_{j=1}^n \theta_{nj}(X_{n-j} - \hat{X}_{n-j}),$$

where $\hat{X}_n$ is the best linear predictor of X_n in terms of $X_{n-1}, \dots, X_0$, are given by

$$\theta_{nj} = \frac{1}{\Gamma(d)} \frac{\Gamma(j+d)}{\Gamma(j+1)} \frac{\Gamma(n+1)}{\Gamma(n+1-d)} \frac{\Gamma(n-j+1-d)}{\Gamma(n-j+1)}.$$

Hint: See Proposition 2.1 in Kokoszka and Taqqu [574].

Exercise 2.10 Prove the relation (2.4.17). *Hint*: See Kokoszka and Taqqu [571], Corollary 3.1.

Exercise 2.11 If $\{X(t)\}_{t \in \mathbb{R}}$ is Gaussian with mean 0 and $X(0) = 0$, show that stationarity of the increments is equivalent to (2.5.3).

Exercise 2.12* Show that for $0 < \gamma < 1$ in Proposition 2.5.6, (c), one has $H < 1/\gamma$ and that the conclusion in Proposition 2.5.6, (f) holds when $\mathbb{E}|X(1)| < \infty$. *Hint:* See Maejima [664], Vervaat [990].

Exercise 2.13 If the assumptions of Proposition 2.5.6, (f), hold and, in addition, the process X is right-continuous, show that almost surely, $X(t) = X(1)t$ for all $t \in \mathbb{R}$.

Exercise 2.14 Show that the finite-dimensional distributions of a standard FBM $\{B_H(t)\}_{t \in \mathbb{R}}$ converge as $H \rightarrow 0$ to those of the Gaussian process $\{(\xi_t - \xi_0)/\sqrt{2}\}_{t \in \mathbb{R}}$, where $\xi_t, t \in \mathbb{R}$, are i.i.d. $\mathcal{N}(0, 1)$ random variables. *Hint:* Show the convergence of the covariance functions.

Exercise 2.15 Show that the log-FSM (2.6.36) is well defined, has stationary increments and is self-similar with $H = 1/\alpha$.

Exercise 2.16 Show that the linear series (2.9.2) is well defined and α -heavy tailed under the assumptions (2.9.5) and (2.9.6), and the innovations ϵ_k being symmetric i.i.d. α -heavy-tailed.

Exercise 2.17 Complete the proof of Proposition 2.10.1 by showing (2.10.14).

Exercise 2.18* (i) Use the arguments developed in Section 2.10.3 to show the asymptotic normality of the estimator of the constant c_4 in (2.10.11). (ii) Similarly, derive the joint asymptotics of the estimators of d and c_4 .

Exercise 2.19* Proposition 2.10.3, (iv), is stated by using time-domain conditions and proved in the time domain. State and prove this result by using conditions and arguments in the spectral domain. *Hint:* Note that, under suitable assumptions, the condition (2.10.19) can be expressed as $\hat{\psi}(x) = Cx^Q + o(x^Q)$, as $x \rightarrow 0$.

Exercise 2.20 Show that FGN satisfies the assumptions of Proposition 2.11.5, part (a), when $H > 1/2$, and part (b), when $H < 1/2$.

Exercise 2.21** Can Proposition 2.11.4 be extended to LRD series? This is essentially equivalent to showing that the convergence of the series in Propositions A.2.1 and 2.2.14 is also uniform over λ .

2.13 Bibliographical Notes

Section 2.1: We begin with a few historical remarks. The origins of long-range dependence can be traced to the works of:

- Hurst [501] who observed long-range dependence in analyzing the Nile river historical levels;

- Cox and Townsend [266] who discovered slowly decaying correlations in textile yarn measurements;
- Whittle [1007] who observed power-law correlations in the spatial context for yields in terms of plot sizes.

The books by Beran [121] and Beran, Feng, Ghosh, and Kulik [127] contain nice historical overviews of long-range dependence, with even earlier references discussing phenomena related to long-range dependence. See also Graves, Gramacy, Watkins, and Franzke [426].

Section 2.2: The results of this section relate the various definitions of long-range dependence under quite general assumptions involving slowly varying functions. As indicated in Remark 2.2.19, some of the relationships have been assumed erringly in the literature. As it can be seen from the section, most facts about slowly varying functions are drawn from the excellent and exhaustive monograph by Bingham et al. [157]. For other discussions of the various definitions of LRD, see also Terdik [955], Heyde [472], Ma [657], Guégan [437], Inoue [520]. Other conditions for relating the definition of LRD in the time and spectral domains are studied in Inoue [518].

Propositions 2.2.14 and 2.2.16 are sometimes referred to, respectively, as Abelian and Tauberian theorems (see Sections 4.3 and 4.10 in Bingham et al. [157]). Abelian theorems generally refer to results on transformations of functions and Tauberian theorems to results on functions based on their transformations, which are generally thought to require more stringent assumptions (see, for example, the discussion in the beginning of Chapter 4 in Bingham et al. [157]). Since it is a bit arbitrary what to call a function or its transformation in Propositions 2.2.14 and 2.2.16, the terminology of the Abelian and Tauberian theorems is not exactly precise.

Section 2.3: ARMA(p, q) models in Example 2.3.3 are studied using standard techniques in time series analysis (Brockwell and Davis [186]), while Markov chains in Example 2.3.4 are treated by using algebraic techniques (Feller [361], Chapter XVI). On the other hand, a Markov chain with a countable state space can exhibit long-range dependence when a return time from a state to itself has infinite variance – see Carpio and Daley [204], Oğuz and Anantharam [777].

The special role of antipersistent series seems to have been recognized early, for example, by Mandelbrot and Van Ness [682], Mandelbrot and Wallis [684]. See Bondon and Palma [174] for more recent work focusing on antipersistent series.

Section 2.4: FARIMA(p, d, q) models were introduced simultaneously by Hosking [484] and Granger and Joyeux [424]. This class of models has become centerpiece in modeling long-range dependent time series. Though see also Veitch, Gorst-Rasmussen, and Gefferth [983]. Strictly stationary solutions of ARMA equations with fractional noise are studied by Vollenbröker [992].

An interesting extension of stationary FARIMA models consists of the Gegenbauer and Gegenbauer ARMA (GARMA) models. For example, the stationary Gegenbauer time series $\{X_n\}$ satisfies the equation

$$(I - 2uB + B^2)^{d/2}X_n = Z_n, \quad n \in \mathbb{Z},$$

where $\{Z_n\} \sim WN(0, \sigma_Z^2)$ and either $|u| < 1, d < 1$ or $u = \pm 1, d < 1/2$. The spectral density of the Gegenbauer process can be expressed as

$$f_X(\lambda) = \frac{\sigma_Z^2}{2\pi} 2^{-d} |\cos(\lambda) - u|^{-d}.$$

Note that when $|u| < 1$ and $0 < d < 1$, the spectral density no longer explodes at $\lambda = 0$ as for the usual long-range dependence but at the so-called *G frequency* (Gegenbauer frequency) $\lambda_0 = \arccos u \neq 0$. In this case, the series also naturally exhibits seasonal variations with the period corresponding to the G frequency. One can show that for the same range of parameter values u and d , the autocovariance function satisfies

$$\gamma_X(n) \sim Cn^{d-1} \cos(\lambda_0 n), \quad \text{as } n \rightarrow \infty,$$

that is, it decays slowly as for the usual long-range dependence but is also characterized by the oscillating behavior. See Gray, Zhang, and Woodward [427, 428]. The GARMA series satisfy the equation of the Gegenbauer series above with the added AR and MA polynomials. For recent work on GARMA series and extensions, see McElroy and Holan [705, 706], Caporale, Cuñado, and Gil-Alana [201], Dissanayake, Peiris, and Proietti [317].

Section 2.5: The study of general self-similar processes was initiated by Lamperti [589]. See also Vervaat [990, 991], Maejima [664]. Self-similar processes are the focus of the book by Embrechts and Maejima [350]. Extensions of the notion of self-similarity for univariate processes are considered in van Harn and Steutel [971], Eglói [346].

Section 2.6: The notes below are separated according to the self-similar processes introduced in the section.

Though introduced as early as in Kolmogorov [575], Pinsky and Yaglom [811], Yaglom [1019], fractional Brownian motion (FBM) was popularized in great part by Mandelbrot and Van Ness [682] who saw the practical potential of the FBM model. What could be regarded as FBM with index $H = 0$ is introduced in Fyodorov, Khoruzhenko, and Simm [377]. Alternative forms of FBM, the so-called type I and type II FBMs, are considered in Marinucci and Robinson [690]. n th-Order FBM is introduced in Perrin, Harba, Berzin-Joseph, Iribarren, and Bonami [807]. See also the notes to Chapter 7 on FBM below.

Bifractional Brownian motion was introduced by Houdré and Villa [489], and has since been studied by a number of researchers, including Russo and Tudor [873], Tudor and Xiao [965], Lei and Nualart [603].

The Rosenblatt process was introduced by Taqqu [943] in connection to non-central limits theorems, naming it after Rosenblatt [860] who considered its marginal distribution. See Chapters 4 and 5 for more information.

Linear fractional stable motion appears first in Astrauskas [53]. Its asymptotic dependence structure is analyzed in Astrauskas et al. [54]. Various models are considered in Samorodnitsky and Taqqu [881, 882]. The linear multifractional stable motion is treated in Stoev and Taqqu [925, 928]. Many other stable self-similar processes, including the Telecom process, can be found in Pipiras and Taqqu [820].

The material on linear fractional Lévy motion is taken from Cohen and Iatas [253], Section 4.4. See also Marquardt [692].

Fractional Ornstein–Uhlenbeck Lévy processes are analyzed in Wolpert and Taqqu [1009]. See also Taqqu and Wolpert [950]. Self-similar processes with stationary increments generated by point processes are studied in O’Brien and Vervaat [776].

Section 2.7: The Lamperti transformation is named after Lamperti [589]. Other interesting work related to the Lamperti transformation includes Burnecki, Maejima, and Weron [195], Flandrin, Borgnat, and Amblard [367], Genton, Perrin, and Taqqu [386], Matsui and Shieh [698].

Section 2.8: Fractional Gaussian noise was introduced in Mandelbrot and Van Ness [682]. As indicated in the section, the fundamental limit theorem is due to Lamperti [589].

The convergence of the partial sums of linear time series to fractional Brownian motion (cf. Proposition 2.8.8) was considered by Davydov [291], Gorodetskii [418].

Section 2.9: Infinite variance long memory time series are considered in Kokoszka and Taqqu [571, 572, 573], Peng [805], Burnecki and Sikora [194]. Long-range dependence and heavy tails also appear in the slightly different context of stochastic volatility models (Kulik and Soulier [581], McElroy and Jach [704]). Other definitions of long-range dependence with infinite variance are considered in Magdziarz [670], Damarackas and Paulauskas [286], Paulauskas [795, 796].

The proof of Proposition 2.9.3 is based on Astrauskas [53]. See also Barbe and McCormick [95], Peligrad and Sang [801, 802], Duffy [333] for more recent work.

Univariate stable distributions is the focus of a fascinating monograph by Uchaikin and Zolotarev [967], as well as of the classic monograph by Zolotarev [1029]. For a more recent treatment of stable distributions and processes, see Rachev and Mittnik [839], Nolan [755].

Section 2.10: As indicated in the section, the R/S method originated with Hurst [501], and has since attracted much attention, especially in the geophysics literature (e.g., Lawrence and Kottegoda [599], Montanari, Rosso, and Taqqu [731], Dmowska and Saltzman [319]). Mandelbrot and Wallis [683, 684, 686, 685] have a number of papers on the subject. A general overview about applications to hydrology can be found in Montanari [729]. Klemeš [563], however, raises doubts about the presence of long memory in hydrological time series.

The theoretical properties of the R/S method were studied by Mandelbrot [678]. See also Feller [359] who proves convergence in the case of i.i.d. data, and Hamed [448], Li, Yu, Carriquiry, and Kliemann [624] for more recent improvements. Several methods related to and improving on R/S , such as V/S , KPS and others, are considered Giraitis, Kokoszka, Leipus, and Teyssière [405].

The regression in the spectral domain was introduced by Geweke and Porter-Hudak [387]. Theoretical properties of the resulting estimator of the long-range dependence parameter were studied in depth by Robinson [852]. Other related studies include Agiakloglou, Newbold, and Wohar [15], Delgado and Robinson [299], Hurvich, Deo, and Brodsky [510], Hurvich and Deo [507], Velasco [985, 986], Kim and Phillips [555], Arteche and Orbe [49, 50]. Extensions include Andersson [31], Andrews and Guggenberger [32], Guggenberger and Sun [439], Feng and Beran [362]. See also Section 10.9 below with some related notes to Chapter 10.

The wavelet estimation methods for long-range dependent series were popularized in a series of papers by Abry and Veitch [8], Veitch and Abry [981]. But useful connections

between wavelets, self-similarity and long-range dependence were recognized earlier in, for example, Flandrin [366], Tewfik and Kim [956], Dijkerman and Mazumdar [315], Wornell [1011], Abry, Veitch, and Flandrin [9]. Some earlier related work can be traced at least to Allan [19]. The work by Abry and Veitch were followed by a number of other interesting studies including Bardet [96, 97], Roughan, Veitch, and Abry [870], Veitch, Taqqu, and Abry [982], Craigmile, Guttorp, and Percival [268], Moulines, Roueff, and Taqqu [735, 736, 737], Teyssière and Abry [957], Bardet and Bibi [98], Bardet, Bibi, and Jouini [99], Faÿ, Moulines, Roueff, and Taqqu [357], Roueff and Taqqu [866, 867], Clausel, Roueff, Taqqu, and Tudor [248, 249, 250].

For a succinct introduction to the wavelet estimation methods for long-range dependent series, see Abry, Flandrin, Taqqu, and Veitch [10]. For visualization tools that can be used in the local analysis of self-similarity, see Stoev, Taqqu, Park, Michailidis, and Marron [929] and Park, Godtliessen, Taqqu, Stoev, and Marron [793]. For an analysis of Ethernet data, see Stoev et al. [929]. Stoev and Taqqu [924] describe the wavelet estimation of the Hurst parameter in stable processes. The case of FARIMA time series with stable innovations is considered in Stoev and Taqqu [927].

Section 2.11: The circulant matrix embedding method seems to go back to Davies and Harte [290]. It was studied more extensively by Dembo, Mallows, and Shepp [301], Dietrich and Newsam [313, 314], Lowen [646], Gneiting [411], Stein [920], Craigmile [267], and Percival [806]. Extensions to multivariate (vector) time series were considered by Chan and Wood [214], Helgason, Pipiras, and Abry [463], and those to random fields by Dietrich and Newsam [313], Wood and Chan [1010], Chan and Wood [213], Stein [921, 922], Gneiting, Ševčíková, Percival, Schlather, and Jiang [413], and Helgason, Pipiras, and Abry [466].

Simulation of stationary Gaussian series from given spectral densities is considered in e.g., Chambers [211], Azimmoheeni, Soltani, and Khalafi [64]. A turning-band method for the simulation of fractional Brownian fields can be found in e.g., Biermé and Moisan [149]. Midpoint displacement methods are discussed in Brouste, Istas, and Lambert-Lacroix [191].